

# Racial Discrimination and Innovation: Evidence from U.S. Inventors, 1882–1936\*

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## Abstract

Does racial discrimination affect which inventors are granted patents? We study this question in the late nineteenth- and early twentieth-century United States. We begin by documenting that individuals with Black-sounding surnames were one-fourth as likely to patent as those with white-sounding surnames, even conditioning on race. To test whether discrimination in the patenting process contributed to this gap, we use lynching episodes—acts of extreme violence over this period that disproportionately targeted Black Americans—as shocks to the racial salience of victims’ surnames. Following an episode, patents granted to inventors sharing the victim’s surname decline over the following five years, with the largest effect immediately after the shock. The evidence is consistent with lynching episodes changing the racial signal conveyed by surnames. A back-of-the-envelope calculation suggests that removing surname-based racial signals would reduce the patenting gap between Black- and white-sounding inventors by 24 percent. Our findings highlight that racial discrimination can shape patenting not only through barriers to entering innovation, but also among individuals already engaged in the innovation process.

**Keywords:** Racial Discrimination, Innovation, Lynching Episodes

**JEL Classification:** J15, N31, N32, O11, O31

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# 1 Introduction

At the beginning of the twentieth century, Black Americans were one-sixth as likely as white Americans to hold a patent, a gap that has shown little improvement to date (Andrews and Rothwell, 2023; Akegiti and Goldschlag, 2025). Barriers that prevent talented individuals from entering innovation can generate aggregate losses through the misallocation of talent; because innovation also generates substantial knowledge spillovers, lost inventive activity may have broader consequences for economic growth (Bloom, Schankerman and Van Reenen, 2013; Hsieh, Hurst, Jones and Klenow, 2019). Moreover, patents confer property rights with potentially high private returns. Persistent disparities in access to these rights may therefore generate or reinforce economic inequality. Yet evidence on the mechanisms underlying racial gaps in patenting, both historically and today, remains limited.

Racial discrimination may contribute to this gap at several stages. It may shape who becomes an inventor through unequal access to education, exposure to innovation, and labor-market opportunities (Bell, Chetty, Jaravel, Petkova and Van Reenen, 2019a,b). It may also affect whether individuals develop and commercialize their inventions or seek patent protection by limiting access to professional networks, wealth, and credit (e.g., Cook, Gerson and Kuan, 2022). This paper studies whether discrimination also affects patenting outcomes after an invention has been developed and the patenting process has begun. The analysis focuses on the late nineteenth- and early twentieth-century United States.

Identifying discrimination at this stage is challenging for several reasons. First, it is difficult to separate barriers encountered during the patenting process from the many forms of racial exclusion that shape who becomes an inventor and who applies for a patent. Second, Black inventors account for only a small share of patentees, limiting statistical power and making differential treatment difficult to estimate precisely. Third, because the Patent Office did not record the race or ethnicity of applicants or inventors, even measuring racial disparities at this stage is difficult.

The analysis proceeds in four steps. First, we draw on the universe of patents granted by the United States Patent Office between 1882 and 1936 and digitized by Berkes (2018). We document that individuals with Black-sounding names—measured by a Black Surname Index (BSI) capturing how disproportionately common a surname is among Black relative to white individuals—are under-represented among patentees relative to their frequency in the population. Leveraging patent records linked to the U.S. population census by Bazzi, Berkes, Fiszbein and Fons-Rosen (2022), we show that this association is not explained by observable inventor characteristics, including occupation and geographic location. Nor is it accounted for by inventors’ recorded race: within the white subsample, individuals with higher-BSI are less likely to patent, while estimates for Black individuals are imprecise but consistent with the same pattern.

In the second step of the analysis, we develop an identification strategy that uses shocks to the racial salience of surnames. Between 1882 and 1936, 3,563 lynching episodes were recorded in the United States, the vast majority of which targeted Black Americans (Hines and Steelwater, 2006; Seguin and Rigby, 2019). Although these episodes were geographically concentrated in the South,

they received nationwide newspaper coverage, with reports typically covering the victim’s name, their race, and their alleged offense. Such coverage could therefore change the racial signal conveyed by the victim’s surname to individuals who encountered it, including participants in the patenting process.

Using a difference-in-differences design, we compare changes in patenting before and after a lynching episode among individuals sharing the victim’s surname with changes among individuals bearing matched control surnames. All specifications include surname fixed effects, which absorb time-invariant characteristics associated with a surname—such as socioeconomic background or its baseline racial signal—and matched-pair-by-year fixed effects, which compare each treated surname with its matched control in the same year. The identifying assumption is that, absent the episode, patents granted to inventors bearing a treated surname would have evolved in parallel with those granted to inventors bearing the matched control surname. We assess the plausibility of this assumption using event-study estimates. Our main analysis focuses on surnames rather than first names because surnames are harder to change—inventors could plausibly alter a first name in response to an episode—and because they are more informative about race in this period.

Following a lynching episode, fewer patents are granted to individuals who share the victim’s surname. Patents granted to treated surnames decline by approximately 7 percent relative to matched controls over the five years following the shock. The effect is largest in the year immediately after the episode and gradually reverts to zero within five years. It persists when we restrict the analysis to patents whose applications were filed before the episode, indicating that it cannot be explained solely by a decline in new applications. Because rejected and abandoned applications were not preserved, however, we cannot distinguish among delayed issuance, denial, and post-filing abandonment by inventors or their attorneys.

We then turn to inventor-level data, which allow us to include inventor fixed effects and thereby control for time-invariant individual characteristics, including race. We focus on the intensive margin of patenting by restricting the sample to inventors who were granted at least one patent both before and after the episode. Although this restriction conditions on post-episode patenting, it holds the set of inventors fixed, allowing us to test whether patenting declined among continuing inventors rather than because of changes in the composition of the inventor pool.

In the third step, we examine the mechanisms behind these results. We show that the decline is not larger among inventors living closer to the episode and is not driven by the states or regions where lynching episodes were concentrated, pointing away from direct local exposure to racial violence as the main mechanism. The inventor-level results provide complementary evidence: the aggregate decline is not confined to inventors recorded as Black, but is also negative among white inventors bearing treated surnames.

Instead, we find evidence consistent with an information-transmission channel. The effect is concentrated among episodes receiving coverage in newspapers based in or near Washington, DC, and is absent among episodes receiving no such coverage. The decline is also larger for episodes occurring closer to Washington. Together, these findings are consistent with information about the episode reaching the patenting environment. Several channels could generate this response: examiners may

have evaluated applications bearing treated surnames less favorably or taken longer to grant them; employers, assignees, or financial intermediaries may have become less willing to provide support; and inventors or their attorneys may have stopped pursuing applications in expectation of differential treatment.

We then ask whether lynching episodes specifically increased the racial salience of treated surnames or instead attached to them a more general stigma associated with violence. Three findings support the racial-signaling interpretation. First, we find no detectable effect following episodes involving white victims. Second, the decline is concentrated among surnames whose initial racial signal was not unambiguously Black—for which a lynching episode plausibly conveyed the most new information about racial identity. Finally, we find no systematic relationship between the magnitude of the decline and the alleged offense attributed to the victim, whereas a stigma-related signal would plausibly have been stronger following allegations of more serious violent offenses.

In the fourth and final step, we provide an illustrative quantification of the role of surname-based racial signals in explaining the patenting gap across the BSI distribution. We use the estimated effect to calculate how patenting would change if surnames no longer conveyed a racial signal. First, this would have distributional effects, leading the patenting gap between the lowest- and highest-BSI quintiles to fall by approximately 24%. Under the stronger assumption that patents lost by one inventor are not replaced by patents from others, the total number of patents would increase by approximately 11%. If instead all inventors were perfectly substitutable in the patents they are granted, there would be no overall effect on the volume of innovation.

Together, our findings provide novel evidence that racial discrimination can shape patenting outcomes not only by creating barriers to entering innovation, but also by affecting individuals who have already entered the patenting process. More broadly, they show that racial signals conveyed by names can influence outcomes even in settings where decisions are intended to depend primarily on technical information. Methodologically, the paper shows how surname-specific shocks can be used to isolate changes in the informational content of names, overcoming the challenge that names simultaneously proxy for race, socioeconomic background, and other characteristics. Because the approach does not require observing inventors’ actual race, it is particularly well suited to historical settings in which such information is incomplete or unavailable.

***Related Literature.*** First, we contribute to the literature on racial discrimination in high-stakes evaluation and allocation decisions. A large body of work documents discrimination in labor markets (among others, Bertrand and Mullainathan, 2004; Oreopoulos, 2011; Moser, 2012; Edelman, Luca and Svirsky, 2017; Kline, Rose and Walters, 2022). Related studies show that discriminatory treatment can also arise in decisions made by public officials, including judges (Arnold, Dobbie and Yang, 2018), police officers (Goncalves and Mello, 2021), public service workers (Einstein and Glick, 2017; Giulietti, Tonin and Vlassopoulos, 2019), and elected politicians (Butler and Broockman, 2011). We extend this literature to the patent system and examine discrimination after individuals have entered innovation and begun the patenting process. To the best of our knowledge, we provide the first causal evidence that racial discrimination can shape patenting outcomes at this stage, adding to a related body of work

providing evidence consistent with gender bias at the Patent Office (Jensen, Kovács and Sorenson, 2018; Avivi, 2024).

Second, we contribute to the literature on discrimination against Black Americans in the pre-Civil Rights era. In a seminal contribution, Cook (2014) shows that Black patenting declines following major discriminatory episodes, including lynching episodes, race riots, and segregation laws. We build on this work by isolating a channel operating during the patenting process through which discriminatory episodes affected innovation. A broader literature documents the pervasive effects of racial discrimination on employment segregation (Aneja and Xu, 2022), wealth inequality (Akbar, Hickly, Shertzer and Walsh, 2022; Derenoncourt, Kim, Kuhn and Schularick, 2024), housing segregation (Logan and Parman, 2017; Shertzer and Walsh, 2019; Derenoncourt, 2022), and mortality and violence (Cook, 2014; Black, Sanders, Taylor and Taylor, 2015; Cook, Logan and Parman, 2018; Albright, Cook, Feigenbaum, Kincaide, Long and Nunn, 2021). Andrews and Rothwell (2023) examine the contributions of Black Americans during the Golden Age of Innovation; our paper identifies one mechanism through which discrimination limited those contributions.

Third, we add to the literature that uses names to proxy for otherwise unobserved individual characteristics, including race (Fryer and Levitt, 2004; Abramitzky, Boustan and Connor, 2024), parental investment and individualism (Bazzi, Fiszbein and Gebresilasse, 2020; Knudsen, 2021), religiosity (Andersen and Bentzen, 2022; Berkes, Coluccia, Dossi and Squicciarini, 2024), socioeconomic background (Biavaschi, Giuliatti and Siddique, 2017; Olivetti, Paserman, Salisbury and Weber, 2020), geographic origin (e.g., Ochsner and Roesel, 2020), nationalism (e.g., Jurajda and Kovač, 2021), and immigrant assimilation (e.g., Fouka, 2020). To the best of our knowledge, we are the first to use name-specific shocks as a source of identification, extending this literature beyond the use of names as proxies for otherwise unobserved characteristics.

**Outline of the Paper.** The remainder of the paper is organized as follows. Section 2 describes the historical and institutional background. Section 3 presents the data. Section 4 documents stylized facts. Section 5 outlines the research design, and Section 6 presents the main results. Section 7 explores mechanisms, Section 8 provides a quantification of the effect, and Section 9 concludes.

## 2 Historical Background

This section provides background on the patenting process at the U.S. Patent Office (Section 2.1) and on lynching episodes in the United States (Section 2.2).

### 2.1 The U.S. Patent Office and the Patenting Process

The United States Constitution empowers Congress to secure for inventors the exclusive right to their discoveries for a limited time. The Patent Act of 1836 reorganized the U.S. patent system and created a distinct Patent Office with formal examining authority, laying the foundations for what is often described as the first “modern” patent system in the world (Khan and Sokoloff, 2004; Khan, 2020). Relative to its European counterparts, the American patent system had two distinctive features. First, patent applications were subject to substantive examination for novelty to assess originality. Following

the 1836 Act, these examinations were conducted by salaried public officers—patent examiners—who were legally prohibited from applying for patents themselves. Second, the Patent Office maintained low application fees throughout the nineteenth and early twentieth centuries, ensuring broad access to intellectual property protection (Sokoloff and Khan, 1990).

Until 1941, the U.S. Patent Office operated from a single location in Washington, DC. During the period we study, the patenting process followed a standardized three-stage procedure. First, applicants submitted patent applications by mail. The Patent Office then assigned each application to one or more technological classes based on its content. Unlike in the modern system, technological classes were assigned exclusively to individual examiners: each examiner covered multiple classes, and no class was reviewed by more than one examiner.<sup>1</sup> Both patent applications and granted patents reported the applicant’s first and last name, city of residence, assignee (when applicable), and a detailed description of the invention. In the final stage, examiners either granted the application or rejected it; applicants could also abandon an application during examination, for instance by failing to respond to an examiner’s objections or to pay the final fee. The average delay between filing and issuance was approximately 1.5 years. When a patent was granted, information on the issued patent was recorded and preserved. When an application was rejected or abandoned, all associated documents were eventually destroyed, leaving no surviving archival record (Andrews, 2021). Our sample, therefore, consists only of issued patents.

Our empirical strategy relies on the premise that participants in the patenting process typically did not directly observe applicants’ race. This assumption is supported by the fact that the Patent Office operated entirely by written correspondence: examiners in Washington, DC evaluated applications submitted by mail and rarely, if ever, met applicants in person. The same was often true of other parties involved in securing patent protection, such as patent attorneys and prospective assignees, who frequently dealt with inventors at a distance. As the city of residence reported on patent applications was typically too coarse to convey meaningful information about race, applicants’ first and last names were the main information available to these actors that could plausibly signal race.<sup>2</sup>

## 2.2 Lynching Episodes in the United States

A lynching is an extrajudicial public execution carried out by a mob, typically intended to punish, intimidate, or kill an alleged transgressor. In the United States, lynching episodes were most often associated with violent collective action by white mobs against Black Americans, particularly during Reconstruction and the early twentieth century (Hines and Steelwater, 2006; Wells-Barnett, 2019). A large body of scholarship identifies economic, political, and social factors as key drivers of these episodes (Olzak, 1990; Tolnay and Beck, 1995). Beyond their immediate political and economic consequences (e.g., Cook, 2014; Jones, Troesken and Walsh, 2017), lynching episodes have been shown to exert

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<sup>1</sup>In 1910, the Patent Office employed 41 examiners. On average, each examiner was responsible for 3.3 technological classes, with no overlap across classes (Coluccia, 2025).

<sup>2</sup>While earlier research emphasized that distinctively Black names arose primarily during the Civil Rights era (Fryer and Levitt, 2004), more recent evidence indicates that they were already in use and socially recognized by the late nineteenth century (Cook, Logan and Parman, 2014).

persistent effects extending well into the present (Williams, 2017). In this paper, we conceptualize lynching episodes as salient public events that plausibly updated beliefs about the racial identity associated with the names of the victims.

A typical lynching episode followed a criminal accusation and arrest, after which a mob assembled and physically assaulted the alleged transgressor, often resulting in death. Alleged offenses ranged from murder, arson, and robbery to sexual violence (Seguin and Rigby, 2019). Using data from Hines and Steelwater (2006) and Seguin and Rigby (2019), Figure Ia reports the number of lynching episodes between 1882 and 1936. As shown in the figure, Black Americans were the primary targets. These episodes were heavily concentrated in the South, although they also occurred in other regions (cf. Figure Ib).<sup>3</sup>

Lynching episodes received extensive coverage in the mass media, primarily through newspapers (Perloff, 2000; Weaver, 2019). This feature is central to our empirical strategy because it implies that patent examiners—despite being geographically distant from many lynching locations—were likely exposed to information about these events.<sup>4</sup> In Figure A.2, we present two illustrative newspaper articles published in Washington, DC, both of which report on lynching episodes and explicitly mention the victim’s first name, surname, and race.<sup>5</sup> These episodes were reported well beyond the local area: Figure A.3 shows articles from geographically diverse newspapers covering the same lynching episode in Lafayette County, Mississippi.

### 3 Data

This section describes the data and the key variables. Section 3.1 introduces the Black Surname Index (BSI), Section 3.2 describes the data on lynching episodes, and Section 3.3 presents the patent and inventor data.

#### 3.1 Quantifying the Racial Informativeness of Surnames

Building on prior work, we construct a measure of the racial signal conveyed by surnames (Fryer and Levitt, 2004; Abramitzky et al., 2024; Fouka, 2020). Using full-count confidential U.S. population census data, we recover information on surnames for the universe of the U.S. population in 1900 (Ruggles, Fitch, Goeken, Hacker, Nelson, Roberts, Schouweiler and Sobek, 2021). We use these data to construct a Black Surname Index (BSI), which captures the extent to which a surname is disproportionately common among the Black population. The underlying idea is that surnames that

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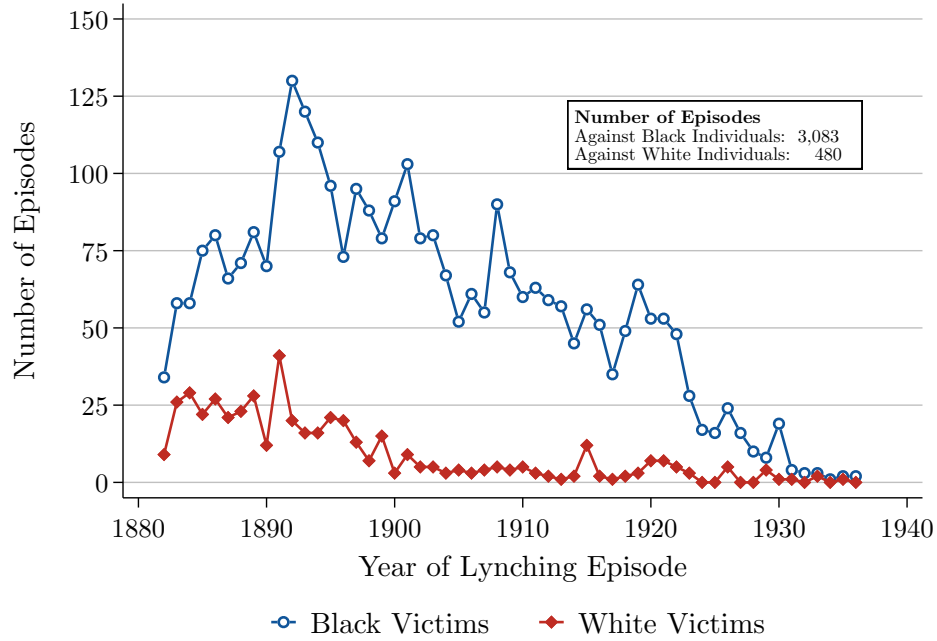
<sup>3</sup>Figure A.1 reports lynching episodes separately by the race of the victim. Lynching episodes targeting Black Americans were overwhelmingly concentrated in the South, whereas lynching episodes in other regions disproportionately targeted non-Black individuals, often European immigrants. By 1900, lynching episodes targeting non-Black populations had largely ceased. By contrast, more than fifty lynching episodes targeting Black individuals occurred annually until the early 1920s, coinciding with the core period of our analysis.

<sup>4</sup>Newspaper circulation was high throughout the period we study. Newspaper circulation per capita in Washington, DC, doubled from 0.37 in 1896 to 0.78 in 1928 (own calculations based on data from Gentzkow, Shapiro and Sinkinson (2004)).

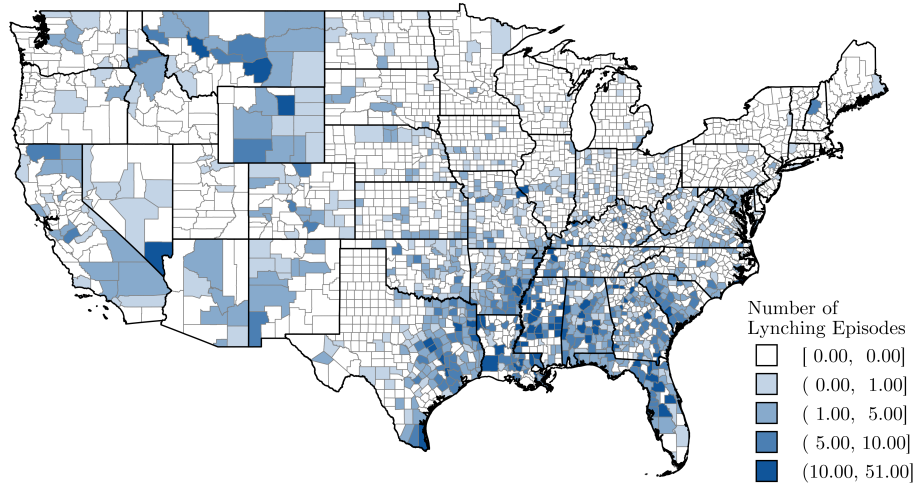
<sup>5</sup>We manually verify newspaper coverage of lynching episodes recorded in Hines and Steelwater (2006). Among a randomly selected subsample, newspapers digitized at [newspapers.com](http://newspapers.com) cover 89% of lynching episodes, while newspapers based in Washington, DC cover 62%. Further details are provided in Appendix D.

**Figure I.** Time and Spatial Variation of Lynching Episodes

(a) Number of Lynching Episodes over Time



(b) Number of Lynching Episodes by County



*Notes.* This Figure reports the number of lynching episodes by race over time (Panel Ia) and their distribution across US counties (Panel Ib). The underlying data are from Seguin and Rigby (2019) and Hines and Steelwater (2006). The sample excludes episodes where the race of the victim is missing. Referenced on pages: 6, 9.

are relatively more frequent among Black individuals convey a stronger racial signal.<sup>6</sup> Formally, the

<sup>6</sup>Throughout the paper, we refer to the Black and white populations with a slight abuse of language. By “white” we mean all individuals who do not report their race as Black in the U.S. census. This group includes individuals identified as white, Asian, and other races. Since Black and white individuals jointly account for 99.57% of the population, we adopt this terminology for brevity.

index is defined as:

$$\text{BSI}_s \equiv \frac{\Pr(\text{Surname} = s \mid \text{Black})}{\Pr(\text{Surname} = s \mid \text{Black}) + \Pr(\text{Surname} = s \mid \text{White})} \quad (1)$$

where  $\Pr(\text{Surname} = s \mid \text{Black})$  denotes the share of Black individuals with surname  $s$ , and  $\Pr(\text{Surname} = s \mid \text{White})$  is defined analogously for white individuals.<sup>7</sup>

The BSI ranges from zero to one. It equals zero when a surname is borne exclusively by white individuals, and one when it is borne exclusively by Black individuals. A value of 0.5 indicates equal relative frequency across the Black and white populations. For example, if Black individuals are three times as likely as white individuals to carry surname  $s$ , then  $\text{BSI}_s = .3/ (.3 + .1) = .75$ . Values above (below) 0.5 therefore indicate that a surname is relatively more (less) common among Black individuals.<sup>8</sup> Table A.1 reports a subset of surnames and their associated BSI values.

Figure A.5 displays the distribution of the BSI in the U.S. population, excluding first-generation immigrants. Figure A.6 shows that surnames convey substantial information about race: even in the absence of any additional characteristics, an individual with a BSI above 0.8 has roughly a 40% probability of being Black. Figure A.4 then compares the performance of the BSI- and BNI-based classifiers. While the two classifiers have similar precision, the BSI-based classifier exhibits substantially higher recall, making surnames more informative than first names in our setting.

### 3.2 Data on Lynching Episodes

Data on lynching episodes come from Hines and Steelwater (2006) and Seguin and Rigby (2019), which rely on newspaper reports as their primary sources. These data include the date of each episode, the victim’s first name, surname, and race, as well as the alleged crime. A detailed discussion of the construction and validation of the lynching-episode data is provided by Cook (2012). Using these data, we construct a treatment indicator that, for each surname, takes the value one in the first year in which an individual bearing that surname is the victim of a lynching episode, and zero otherwise.<sup>9</sup> The period during which lynching episodes occurred and were recorded defines our estimation window, which spans 1882 to 1936.

A potential concern is that lynching episodes may not have altered the perceived racial signal of surnames if victims disproportionately carried surnames that were already strongly associated with Black individuals. Figure A.7 addresses this concern by comparing the BSI distribution of lynching victims, outlined in black, to that of the overall population. While the distribution of victims exhibits excess mass at higher BSI values, it largely overlaps with the population distribution. Throughout

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<sup>7</sup>We focus on surnames rather than first names, as in Fryer and Levitt (2004), for two reasons. First, inventors could plausibly change first names in response to lynching episodes if they anticipated that examiners updated their beliefs about the racial signal conveyed by names following such events. Surnames, by contrast, are considerably harder to change. Second, although first and last names have similar overall classification accuracy (Figure A.4a), surname-based classifiers exhibit markedly higher recall rates (Figure A.4b), making them more suitable for our empirical setting. Nevertheless, Table B.1 shows that these patterns also hold when using a Black Name Index based on first names.

<sup>8</sup>Because of bunching in the lower tail of the distribution, surnames with the lowest BSI are randomly ranked.

<sup>9</sup>For example, if individuals named “Smith” are victims of lynching episodes in 1898, 1911, and 1925, the variable  $\text{Treated}_{\text{Smith}}$  equals one in 1898.

the analysis, we therefore explicitly examine heterogeneity in treatment effects by the racial signal conveyed by surnames.

Lynching episodes received extensive coverage in contemporary media (Weaver, 2019). For our empirical strategy, it is particularly important that newspapers in Washington, DC—the location of the U.S. Patent Office—also reported on these events. To validate this assumption, we manually verify newspaper coverage of lynching episodes across both DC-based and outlets from across the nation. We find that 62% of the lynching episodes in our sample appear in DC newspapers, while the vast majority of episodes are reported in newspapers across the nation (Appendix D).<sup>10</sup>

Lynching episodes were most prevalent in the 1890s and 1900s and overwhelmingly targeted Black Americans. Of the 3,563 recorded episodes, 3,083 (87%) involved Black victims (Figure Ia). These episodes were geographically concentrated in Southern regions, particularly along the Mississippi River, where cotton cultivation was most intensive. This geographic concentration strengthens our empirical design: in the absence of media transmission, it is unlikely that localized outbreaks of violence could directly affect patent examiners in Washington, DC. Moreover, since most inventors resided in Northern states, direct effects on inventors themselves are similarly unlikely. Table A.2 reports summary statistics for lynching episodes in the final dataset.

### 3.3 Patent and Inventor Data

Throughout the paper, we measure innovation activity using patent data, a standard proxy in the literature (e.g., Griliches, 1998). We draw on two complementary data sources: patent records constructed from digitized historical documents, and a linked dataset that matches these records to the full-count population census in order to observe inventors’ demographic characteristics.

#### 3.3.1 Patent Data

Our main analysis relies on information from granted patents. Original patent documents were digitized by Berkes (2018).<sup>11</sup> Since patent specifications list the surname of each inventor, we construct a balanced panel of surnames observed annually between 1882 and 1936. For each surname-year pair, we compute the total number of patents granted to inventors bearing that surname. For patents with multiple inventors, we allocate equal shares to each inventor.

Patents differ widely in their innovativeness and economic importance. Unlike in modern settings, citation-based measures are unavailable, as prior art citations were not required and patent documents lacked a dedicated citation section before 1947 (Andrews, 2021). We therefore proxy for patent importance using the text-based novelty measure developed by Kelly, Papanikolaou, Seru and Taddy (2021), which compares the language of a patent to that of patents filed both before and after it.

Panel A of Table I reports descriptive statistics for the resulting surname-level dataset. The sample comprises 93,276 inventor surnames, which are associated, on average, with 16 patents and 1.6 break-

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<sup>10</sup>The appendix also describes how we construct search-based newspaper-coverage measures for the full analysis sample using automated searches of the `newspapers.com` archive. For each episode, we record whether the search returns at least one result nationwide, and in Washington, D.C. and the neighboring states of Virginia or Maryland.

<sup>11</sup>Because rejected applications and their associated records were not preserved, we do not observe the universe of submitted applications. Our baseline outcome therefore measures the number and timing of granted patents, rather than application-level grant rates.

**Table I.** Descriptive Statistics

|                                              | (1)       | (2)       | (3)          | (4)           | (5)        | (6)          |
|----------------------------------------------|-----------|-----------|--------------|---------------|------------|--------------|
| <b>Panel A. Patent Data</b>                  |           |           |              |               |            |              |
|                                              | Mean      | Std. Dev. | Median       | Min.          | Max.       | Observations |
| N. of Patents                                | 16.355    | 125.440   | 2.000        | 1.000         | 15054.000  | 93,276       |
| N. of Breakthrough Patents                   | 1.643     | 12.741    | 0.000        | 0.000         | 1279.000   | 93,276       |
| Black Surname Index                          | 0.130     | 0.243     | 0.000        | 0.000         | 1.000      | 93,276       |
| Surname Frequency                            | 659.629   | 6679.019  | 60.000       | 1.000         | 870172.000 | 93,276       |
| <b>Panel B. Inventor Characteristics</b>     |           |           |              |               |            |              |
|                                              | Inventors |           |              | Non-Inventors |            |              |
|                                              | Mean      | Std. Dev. | Observations | Mean          | Std. Dev.  | Observations |
| <i>Panel B.1. Patenting and Demographics</i> |           |           |              |               |            |              |
| N. of Patents                                | 1.992     | 3.613     | 64,602       | 0.000         | 0.011      | 28,918,338   |
| Black Individual                             | 0.020     | 0.140     | 64,602       | 0.092         | 0.290      | 28,918,338   |
| Immigrant                                    | 0.203     | 0.402     | 64,602       | 0.211         | 0.408      | 28,918,338   |
| Male                                         | 0.968     | 0.176     | 64,602       | 0.510         | 0.500      | 28,918,338   |
| Literate                                     | 0.982     | 0.134     | 64,602       | 0.907         | 0.291      | 28,918,338   |
| <i>Panel B.2. Area of Residence</i>          |           |           |              |               |            |              |
| South                                        | 0.112     | 0.315     | 64,602       | 0.277         | 0.447      | 28,918,338   |
| West                                         | 0.118     | 0.323     | 64,602       | 0.081         | 0.273      | 28,918,338   |
| Midwest                                      | 0.400     | 0.490     | 64,602       | 0.342         | 0.475      | 28,918,338   |
| Northeast                                    | 0.362     | 0.481     | 64,602       | 0.295         | 0.456      | 28,918,338   |
| <i>Panel B.3. Occupation</i>                 |           |           |              |               |            |              |
| Professional                                 | 0.102     | 0.302     | 64,602       | 0.025         | 0.157      | 28,918,338   |
| Manager                                      | 0.177     | 0.382     | 64,602       | 0.041         | 0.198      | 28,918,338   |
| Skilled Blue Collar                          | 0.222     | 0.416     | 64,602       | 0.071         | 0.256      | 28,918,338   |
| Farmer                                       | 0.087     | 0.281     | 64,602       | 0.123         | 0.329      | 28,918,338   |
| Clerical                                     | 0.033     | 0.180     | 64,602       | 0.020         | 0.139      | 28,918,338   |
| Sales                                        | 0.056     | 0.230     | 64,602       | 0.024         | 0.153      | 28,918,338   |
| Unskilled Blue Collar                        | 0.076     | 0.264     | 64,602       | 0.066         | 0.248      | 28,918,338   |
| Services                                     | 0.024     | 0.154     | 64,602       | 0.052         | 0.223      | 28,918,338   |

*Notes.* The statistics reported in Panel A are computed at the surname level from the raw patent data. Panel B is based on the linked panel-census sample. Columns (1–3) report data for the sample of inventors, and columns (4–6) display the same statistics for the remaining population. Demographic data are from the full-count population census; Panel B.2 reports the Census Bureau-designated regions; Panel B.3 groups the occupations listed in the harmonized IPUMS classification. Referenced on pages: 11, 9.

through patents. The average delay between application filing and patent issuance is approximately 560 days. The mean Black Surname Index among inventor surnames is 0.13, substantially below the population average of 0.37. Figure A.8 shows the geographic distribution of patenting activity across states. Innovation activity is largely concentrated outside the South, reinforcing the geographic separation between innovation and lynching episodes.

### 3.3.2 Sample of Patent Records Linked to Full-Count Census

We additionally employ a newly constructed dataset that links inventors listed on patent records to individuals in the full-count population census (Bazzi et al., 2022).<sup>12</sup> This linkage allows us to observe a rich set of inventor characteristics—including race—that are not recorded on patent documents. For each census year  $t$ , inventors appearing on patents filed between  $t - 5$  and  $t + 4$  are linked to the census conducted in year  $t$ . The matching procedure exploits the geocoded residence information reported on patent records and prioritizes geographically proximate matches to reduce the incidence of false positives.

Using the linked data, we construct a balanced panel of inventors observed over a ten-year window around each census decade. When an inventor is matched to multiple census records, we weight each match by the inverse of the number of matches. The final linked sample covers patent records issued between 1895 and 1935.<sup>13</sup>

Panel B of Table I summarizes inventor characteristics and compares them with those of the general population, following the approach of Akcigit, Grigsby and Nicholas (2017). Approximately 98% of inventors are white, and 97% are male. Black Americans are thus almost five times less likely than whites to ever hold a patent. They are substantially less likely to reside in the South and more likely to live in the Northeast. Inventors are also positively selected on occupation: they are roughly three times more likely to work in professional, managerial, or skilled manufacturing occupations, and substantially less likely to be employed in agriculture.

## 4 Black-Sounding Surnames and Patenting

In this section, we document two novel stylized facts. First, individuals with Black-sounding surnames hold fewer patents than individuals with white-sounding surnames (Section 4.1). Second, this negative association also holds in the subsets of Black and white individuals (Section 4.2). We conclude the section by presenting robustness checks and explaining why these correlations should not be interpreted causally, thereby motivating our difference-in-differences design (Section 4.3).

### 4.1 Surname-level Evidence

To examine the relationship between Black-sounding surnames and patenting, we begin with a surname-level analysis that does not rely on linking patent records to the census. This allows us to study the full universe of inventors, including those who cannot be matched to a census record.

We divide the sample into quintiles based on the Black Surname Index (BSI). The first quintile contains surnames more commonly held by white individuals, while subsequent quintiles correspond to increasingly distinctively Black surnames. We then estimate the following Poisson regression:

$$\log \mathbb{E}(\text{Patents per Capita}_{i,c,t}) = \mathbf{X}_{c,t} + \sum_{k=2}^5 \beta_k \cdot I(q_i^{\text{BSI}} = k), \quad (2)$$

---

<sup>12</sup>Due to computational constraints, we use a 20% sample of the full-count census population, corresponding to approximately 30 million individuals. Details are provided in Appendix A.

<sup>13</sup>We start our individual-level analysis in 1900 since the 1890 U.S. census was lost in a fire.

where  $i$ ,  $c$ , and  $t$  denote the surname, county, and year.<sup>14</sup> The dependent variable is the number of patents normalized by surname frequency in the 1900 census.  $\mathbf{X}_{c,t}$  is a vector of fixed effects that varies across specifications. The term  $q_i^{\text{BSI}}$  indicates the BSI quintile of surname  $i$ . Standard errors are clustered at the surname level. Because some surnames are much more common than others and we want to avoid results driven by relatively rare surnames, we weight surnames by their absolute frequencies. The first quintile serves as the omitted category.

The results are summarized in Figure II. We begin by showing the raw average patenting rate by BSI interval in Figure IIa. The patenting rate among individuals with surnames that are never observed among Black individuals is approximately 5%. This figure declines to about 2.5% for individuals with a BSI between 0 and .25, and falls further to less than 1.5% among individuals with surnames that are almost exclusively observed within the African American community.

In Figure IIb, we display the results of equation (2). The blue circles report the unconditional correlation: the number of patents per capita declines as surnames become more Black-sounding. These differences are statistically significant and economically large, especially for surnames in the highest BSI quintile. The general pattern is robust to the inclusion of additional fixed effects. The red markers and green squares report estimates including, respectively, year fixed effects and county-by-year fixed effects.

Since Black-sounding surnames are not randomly assigned, unobserved factors may affect both the BSI and innovation outcomes. One natural omitted variable is geography: Black-sounding surnames are disproportionately concentrated in economically poorer regions, especially the South. Poorer economic conditions may, in turn, hamper human capital accumulation and ultimately reduce the likelihood of producing, or even filing, patents.<sup>15</sup> In Table B.2, we report the correlation between the BSI and a range of observable characteristics, using both surname-level and individual-level data. Individuals with more Black-sounding surnames are, as expected, more likely to be Black and less likely to be foreign-born. They are also less likely to be married and less likely to be able to read and write (Panel A). Unsurprisingly, Black-sounding surnames are more concentrated in Southern states (Panel B). Finally, they are more common among farmers and low-skilled laborers, and less common among skilled manufacturing workers, professionals, and managers (Panel C). This analysis suggests that the Black Surname Index is not a one-dimensional signal of applicants' race, but instead bundles several characteristics that may themselves help explain why individuals with more Black-sounding surnames hold fewer patents.

## 4.2 Individual-level Evidence

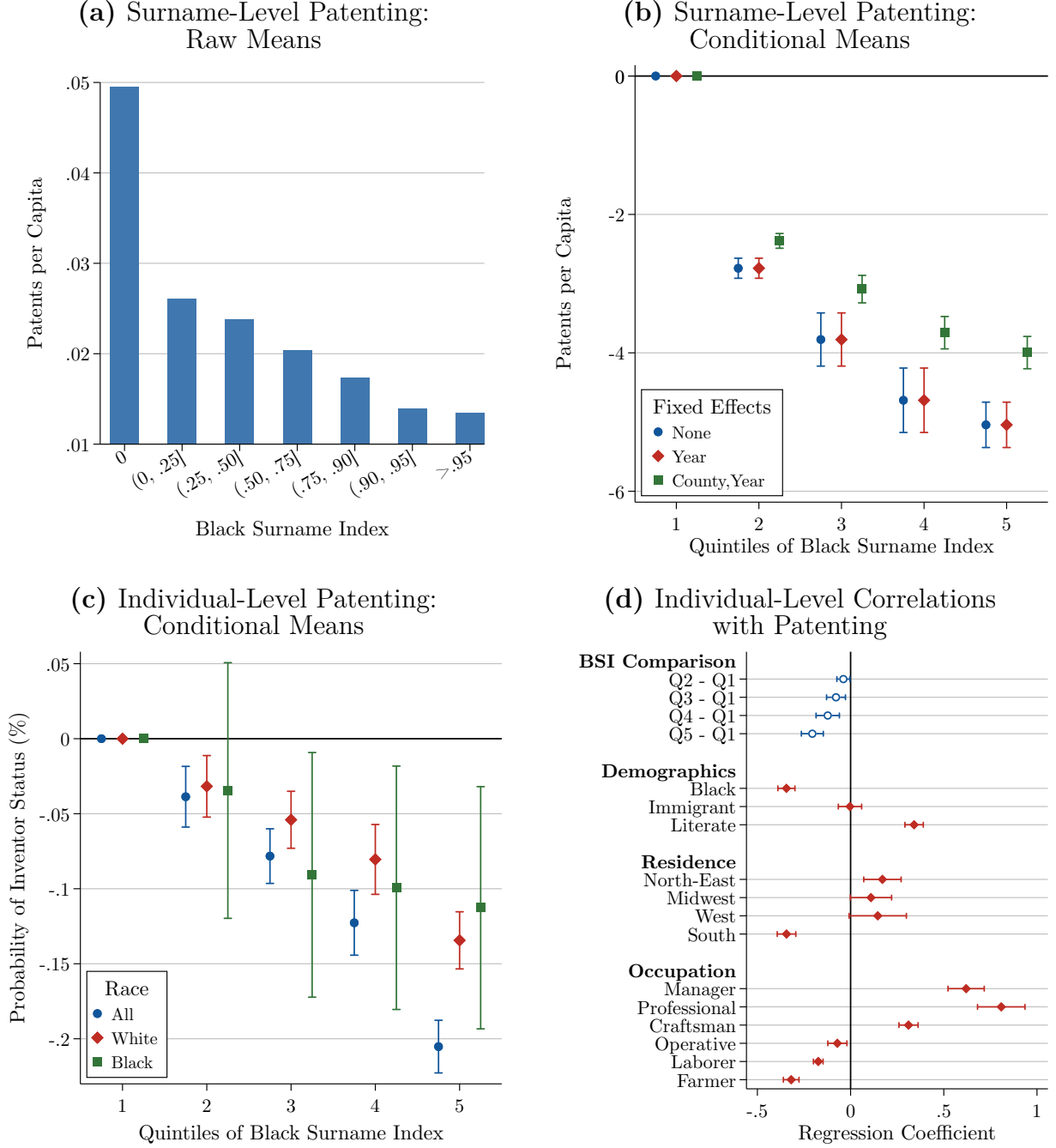
In Figure IIc, we turn to individual-level inventor data linked to the population census. We estimate a regression similar to (2), except that  $i$  denotes an individual, the sample is pooled over time, and the dependent variable is an indicator equal to one if individual  $i$  is an inventor and zero otherwise. Results are similar when we instead use the number of patents as the outcome. At the individual

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<sup>14</sup>We estimate Poisson pseudo-maximum-likelihood models because the dependent variables are non-negative, contain a large mass at zero, and because it is highly right-skewed.

<sup>15</sup>Note, however, that Figure IIb shows that the results are not sensitive to the inclusion of finer geographic fixed effects.

**Figure II.** Black-Sounding Surnames and the Production of Innovation



*Notes.* This figure reports the correlation between patenting activity and the Black Surname Index, estimated according to equation (2). In Panel A, the BSI is grouped into intervals; in Panels B–D, it is divided into quintiles, with the first quintile serving as the baseline category. In Panels A and B, the unit of observation is a surname-by-county pair at yearly frequency. The dependent variable is the number of patents obtained by inventors with that surname, normalized by surname frequency. In Panel B, the blue circles report the unconditional correlation, while the red diamonds and green squares report the correlation including, respectively, year and county and year fixed effects. Surnames are weighted by their frequency. Panels C and D use the individual-level linked census sample. Referenced on page: 12.

level, the blue dots replicate the pattern documented in the surname-level analysis.

A key advantage of the inventor sample is that we can distinguish between Black and white inventors. This distinction allows us to separate two explanations for the gradient documented above.

If the gradient reflected only characteristics of Black individuals—whether discrimination based on observed race or unobserved differences correlated with race—it should be absent among white individuals, whose surnames happen to be Black-sounding. If, instead, part of the gradient reflects the racial signal conveyed by the surname itself, it should appear within racial groups. The red markers and green squares in Figure IIc report results separately for white and Black individuals. The negative gradient is clearly present within the white subsample. The estimates for Black individuals, though imprecise given the small number of Black inventors, are consistent with the same pattern. These results indicate that the correlation between the BSI and patenting is not driven solely by Black individuals and are consistent with surnames themselves carrying a signal that matters for patenting.

How does the correlation between the BSI and patenting compare to other predictors of innovation? In Figure IId, we report the results of a series of regressions of an indicator equal to one for inventors and zero otherwise on a set of categorical characteristics. In the first four rows, the main regressor is an indicator equal to one if an individual’s BSI falls in the  $k$ -th quintile and zero if it falls in the first quintile. The corresponding coefficient therefore compares the probability of inventing for individuals in the  $k$ -th BSI quintile relative to individuals in the first quintile. The red dots report analogous coefficients for selected demographic and occupational characteristics. Moving from the first to the fifth BSI quintile is associated with a decline in the probability of inventing that is comparable in magnitude to the patenting gap associated with being Black, living in the South, or working as a farmer. Importantly, the negative correlation between the BSI and the probability of inventing remains statistically significant even when controlling for all of these characteristics, as shown in Table B.3.

### 4.3 Robustness Checks

We conduct several robustness checks to assess whether the baseline correlation between the Black Surname Index (BSI) and patenting is sensitive to alternative samples, geographic composition, and measures of patent quality.

Throughout our study period, the United States received a large inflow of immigrants, primarily from Europe. Since nearly all of these immigrants were white, their surnames may distort the informational content of the Black Surname Index. To address this concern, Table B.4 presents regression estimates of the correlation between patenting and both the baseline BSI and a modified version that excludes first-generation immigrants. The negative correlation remains when this group is excluded.

Because Black-sounding surnames are more common in the South (Table B.2), one concern is that the BSI may proxy for geographic origin rather than race. If so, we would expect the negative correlation between the BSI and patenting to be concentrated in Southern states. Figure B.1 plots the estimated correlation between the BSI and patents per capita by state, highlighting Southern states in red. We find no statistically significant difference between the South and the rest of the country, suggesting that the BSI is unlikely to proxy primarily for Southern origin.

The measure developed by Kelly et al. (2021) captures patent importance by comparing each patent’s text to that of preceding and subsequent grants, thereby yielding a continuous measure. Following the authors, we classify patents in the top 5% of this distribution as “breakthrough” in our

baseline analysis. Table B.5 shows that the results are robust to alternative thresholds, namely the top 20% (Panel A) and the top 10% (Panel B). The table also confirms a positive correlation between breakthrough innovation and the BSI when first-generation immigrants are excluded (columns 5–8). Because the importance measure compares patents within a five-year window, Kelly et al. (2021) recommend partialling out year fixed effects to account for changes in language and in the patenting process. Our baseline follows this approach, but Table B.6 shows that the results are quantitatively similar even without this adjustment.

We replicate the baseline sensitivity analysis at the individual level in Table B.3. Columns (1–4) report results using the baseline BSI, while columns (5–8) use the version that excludes first-generation immigrants. Panel A presents results for the full population, and Panels B and C focus on the white and Black sub-samples, respectively. We continue to find a negative correlation between patenting and the BSI when controlling for region of residence (columns 2 and 6), occupation (columns 3 and 7), and demographic characteristics (columns 4 and 8), including within the white sub-sample.

## 5 Empirical Design

This section outlines our identification strategy, which uses the timing of lynching episodes as shocks to the racial salience of victims’ surnames. We begin by describing the identifying assumption underlying this approach (Section 5.1), and then explain how we construct a comparable control group through propensity-score matching (Section 5.2). We then present the matched difference-in-differences specification used in the main analysis (Section 5.3), and conclude by describing the individual-level design (Section 5.4).

### 5.1 Identification Strategy

Our identification strategy is based on changes in the racial signal conveyed by surnames. We argue that lynching episodes generated such changes because newspapers nationwide often reported the victim’s name and race, thereby increasing the racial salience of the former. Because patent applications conveyed applicants’ names but not their race, these episodes plausibly shifted the racial signal that surnames carried to those who evaluated or supported an application. The identifying assumption is that, absent the episode, granted patents to inventors bearing the treated surname would have evolved in parallel with granted patents to inventors bearing the matched control surname.

Under this assumption, the design identifies the reduced-form effect of a surname becoming associated with a lynching episode on subsequent patent grants. Interpreting this effect as arising from a change in the surname’s racial signal additionally requires that the episode did not differentially affect inventors bearing the treated surname through other channels, such as local disruption or a general stigma associated with violence. We examine these alternative channels in Section 7.

Treated surnames are, on average, more Black-sounding than surnames in the population as a whole, but they are observed throughout the BSI distribution. Table II further shows that surnames exposed to treatment differ from the rest of the population along nearly all observable characteristics (column 3). In particular, they are more common and therefore account for a larger share of granted

patents. More common surnames are also more likely to be treated earlier. These patterns caution against direct comparisons between exposed and unexposed surnames. To construct a more comparable control group, we adopt the matched difference-in-differences procedure used, among others, by Goldschmidt and Schmieder (2017), Jäger and Heining (2022), and Fenizia and Saggio (2024).

## 5.2 Matching Algorithm

We use nearest-neighbor propensity score matching to identify an appropriate control for each treated surname. We first group surnames by the year in which they are first treated. For each group  $\tau^*$ , we estimate a probit regression on the cross-section consisting of surnames treated at time  $\tau^*$  and surnames that are never treated. The regression relates treatment status, equal to one for treated surnames and zero for never-treated surnames, to a vector of surname-specific characteristics that includes surname frequency, the total number of patents before the start of the sample period (1850–1882), the Black Surname Index, the population share across census divisions, the employment share across occupations, the share of Black individuals, the share of immigrants, the share of young individuals, the share of men, the share of married individuals, and the share of literate individuals. All variables except the BSI are expressed in zero-inflated logs. We then compute the propensity score for each surname using the predicted values from the probit regression. For each treated surname, we select as control the never-treated surname with the closest propensity score.<sup>16</sup> Each treated surname is matched to one control surname.

Column (5) of Table II reports average values of surname-specific characteristics for the control surnames selected by this procedure, while columns (6–7) compare these values with those of the treated surnames. Treated and control groups are broadly comparable along all observable characteristics. Even when the difference between the two groups is statistically significant, its magnitude is small relative to the corresponding difference between treated surnames and the full set of untreated surnames shown in column (3). This exercise, therefore, yields a control group that is plausibly comparable to the treated group.<sup>17</sup>

## 5.3 Difference-in-Differences Design

To estimate the impact of lynching episodes on innovation, we estimate the following Poisson pseudo-maximum-likelihood difference-in-differences regression on the sample of treated surnames and their

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<sup>16</sup>Formally, let  $s$  denote a treated surname, and let  $\mathcal{S}$  denote the set of never-treated surnames. Surname  $s$  is associated with propensity score  $\pi(s)$ . Its matched control is defined as

$$\tilde{s} = \arg \min_{s' \in \mathcal{S}} |\pi(s) - \pi(s')|.$$

For example, the surname “Morgan” has a BSI of 0.13 and is matched to the surname “Armstrong,” whose BSI is 0.14. Table II provides a formal comparison of the characteristics of treated and control surnames.

<sup>17</sup>The sample of surnames in Table II (1,102 treated + 1,102 control = 2,204) differs from the baseline estimation sample of 652 treated + 652 control = 1,304 surnames (see column 1 of Table III) because, in the analysis, we only keep surnames that are observed over five years before and after the treatment time. Table C.1 reports the balance comparisons for the surnames included in the estimation sample. As in Table II, we find that the matching algorithm substantially improves the comparability of treated and matched control surnames (columns 6–7) relative to a comparison between treated and untreated surnames (columns 3–4).

**Table II.** Comparison Between Treated and Control Surnames

|                                             | Treated Surnames | All Non-Treated Surnames |            |         | Matched Non-Treated Surnames |            |         |
|---------------------------------------------|------------------|--------------------------|------------|---------|------------------------------|------------|---------|
|                                             | Mean             | Mean                     | Difference |         | Mean                         | Difference |         |
|                                             | (1)              | (2)                      | (3)        | (4)     | (5)                          | (6)        | (7)     |
| <b>Panel A. Demographic Characteristics</b> |                  |                          |            |         |                              |            |         |
| Black Surname Index                         | 0.521            | 0.235                    | 0.286***   | (0.007) | 0.500                        | 0.022*     | (0.013) |
| Share of Blacks Individuals                 | 0.145            | 0.056                    | 0.089***   | (0.003) | 0.137                        | 0.008      | (0.006) |
| Number of Individuals                       | 9.180            | 7.169                    | 2.011***   | (0.043) | 9.058                        | 0.122      | (0.147) |
| Total Number of Patents                     | 2.515            | 1.310                    | 1.206***   | (0.039) | 2.491                        | 0.024      | (0.149) |
| Share of Immigrants                         | 0.069            | 0.112                    | -0.043***  | (0.002) | 0.072                        | -0.004     | (0.004) |
| Share of Males                              | 0.409            | 0.410                    | -0.000*    | (0.000) | 0.408                        | 0.001*     | (0.001) |
| Share of Young                              | 0.306            | 0.294                    | 0.012***   | (0.001) | 0.301                        | 0.004***   | (0.002) |
| Share of Married                            | 0.314            | 0.322                    | -0.008***  | (0.001) | 0.315                        | -0.001     | (0.001) |
| Share of Literate                           | 0.510            | 0.540                    | -0.030***  | (0.001) | 0.514                        | -0.004     | (0.003) |
| <b>Panel B. Area of Residence</b>           |                  |                          |            |         |                              |            |         |
| South                                       | 0.337            | 0.197                    | 0.141***   | (0.005) | 0.317                        | 0.020**    | (0.008) |
| West                                        | 0.047            | 0.053                    | -0.006***  | (0.001) | 0.049                        | -0.002     | (0.001) |
| Midwest                                     | 0.255            | 0.317                    | -0.062***  | (0.003) | 0.259                        | -0.004     | (0.005) |
| North-East                                  | 0.207            | 0.273                    | -0.066***  | (0.004) | 0.222                        | -0.015**   | (0.007) |
| <b>Panel C. Occupational Structure</b>      |                  |                          |            |         |                              |            |         |
| Professionals                               | 0.016            | 0.018                    | -0.001***  | (0.000) | 0.017                        | -0.001     | (0.000) |
| Farmers                                     | 0.082            | 0.072                    | 0.010***   | (0.001) | 0.080                        | 0.002      | (0.001) |
| Managers                                    | 0.018            | 0.021                    | -0.004***  | (0.000) | 0.018                        | -0.001**   | (0.000) |
| Clerical                                    | 0.011            | 0.013                    | -0.002***  | (0.000) | 0.012                        | -0.001*    | (0.000) |
| Sales                                       | 0.014            | 0.017                    | -0.003***  | (0.000) | 0.015                        | -0.001*    | (0.000) |
| Skilled Blue Collar                         | 0.035            | 0.044                    | -0.009***  | (0.000) | 0.036                        | -0.001*    | (0.001) |
| Unskilled Blue Collar                       | 0.041            | 0.048                    | -0.008***  | (0.000) | 0.041                        | -0.001     | (0.001) |
| Services                                    | 0.032            | 0.028                    | 0.004***   | (0.000) | 0.032                        | -0.000     | (0.001) |
| Laborers                                    | 0.118            | 0.102                    | 0.016***   | (0.001) | 0.117                        | 0.002      | (0.001) |
| Number of Observations                      | 1,102            | 7,870                    | 8,972      |         | 1,102                        | 2,204      |         |

*Notes.* This table reports sample means for surname-level characteristics. One observation is one surname that is held by at least 100 individuals in the 1900 U.S. Census. Column (1) reports statistics for the surnames of victims of lynching episodes. Columns (2) and (3–4) report, respectively, statistics for the universe of untreated surnames and the differences between treated and untreated surnames. Columns (5) and (6–7) restrict the sample to untreated surnames matched to treated surnames. Standard errors are reported in parentheses and clustered by surname. Table C.1 reports the same comparisons for the subsample of surnames included in the estimation sample of column (1) in Table III. Referenced on pages: 15, 16, 19.

\*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

matched controls:

$$\log \mathbb{E}(y_{i,t}) = \alpha_i + \alpha_{m(i) \times t} + \sum_{\substack{k=-5 \\ k \neq 0}}^{+5} \beta_k \cdot I(t = \tau_{m(i)}^* + k) \cdot Treated_i, \quad (3)$$

where  $i$  and  $t$  denote surname and issue year, and  $m(i)$  identifies the match associated with surname  $i$ . The dependent variable  $y_{i,t}$  is the number of granted patents. Because the outcome is a non-negative count with a large mass at zero, we estimate the model by Poisson pseudo-maximum likelihood. The operator  $\mathbb{E}(\cdot)$  denotes the conditional expectation. The terms  $\alpha_i$  and  $\alpha_{m(i) \times t}$  denote surname fixed effects and match-by-year fixed effects, respectively. By including match-by-year fixed effects, the identifying variation compares treated and control surnames within each match over time. The variable  $Treated_i$  is an indicator equal to one if surname  $i$  is treated and zero otherwise. By omitting period  $k = 0$ , the coefficients  $\beta_k$  capture changes in outcomes between treated and control surnames relative to the treatment year.

As shown by Fenizia and Saggio (2024), this matched design helps alleviate concerns that arise under staggered treatment timing (Goodman-Bacon, 2021). In the robustness section, however, we show that our estimates remain quantitatively similar when using alternative estimators designed to avoid the pitfalls of the two-way fixed-effects estimator (Cengiz, Dube, Lindner and Zipperer, 2019; Sun and Abraham, 2021).<sup>18</sup> While surnames can be treated multiple times, equation (3) treats exposure as absorbing. To reduce the risk that the estimated average treatment effect conflates the effects of multiple lynching episodes, we restrict the estimation sample to a five-year window around the treatment date  $\tau^*$ .

This design allows for level differences between treated and control groups, provided that they would have followed parallel trends absent treatment. We assess the empirical plausibility of this assumption in the results section using event-study estimates.

#### 5.4 Individual-Level Analysis

We also estimate equation (3) at the individual inventor level. The matching algorithm follows the same logic as in the surname-level analysis described in Section 5.2. The only difference is that the right-hand-side variables are individual characteristics rather than population shares. Table C.2 reports balance checks for the inventor-level sample.

The estimating equation mirrors (3), but  $i$  now indexes inventors rather than surnames. Because we do not observe when an inventor becomes active or retires, we restrict the sample to inventors who are granted at least one patent both before and after their treatment period  $\tau^*$ . We impose the same restriction on their matched controls. As a result, this analysis focuses on the intensive margin, that is, on the patenting activity of individuals who were already inventors before the shock.<sup>19</sup>

## 6 Main Results

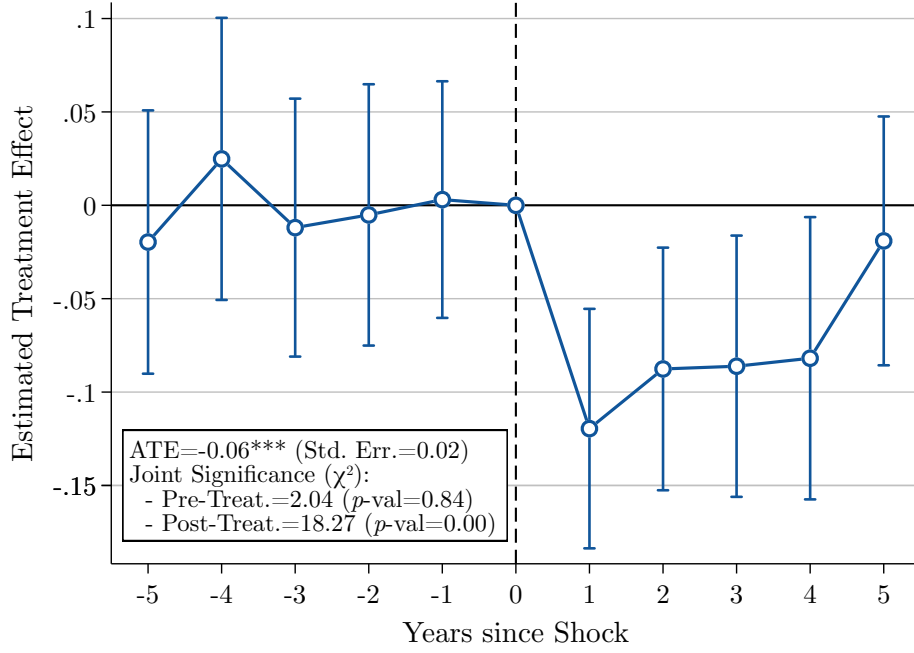
This section provides evidence that lynching episodes reduce patented innovation by individuals who share their surname with the victims. We begin with surname-level estimates (Section 6.1), then

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<sup>18</sup>We select alternative estimators that both allow for the inclusion of match-by-year fixed effects, which are required by the matched difference-in-differences design, and can be estimated as a Poisson regression.

<sup>19</sup>This restriction conditions on post-treatment patenting and therefore changes the population represented by the inventor-level estimates. The results should be interpreted as describing patenting among persistently active inventors who receive patents both before and after the treatment date, rather than the effect among all inventors active before treatment.

**Figure III.** Dynamic Difference-in-Differences Effect of Lynching Episodes on Innovation



*Notes.* This Figure displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed at a yearly frequency over a ten-year window around the time when someone with the given surname is the victim of a lynching episode. Each treated surname is matched to a control non-treated one as described in Section 5.2. The sample includes all surnames reported in columns (1) and (5) of Table II that are granted at least one patent within a five-year window before and after the match-specific treatment date. The dependent variable is the total number of patents granted to individuals with the given last name in year  $t$ . The dots report the estimated coefficients of a set of interaction terms between a surname-specific lynching indicator variable and dummies that correspond to the number of years since the lynching episode. The event time is shown on the x-axis. The baseline category is the year when someone with a given surname is treated. The model is a Poisson pseudo-likelihood regression, as in (3), and it includes surname and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 19.

confirm the pattern using inventor-level data (Section 6.2), and finally present robustness checks (Section 6.3).

### 6.1 Surname-Level Evidence

Figure III plots the dynamic treatment effects from equation (3). We find a sharp decline in patenting after treatment. The effect remains statistically significant for up to four years and then fades, becoming indistinguishable from zero. In magnitude, the shock implies an immediate 11% decline in patents granted to the treated surname relative to its matched control, corresponding to roughly two fewer patents in the year after the episode. We find no statistically significant pre-treatment effects.

Columns (1–4) of Table III report estimates from a specification that collapses event time into pre- and post-treatment periods. On average, lynching episodes reduce innovation by about 7% over the five-year window following treatment.

The estimates are similar in magnitude when we introduce additional controls: surname-frequency-quantile fixed effects interacted with the post-treatment indicator (column 2), Black-Surname-Index-quantile fixed effects interacted with the post-treatment indicator (column 3), and surname-level

**Table III.** The Effect of Lynching Episodes on Innovation: Surname-Level Evidence

|                                 | All Patents          |                      |                      |                      |                     | Filed Before Lynching |                     |
|---------------------------------|----------------------|----------------------|----------------------|----------------------|---------------------|-----------------------|---------------------|
|                                 | (1)                  | (2)                  | (3)                  | (4)                  | (5)<br>Breakthrough | (6)                   | (7)<br>Breakthrough |
| Lynching Episode $\times$ Post  | -0.066***<br>(0.020) | -0.068***<br>(0.020) | -0.066***<br>(0.022) | -0.065***<br>(0.021) | -0.178**<br>(0.091) | -0.056*<br>(0.029)    | -0.143<br>(0.160)   |
| Surname FE                      | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                   | Yes                 |
| Match-Year FE                   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                   | Yes                 |
| BSI $\times$ Post               | No                   | No                   | Yes                  | No                   | No                  | No                    | No                  |
| Surname Frequency $\times$ Post | No                   | Yes                  | No                   | No                   | No                  | No                    | No                  |
| Controls $\times$ Post          | No                   | No                   | No                   | Yes                  | No                  | No                    | No                  |
| Number of Surnames              | 1,304                | 1,304                | 1,304                | 1,276                | 486                 | 1,214                 | 280                 |
| Observations                    | 13,262               | 13,262               | 13,262               | 12,992               | 2,604               | 9,282                 | 1,124               |
| R <sup>2</sup>                  | 0.701                | 0.701                | 0.701                | 0.702                | 0.289               | 0.663                 | 0.229               |

*Notes.* This Table displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed at a yearly frequency over a ten-year window around the time when someone with the given surname is the victim of a lynching episode. The sample includes all surnames reported in columns (1) and (5) of Table II that are granted at least one patent within a five-year window before and after the match-specific treatment date. The dependent variable is the total number of patents granted to individuals with the given last name in year  $t$ . Each treated surname is matched to a control non-treated surname. The dependent variable in columns (1–4) and (6) is the total number of patents granted to individuals in year  $t$ ; in columns (5) and (7), it is the number of patents in the top 5% of the patent-importance distribution granted to individuals in year  $t$ . The treatment variable is defined as the interaction between a surname-specific lynching indicator and a post-treatment, match-specific indicator. Each regression is estimated as a Poisson pseudo-likelihood regression, as in (3). All regressions include surname and match-by-year fixed effects. Columns (2)–(4) additionally include, respectively, surname-frequency-quantile fixed effects interacted with a post-treatment indicator, Black-Surname-Index-quantile fixed effects interacted with a post-treatment indicator, and a set of surname-level covariates (measured in 1900) interacted with a post-treatment indicator. Columns (6)–(7) restrict the sample to patents filed before the lynching episode. Standard errors are shown in parentheses and are clustered by surname. Referenced on pages: 16, 19, 23. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

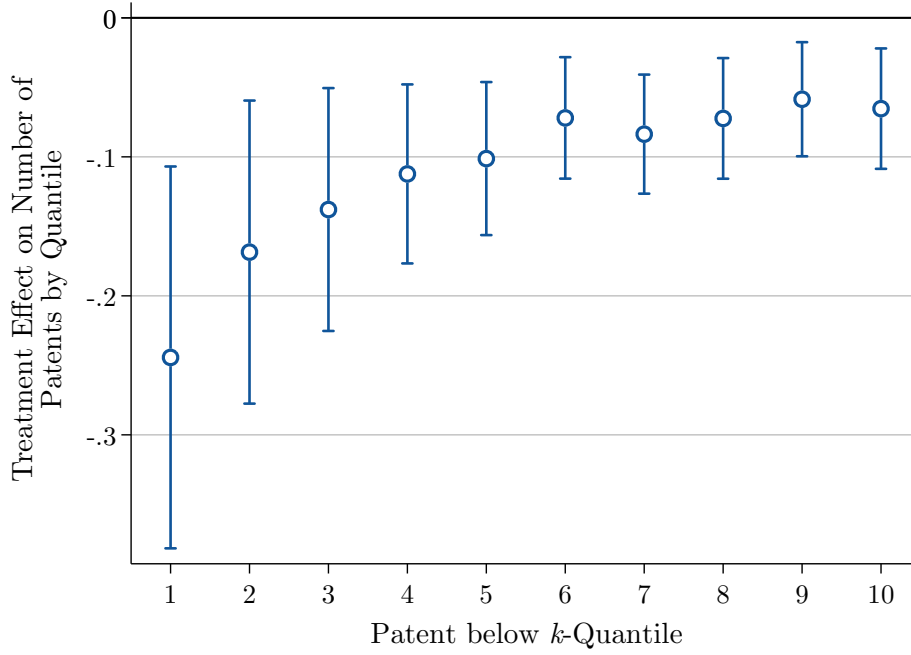
covariates interacted with the post-treatment indicator (column 4).<sup>20</sup> These specifications address the concern that the baseline estimates may be driven by differential post-treatment trends across surnames that vary in frequency, racial signal, or observable characteristics.

In Figure IV, we examine heterogeneity by patent importance. For each threshold  $k$ , we re-estimate equation (3) using as the dependent variable the number of patents below the  $k$ -th quantile of the patent-importance distribution. The outcomes are therefore cumulative: as  $k$  increases, the dependent variable includes all patents below progressively higher importance thresholds. The estimated effects are larger in magnitude at the lower end of the distribution and attenuate as more important patents are included. This pattern indicates that the shock had larger effects on less important patents, while the most important inventions were less affected. The text-based measure captures a patent’s linguistic novelty, not its distance from the legal threshold for patentability. These results therefore show that the effect is concentrated among less important patents but do not establish that discrimination operated specifically at the granting margin.

To assess whether *only* lower-importance innovations are affected by the shock, column (5) re-estimates the main specification using as the dependent variable the number of patents in the top 5%

<sup>20</sup>These controls correspond to the variables used in the matching procedure and are listed in Table II.

**Figure IV.** Heterogeneous Treatment Effect by Patent Importance



*Notes.* This Figure displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed at a yearly frequency over a ten-year window around the time when someone with the given surname is the victim of a lynching episode. Each treated surname is matched to a control non-treated surname as described in Section 5.2. The treatment variable codes an interaction term between a surname-specific lynching indicator variable and a post-treatment match-specific indicator term. Each dot reports the estimated difference-in-differences coefficient from a regression in which the dependent variable is the number of patents below the  $k$ -quantile of the patent-importance distribution granted to inventors with a given surname. Each model is a Poisson pseudo-likelihood regression, as in (3), and includes surname and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 20.

of the patent-importance distribution, measured using the text-based metric developed by Kelly et al. (2021).<sup>21</sup> Lynching episodes also negatively affect high-importance patents, suggesting that the shock reduces not only lower-importance patents but also breakthrough ones.

Columns (6)–(7) restrict attention to patents filed before the lynching episode.<sup>22</sup> The estimated effects are similar in magnitude to those in the full sample, indicating that the baseline results are not driven by a post-episode decline in the initial propensity to file an application. This restriction does not, however, distinguish among applications whose issuance was delayed, applications that were ultimately not granted, and applications that inventors or their attorneys chose not to pursue after filing. Because records of rejected or abandoned applications and correspondence between applicants and examiners are unavailable for this period, we cannot separately identify these channels.

<sup>21</sup>Following Kelly et al. (2021), we partial out issue-year fixed effects from the patent-importance measure and take patents within the top 5% of the resulting distribution so that the results are not driven by changes in patent formats common across applications filed within the same year.

<sup>22</sup>Suppose February 1900 is the first time someone called  $n$  is the victim of a lynching episode. For the treated name, the count includes all patents filed in January 1900 and before by inventors with the same name.

**Table IV.** Effect of Lynching Episodes on Innovation: Individual-Level Evidence

|                                | N. of Patents           |                           |                           | $I(\text{N. of Patents} > 0)$ |                           |                           | Breakthrough      |
|--------------------------------|-------------------------|---------------------------|---------------------------|-------------------------------|---------------------------|---------------------------|-------------------|
|                                | (1)<br>All<br>Inventors | (2)<br>White<br>Inventors | (3)<br>Black<br>Inventors | (4)<br>All<br>Inventors       | (5)<br>White<br>Inventors | (6)<br>Black<br>Inventors | (7)               |
| Lynching Episode $\times$ Post | -0.076***<br>(0.026)    | -0.085***<br>(0.027)      | -0.601**<br>(0.289)       | -0.038*<br>(0.020)            | -0.047**<br>(0.020)       | -0.047<br>(0.173)         | -0.497<br>(0.303) |
| Individual FE                  | Yes                     | Yes                       | Yes                       | Yes                           | Yes                       | Yes                       | Yes               |
| Match-Year FE                  | Yes                     | Yes                       | Yes                       | Yes                           | Yes                       | Yes                       | Yes               |
| Number of Individuals          | 9,015                   | 8,097                     | 63                        | 8,888                         | 7,984                     | 61                        | 507               |
| Observations                   | 64,038                  | 57,236                    | 456                       | 60,108                        | 53,782                    | 412                       | 1,878             |
| R <sup>2</sup>                 | 0.386                   | 0.389                     | 0.371                     | 0.081                         | 0.082                     | 0.086                     | 0.411             |

*Notes.* This Table displays the effect of lynching episodes on patented innovation. The unit of observation is an individual inventor, observed at a yearly frequency over a ten-year window around the census to which he is linked. Each inventor is matched to a control non-treated inventor. The dependent variable in columns (1–3) is the number of patents; in columns (4–6), it is an indicator equal to one when the inventor produces at least one-tenth of a patent and zero otherwise; in column (7), it is the number of patents in the top 5% of the patent-importance distribution. The treatment variable codes an interaction term between a surname-specific lynching indicator variable and a post-treatment match-specific indicator term. The sample consists of all inventors who produced at least one patent before and after someone with their surname is treated, between 1895 and 1925. Each regression is estimated as a Poisson pseudo-likelihood regression, as in (3). All regressions include inventor and match-by-year fixed effects. Columns (1), (4), and (7) are estimated on the universe of matched inventors, whereas in columns (2) and (5)—resp. (3) and (6)—we restrict the sample to include white—resp. Black—inventors only. Standard errors are shown in parentheses and are clustered by surname. Referenced on pages: 22, 25. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

## 6.2 Inventor-Level Evidence

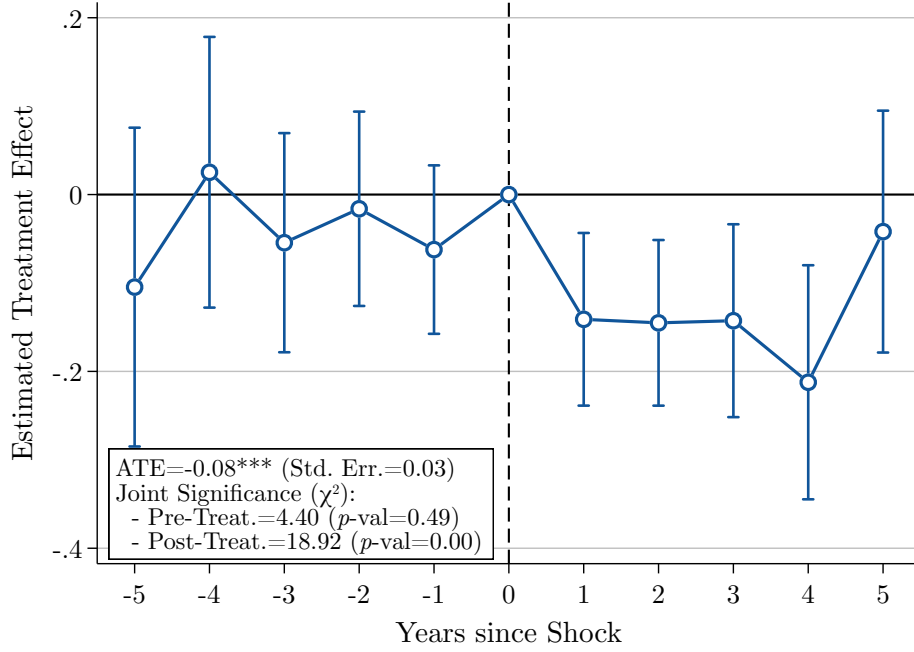
We next turn to inventor-level estimates, which allow us to distinguish inventors by race and confirm the surname-level pattern.

Table IV reports static difference-in-differences estimates. Because the inventor-year panel is sparse and the outcome is heavily right-skewed, we report results for two outcomes: the number of patents (columns 1–3) and an indicator equal to one if the inventor is associated with at least one-tenth of a patent in a given year (columns 4–6).<sup>23</sup> The estimates confirm the surname-level pattern: the number of patents (column 1) declines by about eight percent over the five years following treatment, and the probability of patenting (column 4) falls by roughly four percent.

A key advantage of the inventor-level analysis is that it allows us to distinguish inventors by race. Discrimination based on inventors' observed race would predict no effect among white inventors who happen to bear a treated surname. Instead, the decline is clearly present in this group (columns 2 and 5). The point estimate for Black inventors is larger but very imprecisely estimated given the small number of Black inventors in the linked sample, and we cannot reject equality of the coefficients once we allow for differential post-treatment trends by the Black Surname Index. The data therefore cannot determine whether observed race played an additional role for Black inventors; what they establish is

<sup>23</sup>Inventors listed on patent texts can be matched to multiple census entries. In the analysis, we assign each match a fractional share of the patent equal to the inverse of the number of matches. The binary outcome equals one if an individual is associated with at least one-tenth of a patent in a given year, which is approximately the median in the data.

**Figure V.** Dynamic Individual-Level Difference-in-Differences Effect of Lynching Episodes on Innovation



*Notes.* This Figure displays the effect of lynching episodes on patented innovation. The unit of observation is an individual observed at a yearly frequency over a ten-year window around the census to which he is linked. Each treated inventor is matched to a control non-treated inventor as described in Section 5.4. The dependent variable is the number of patents. The dots report the estimated coefficients of a set of interaction terms between a surname-specific lynching indicator variable and dummies that correspond to the number of years since the lynching episode. The event time is shown on the x-axis. The baseline category is the year when someone with a given surname is treated. The model is a Poisson pseudo-likelihood regression, as in (3), and it includes individual and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 23.

that observed race cannot be the primary driver of the decline.

Finally, we examine whether the negative effect remains when focusing on high-importance patents. In column (7), the point estimate remains negative when we restrict the dependent variable to patents in the top 5% of the patent-importance distribution, although it is imprecisely estimated. Figure V plots event-study estimates from equation (3) at the individual level.<sup>24</sup> Patenting declines sharply following treatment, with effects that remain statistically significant for up to five years. As in the surname-level analysis, we find no statistically significant pre-treatment effects.

### 6.3 Robustness Checks

We report additional robustness analyses in the Appendix. Table C.3 presents alternative standard error calculations for equation (3); statistical significance is preserved throughout. Table C.4 replicates the baseline sensitivity analyses in Table III on the sub-sample of patents filed before treatment. Table C.5 reports analogous regressions for lynching episodes involving white victims. We find no

<sup>24</sup>Figure C.1 reports analogous event-study estimates separately for white and Black inventors.

detectable effect on patenting.<sup>25</sup>

In Figure C.2, we re-estimate equation (3) using recent methods designed for staggered treatment timing. The blue dots report the baseline two-way fixed-effects estimates, while the red and green dots show estimates based on Sun and Abraham (2021) and Cengiz et al. (2019), respectively. The three estimators deliver similar conclusions.

As in the descriptive section, Table C.6 reports the effect of lynching episodes on patent importance using alternative measures. In Panel A, we focus on patents in the bottom 10% of the patent-importance distribution; in Panel B, we restrict attention to patents in the top 10%. Across measures, we confirm that the impact of lynching episodes is concentrated among relatively lower-importance patents. This pattern is robust to partialling out year fixed effects prior to aggregation and to varying the time window used to construct patent importance. We also employ the approach developed by Rambachan and Roth (2023) to assess sensitivity to violations of parallel trends. Figure C.3 shows that the baseline effect remains statistically significant under deviations from parallel trends that are nearly as large as the maximum pre-treatment differences observed in the data.

Finally, Table C.7 reports robustness checks for the inventor-level specifications. We consider outcomes based on (i) the number of patents (columns 1–3), (ii) an indicator for being granted at least one patent in a given year (columns 4–6), and (iii) patents in the top 5% of the patent importance distribution (columns 7–9). In Panel A, we include all inventors; in Panel B, we restrict the sample to white inventors.<sup>26</sup> The baseline results are robust to including geographic (county-level) fixed effects and individual-level controls interacted with a post-treatment indicator.

## 7 Mechanisms

This section examines the mechanisms underlying our main result. We first show that the decline in granted patents is unlikely to reflect the local effects of lynching episodes, but instead a change in the information conveyed by treated surnames within the patenting process (Section 7.1). We then provide evidence suggesting that lynching episodes increased the *racial* salience of treated surnames, rather than attaching to them a broader stigma associated with violence (Section 7.2). Finally, we discuss racial passing in the census and its implications for our inventor-level analysis (Section 7.3).

### 7.1 Local Exposure and Channels within the Patenting Process

In the previous section, we showed that the decline in patenting persists when we restrict the analysis to patents whose applications were filed before the lynching episode. This establishes that at least part of the response occurred after applications had entered the patenting process and rules out an explanation based solely on fewer inventions or new applications. It does not, however, identify which actor or stage of the patenting process generated the decline.

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<sup>25</sup>We reject the null hypothesis that the coefficients for episodes involving white and Black victims are equal at the 5% significance level.

<sup>26</sup>Due to the small sample size, we cannot replicate the same robustness exercises for Black inventors.

**Table V.** Heterogeneity by Distance from Lynching Episodes and the Patent Office

|                                                | Distance Between DC<br>and Lynching Episode |                      | $k$ -Quintile of Distance between<br>Inventor and Lynching Episode |                   |                     |                     |                   |
|------------------------------------------------|---------------------------------------------|----------------------|--------------------------------------------------------------------|-------------------|---------------------|---------------------|-------------------|
|                                                | (1)                                         | (2)                  | (3)                                                                | (4)               | (5)                 | (6)                 | (7)               |
|                                                |                                             |                      | $k = 1$                                                            | $k = 2$           | $k = 3$             | $k = 4$             | $k = 5$           |
| Shock $\times$ Post                            | -0.116**<br>(0.048)                         |                      | -0.086**<br>(0.039)                                                | -0.037<br>(0.051) | -0.110**<br>(0.048) | -0.106**<br>(0.045) | -0.026<br>(0.045) |
| Shock $\times$ Post $\times$ Distance Shock-DC | 0.031<br>(0.037)                            |                      |                                                                    |                   |                     |                     |                   |
| Lynching Episode $\times$ Post $\times$        |                                             |                      |                                                                    |                   |                     |                     |                   |
| Quintile of Distance from DC=1                 |                                             | -0.047<br>(0.035)    |                                                                    |                   |                     |                     |                   |
| Quintile of Distance from DC=2                 |                                             | -0.175***<br>(0.052) |                                                                    |                   |                     |                     |                   |
| Quintile of Distance from DC=3                 |                                             | -0.075<br>(0.047)    |                                                                    |                   |                     |                     |                   |
| Quintile of Distance from DC=4                 |                                             | -0.037<br>(0.050)    |                                                                    |                   |                     |                     |                   |
| Quintile of Distance from DC=5                 |                                             | -0.063*<br>(0.036)   |                                                                    |                   |                     |                     |                   |
| Surname FE                                     | Yes                                         | Yes                  | Yes                                                                | Yes               | Yes                 | Yes                 | Yes               |
| Match-Year FE                                  | Yes                                         | Yes                  | Yes                                                                | Yes               | Yes                 | Yes                 | Yes               |
| Number of Surnames                             | 1,304                                       | 1,304                | 888                                                                | 864               | 868                 | 836                 | 812               |
| Observations                                   | 13,262                                      | 13,262               | 6,736                                                              | 6,806             | 6,932               | 6,732               | 6,386             |
| R <sup>2</sup>                                 | 0.701                                       | 0.701                | 0.460                                                              | 0.512             | 0.539               | 0.506               | 0.526             |

*Notes.* This table reports heterogeneity by the distance between lynching episodes, inventors, and the Patent Office. The unit of observation is a surname observed annually from five years before to five years after the episode. Each treated surname is matched to a never-treated control surname. The dependent variable is the number of patents granted in a given year to inventors bearing that surname. In column (1), the treatment variable includes the baseline interaction and an additional interaction with the distance between the lynching episode and the Patent Office in Washington, DC. Column (2) allows this relationship to vary across quintiles of distance to the Patent Office. In columns (3)–(7), patents are divided into quintiles based on the distance between the inventor’s residence and the location of the lynching episode; each column includes patents produced by inventors in one quintile of this distribution. Each regression is estimated by Poisson pseudo-maximum likelihood, as in (3), and includes surname and match-by-year fixed effects. Standard errors are shown in parentheses and clustered by surname. Referenced on pages: 24, 26. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

We first study whether the effect reflects increased discrimination or economic disruption in inventors’ local environments (Cook, 2014). If direct local exposure were the main mechanism behind our results, the decline should be larger among inventors living closer to the lynching episode. Columns (3)–(7) of Table V test this prediction by estimating the baseline specification separately across quintiles of the distance between inventors and lynching episodes.<sup>27</sup> We find no systematic relationship between the treatment effect and inventor distance. Figure C.4 also shows that the aggregate result is not driven by the states or regions in which lynching episodes were concentrated.

<sup>27</sup>Patent records report inventors’ places of residence, which we use to calculate the distance between each inventor and the location of the lynching episode. Figure A.9 displays the distribution of distances between inventors, lynching episodes, and the Patent Office. Because episodes were concentrated in Southern regions while inventors were primarily located in Northern states, most inventors lived far from the episodes.

The inventor-level results provide a complementary test. If local discrimination based on inventors' observed race drove the decline, we would expect the effect to be concentrated among Black inventors and absent among white inventors who happen to have a treated surname. Instead, the estimated effect is negative among white inventors (column 2 of Table IV). The point estimate is larger for Black inventors but is imprecisely estimated because the linked sample contains few Black inventors. Once we allow post-treatment trends to vary with the Black Surname Index, we cannot reject equality of the estimates for white and Black inventors. Together, these findings are difficult to reconcile with an explanation based primarily on direct local exposure to violence or discrimination.

The evidence is instead more consistent with an information-transmission channel. Columns (3) and (4) of Table VI show that the decline in granted patents is concentrated among episodes receiving coverage in newspapers based in or near Washington, DC, and is absent among episodes without such coverage.<sup>28</sup> Columns (1)–(2) of Table V further show that the decline is larger for episodes occurring closer to Washington. Together, these findings are consistent with information about the episode reaching actors involved in the patenting process.

Several channels could generate this response. Treated surnames may have affected how applications were evaluated within the patenting process, either directly through examiner decisions or indirectly through the expectations and behavior of patent attorneys and agents. Inventors themselves may also have become less willing to pursue pending applications, while firms, assignees, banks, or other intermediaries may have reduced their support. Importantly, each of these channels could operate whenever an actor observed an inventor's surname without knowing the inventor's actual race.<sup>29</sup> The historical records do not allow us to distinguish among these channels: rejected and abandoned applications were not preserved, nor was correspondence among applicants, patent attorneys, and examiners.

## 7.2 Racial Updating or a General Stigma of Violence?

The evidence above suggests that information about lynching episodes reached actors involved in the patenting process. We next ask whether these episodes specifically increased the racial salience of treated surnames or instead attached to them a more general stigma associated with violence. Three pieces of evidence support the racial-updating interpretation.

First, the decline is specific to episodes involving Black victims. If a lynching episode attached a race-neutral stigma of violence to the victim's surname, we would expect similar effects following

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<sup>28</sup>The construction and validation of the newspaper-coverage measures are described in Appendix D. In Table C.8, we examine alternative measures of newspaper coverage. In the baseline exercise, we classify an episode as covered if it is mentioned in at least one newspaper based in Washington, DC, Virginia, or Maryland. We then consider a broader measure based on coverage by any newspaper in the United States and a narrower measure based only on newspapers in Washington, DC. We also report complementary specifications for episodes not covered in each area. Across specifications, negative effects are concentrated among episodes covered in or near Washington, DC.

<sup>29</sup>The patenting environment included not only examiners and other Patent Office personnel, but also patent attorneys and agents. Historical evidence indicates that patent practitioners elsewhere in the country often worked through Washington-based correspondents, who searched Patent Office records and interacted with examiners on their behalf (Lamoreaux and Sokoloff, 2002; Lamoreaux, Sokoloff and Sutthiphisal, 2013). These practitioners could therefore observe an inventor's surname and participate in the prosecution of an application without meeting the inventor or directly observing the inventor's race.

**Table VI.** Heterogeneity by Race of Victim, BSI Quintile, and Newspaper Coverage

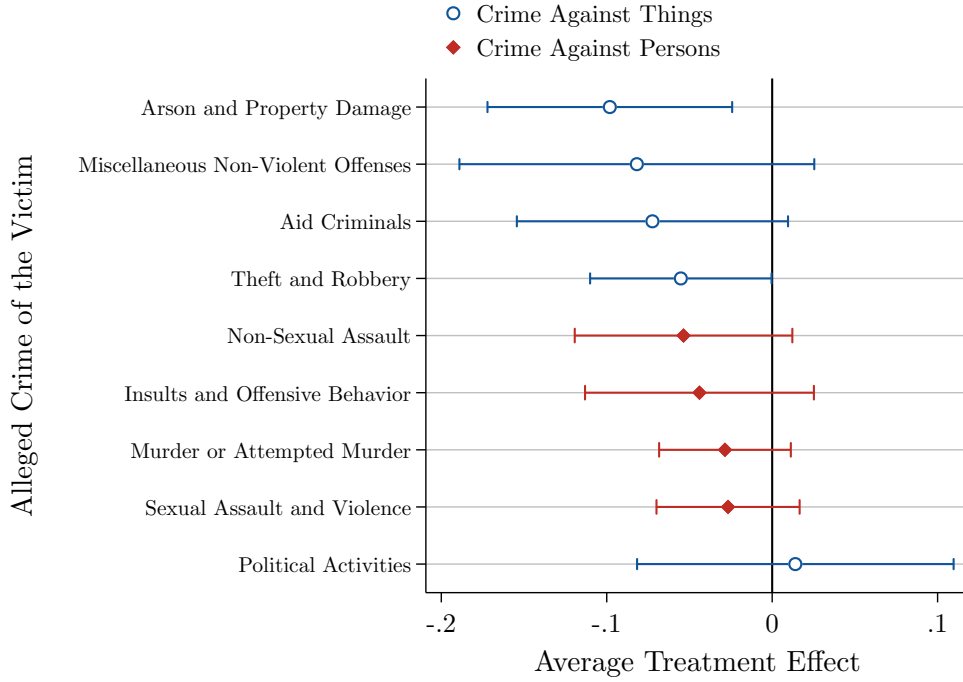
|                                                             | N. of Patents     |                      |                           |                    |
|-------------------------------------------------------------|-------------------|----------------------|---------------------------|--------------------|
|                                                             | White Victims     | Heterogeneity by BSI | Coverage in DC Newspapers |                    |
|                                                             | (1)               | (2)                  | (3)<br>Covered            | (4)<br>Not Covered |
| Lynching Episode with White Victim $\times$ Post            | -0.018<br>(0.033) |                      |                           |                    |
| Lynching Episode $\times$ Post $\times$ Quintile of BSI = 1 |                   | -0.147*<br>(0.075)   |                           |                    |
| Lynching Episode $\times$ Post $\times$ Quintile of BSI = 2 |                   | -0.077<br>(0.050)    |                           |                    |
| Lynching Episode $\times$ Post $\times$ Quintile of BSI = 3 |                   | -0.127***<br>(0.041) |                           |                    |
| Lynching Episode $\times$ Post $\times$ Quintile of BSI = 4 |                   | -0.015<br>(0.029)    |                           |                    |
| Lynching Episode $\times$ Post $\times$ Quintile of BSI = 5 |                   | -0.022<br>(0.052)    |                           |                    |
| Lynching Episode Covered in News $\times$ Post              |                   |                      | -0.060***<br>(0.021)      |                    |
| Lynching Episode Not Covered in News $\times$ Post          |                   |                      |                           | 0.006<br>(0.026)   |
| Surname FE                                                  | Yes               | Yes                  | Yes                       | Yes                |
| Match-Year FE                                               | Yes               | Yes                  | Yes                       | Yes                |
| Number of Surnames                                          | 380               | 1,304                | 946                       | 522                |
| Observations                                                | 3,878             | 13,262               | 9,004                     | 4,948              |
| R <sup>2</sup>                                              | 0.736             | 0.701                | 0.761                     | 0.811              |

*Notes.* This table reports placebo and heterogeneity analyses. The unit of observation is a surname observed annually from five years before to five years after the year in which an individual bearing that surname is the victim of a lynching episode. Each treated surname is matched to a never-treated control surname. The dependent variable is the number of patents granted in a given year to inventors bearing that surname. In column (1), the treatment variable is the interaction between an indicator for surnames of white victims of lynching episodes and a match-specific post-episode indicator. In column (2), the baseline treatment is interacted with quintiles of the Black Surname Index. In columns (3) and (4), episodes are divided according to newspaper coverage: column (3) considers episodes covered by newspapers based in or near Washington, DC, while column (4) considers episodes without such coverage. Each regression is estimated by Poisson pseudo-maximum likelihood, as in (3), and includes surname and match-by-year fixed effects. Standard errors are shown in parentheses and clustered by surname. Referenced on pages: 26, 27, 29. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

episodes involving white victims. Column (1) of Table VI reports analogous regressions for episodes involving white victims. The estimated effect is close to zero, and we reject equality of the estimates for episodes involving Black and white victims.<sup>30</sup> This result is difficult to reconcile with a purely race-neutral stigma of violence.

<sup>30</sup>We reject the null hypothesis that the coefficients for episodes involving white and Black victims are equal at the 5% significance level.

**Figure VI.** Heterogeneity by Alleged Offense of the Victim of a Lynching Episode



*Notes.* This figure reports the effect of lynching episodes on granted patents by the alleged offense attributed to the victim. The unit of observation is a surname observed annually from five years before to five years after the episode. Each treated surname is matched to a never-treated control surname as described in Section 5.2. Each dot reports the estimated treatment effect from a separate regression for one alleged-offense category. Blue dots correspond to alleged offenses against property, while red dots correspond to alleged offenses against persons. Each model is estimated by Poisson pseudo-maximum likelihood, as in (3), and includes surname and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 28.

Second, the effect varies with the surname's initial racial signal. If a lynching episode conveys information about the race associated with a surname, its impact should depend on how informative the surname was beforehand. Column (2) of Table VI and Figure C.5 show that the decline is concentrated among surnames that were initially predominantly associated with White individuals or had intermediate BSI values, while the effect attenuates among surnames already strongly associated with Black individuals. This pattern is consistent with lynching episodes providing new information about a surname's association with Black individuals primarily when that association was not already unambiguous.

Finally, the decline does not vary systematically with the alleged offense attributed to the victim. Figure VI reports estimates of equation (3) separately by offense. If the effect instead reflected stigma associated with the victim's alleged conduct, we might expect larger declines following allegations of serious violent offenses, such as murder or sexual assault, than following property offenses or less serious conduct. We find no statistically significant differences across offense categories. If anything, the estimates for property offenses, including arson, are at least as large as those for murder or sexual assault. This pattern provides little support for an explanation based on the nature or severity of the alleged conduct.

### 7.3 Racial Passing in the Census

Dahis, Nix and Qian (2019) document that some individuals recorded as Black in one census were recorded as white in a subsequent census. Such changes do not affect the surname-level analysis, which does not rely on observing inventors' race. They also do not affect the full-sample inventor-level estimates, since inventors remain in the sample regardless of their recorded census race.

Racial passing may, however, introduce measurement error when we estimate effects separately by recorded race. In particular, some inventors recorded as white may have previously been recorded as Black. This would blur differences in estimated treatment effects between inventors recorded as Black and white and reduce the precision with which we can compare the two groups. It cannot, however, account for the aggregate decline in granted patents, which is also present in the surname-level analysis and in the full inventor sample (Tables IV and C.7).

## 8 Quantification

The empirical results show that patenting declines among individuals who share a surname with a Black victim of a lynching episode. The evidence is consistent with this decline reflecting an increase in the racial salience of the treated surname.

This section uses the reduced-form estimates to assess how much of the patenting gradient across the Black Surname Index (BSI) distribution, documented in Section 4, can be accounted for by the estimated penalty associated with surname-based racial signals. The remaining gradient may arise from differences that predate or extend beyond the patenting process, including unequal access to role models, education, employment opportunities, wealth, and professional networks. We also consider the implications for aggregate patenting and for patenting rates among Black and white individuals.

The exercise requires two assumptions. First, we assume that an episode raises the perceived BSI of surnames in the middle of the distribution—that is, racially ambiguous surnames—to one.<sup>31</sup> Second, we assume that white-sounding surnames do not face a penalty associated with racial signals. We therefore define the counterfactual without such penalties as one in which all surnames are perceived as white-sounding and set their perceived BSI to zero.<sup>32</sup>

We begin by asking how patenting would be distributed across the BSI distribution if surnames did not convey any racial signal. We translate the reduced-form estimate into a semi-elasticity of expected granted patents with respect to perceived BSI. We use the estimate for the third BSI quintile ( $-0.127$ , column 2 of Table VI), in which the mean BSI is approximately 0.5. This quintile contains racially ambiguous surnames and is therefore the group for which a lynching episode is most naturally interpreted as changing the perceived racial identity associated with the surname. Assuming that an episode shifts the perceived BSI of treated surnames in this quintile from 0.5 to one, the estimate

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<sup>31</sup>This assumption is conservative for the implied semi-elasticity. If an episode raises the perceived BSI of treated surnames to a value below one—for example, because only a subset of individuals updates its beliefs—the same reduced-form estimate would imply a larger response of patenting to perceived racial signal.

<sup>32</sup>This assumption follows the approach of Hsieh et al. (2019), who treat white men as the undistorted group and quantify the frictions faced by other demographic groups.

implies a log semi-elasticity of approximately  $-0.254$  per unit increase in perceived BSI. A change in perceived BSI from zero to one would therefore reduce expected granted patents by approximately 22%.<sup>33</sup> This calculation assumes that the smaller reduced-form estimates for high-BSI quintiles in Table VI reflect smaller episode-induced changes in perceived BSI rather than a smaller response of patenting to a given change in perceived BSI.<sup>34</sup>

We apply the estimated semi-elasticity to observed patent counts in each BSI quintile.<sup>35</sup> For the distributional exercise, we rescale the counterfactual counts to hold the total number of patents fixed. Holding the total fixed isolates how removing the estimated penalty would change the distribution of patenting across BSI quintiles.

Patent shares then shift toward inventors with more Black-sounding surnames. In Panel A of Figure VII, the share of patents associated with surnames in the highest-BSI quintile rises from 14.2% to 15.6%, an increase of 1.4 percentage points, or approximately 10%. Equivalently, patenting in the highest-BSI quintile rises from 59% to 69% of patenting in the lowest-BSI quintile, closing approximately 24% of the gap between the two groups.

We next examine how the total number of granted patents would change if additional patents granted to inventors with more Black-sounding surnames were not offset by fewer patents granted to other inventors. This calculation imposes the stronger assumption of no substitution across inventors. Under this assumption, assigning all surnames  $BSI^{\text{cf}} = 0$  would increase aggregate patenting by approximately 11%. Conversely, assigning all surnames  $BSI^{\text{cf}} = 1$  would reduce aggregate patenting by approximately 14%. Panel B of Figure VII summarizes these calculations.

Under the same no-substitution assumption, we also examine the implications for the Black–white patenting gap. We combine patenting across BSI quintiles with the distribution of Black and white individuals across those quintiles and calculate group-specific patenting rates using population counts from the 1900 U.S. Census. This calculation also assumes that the estimated semi-elasticity with respect to perceived BSI applies equally to Black and white individuals within each quintile. When

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<sup>33</sup>Specifically, the implied semi-elasticity is

$$\hat{\theta} = \frac{\hat{\tau}_{Q3}}{1 - \overline{BSI}_{Q3}} = \frac{-0.127}{1 - 0.5} = -0.254.$$

The implied percentage change associated with an increase in perceived BSI from zero to one is

$$100 \times [\exp(-0.254) - 1] = -22.4\%.$$

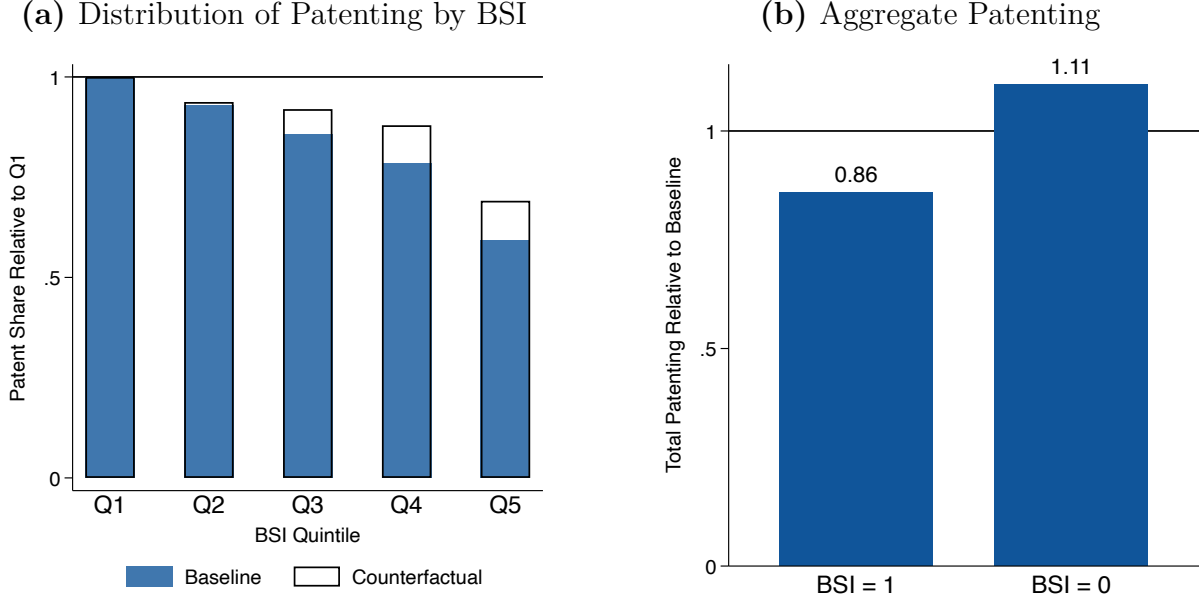
<sup>34</sup>We cannot test this interpretation directly. The first-quintile estimate provides an alternative calibration: assuming that an episode raises perceived BSI in that quintile from approximately zero to one, the implied semi-elasticity is  $-0.147$ , so that increasing perceived BSI from zero to one would reduce expected granted patents by approximately 14%. The counterfactual changes reported below are smaller under this alternative calibration, so the baseline figures should be viewed as the upper end of the range implied by the reduced-form estimates.

<sup>35</sup>Let  $P_q$  denote the observed number of patents in quintile  $q$ , and let  $BSI_q$  denote the average BSI in that quintile. For a counterfactual value  $BSI_q^{\text{cf}}$ , the unnormalized number of patents is

$$\tilde{P}_q^{\text{cf}} = P_q \exp \left[ \hat{\theta} (BSI_q^{\text{cf}} - BSI_q) \right].$$

For the distributional exercise reported in Panel A of Figure VII, we rescale these counts proportionally to hold the total number of patents fixed. For the aggregate and Black–white calculations, we use the unnormalized counts.

**Figure VII. Counterfactual Patenting**



*Notes.* This figure reports the counterfactual exercises described in Section 8. Panel (a) compares the observed distribution of patenting across BSI quintiles with the counterfactual distribution obtained after removing the estimated penalty associated with higher BSI values. Counterfactual patent counts in this panel are normalized to hold the aggregate number of patents fixed. Panel (b) reports aggregate patenting relative to the observed baseline under two uniform counterfactual assignments,  $BSI^{cf} = 0$  and  $BSI^{cf} = 1$ . Panel (b) uses unnormalized counterfactual patent counts and therefore assumes that changes in patents granted to one group of inventors are not offset by changes in grants to other inventors. Both exercises use the estimated semi-elasticity and assume that a lynching episode shifts perceived BSI in the third quintile from 0.5 to one.

all surnames are assigned  $BSI^{cf} = 0$ , Black patenting increases by approximately 17%, while white patenting increases by approximately 11%. The Black-to-white patenting-rate ratio therefore rises by approximately 6%.<sup>36</sup> White patenting also increases because white individuals are represented throughout the BSI distribution, including among surnames with positive BSI values. Because Black individuals are more concentrated among higher-BSI surnames, however, their proportional increase is larger.

The aggregate and Black–white calculations assume that patents lost by one inventor are not replaced by patents granted to others. The estimates in Panel B should therefore be interpreted as upper bounds on the effect on aggregate patenting. By contrast, the distributional exercise in Panel A holds the total number of patents fixed and does not require the no-substitution assumption. All calculations treat the estimated decline as patents forgone rather than delayed. If some grants were merely delayed, the exercises would overstate the permanent effect.

Even if other inventors replaced some of the lost patents, there could still be an effect on the direction of innovation and on which groups benefit from it. Existing evidence suggests that inventors

<sup>36</sup>The unnormalized counterfactual raises Black patenting by 16.7% and white patenting by 10.5%. Holding population counts fixed, the Black-to-white patenting-rate ratio changes by

$$\frac{1.167}{1.105} - 1 = 0.056,$$

or approximately 6%.

from different demographic groups tend to focus on different needs and are disproportionately likely to develop innovations that benefit their own group (Dossi, 2024; Einio, Feng and Jaravel, 2025). Inventors from different backgrounds may therefore not be perfect substitutes in the production of innovation. In our setting, the BSI also varies systematically with geography and socioeconomic background. The loss of patents among inventors with more Black-sounding surnames may therefore affect not only the total volume of innovation, but also the types of inventions produced and the groups they benefit.

## 9 Conclusion

This paper provides evidence that racial discrimination affected patenting outcomes in the late nineteenth- and early twentieth-century United States. Using the universe of patents granted between 1882 and 1936, we first document systematically lower patenting among individuals with more Black-sounding surnames. To study whether racial discrimination affected individuals who had already entered the innovation process, we use lynching episodes as shocks to the racial salience of victims' surnames. Following an episode, patents granted to inventors sharing the victim's surname decline, with the largest effect immediately afterward and a gradual fade over the next five years. The evidence suggests that lynching episodes changed the racial signal conveyed by treated surnames to actors involved in the patenting process.

An illustrative quantification suggests that this channel may account for a meaningful share of the patenting gap. Removing the estimated penalty associated with higher BSI values would reduce the patenting gap between white- and Black-sounding inventors by 24%. Under the no-substitution assumption, aggregate granted patents would increase by up to 11%.

Patenting outcomes are intended to depend on the technical merits of an invention rather than on the perceived identity of its inventor. Our findings show that this was not always the case. Policies aimed at reducing racial disparities in innovation should therefore address not only inequalities arising before individuals become inventors, but also disparities arising after they enter the patenting process. Equal participation in innovation depends not only on who becomes an inventor, but also on whether patent protection is awarded on equal terms once it is sought. In addition, because inventors from different demographic and socioeconomic backgrounds may not be perfectly substitutable, these disparities may affect not only who receives patents and the overall volume of patenting, but also the distribution and direction of inventive activity.

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# ONLINE APPENDIX

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RACIAL DISCRIMINATION AND INNOVATION: EVIDENCE FROM U.S.

INVENTORS, 1882–1936

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September, 2026

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## A Data Sources and Description

### A.I Data Sources

We draw on three main data sources. First, we use the Comprehensive Universe of U.S. Patents (Berkes, 2018). Second, data on lynching episodes are compiled from two sources, Hines and Steelwater (2006) and Seguin and Rigby (2019). Third, de-anonymized individual-level census data are from the IPUMS project (Ruggles et al., 2021). In this section, we briefly discuss the first two data sources and refer interested readers to the cited references for more detailed descriptions.<sup>1</sup>

#### A.I.1 Linked Patent-Census Data

Patent data come from Berkes (2018), who digitized the universe of patents granted by the United States Patent Office (USPTO) over the period 1836–2010 from original documents. Comparing alternative datasets of historical U.S. patents, Andrews concludes that “CUSP is currently the gold standard both in terms of completeness and scope of the types of patent information it contains” (Andrews, 2021, p. 391). In our analysis, we focus on the period 1882–1936.

A major advantage of this historical period is that inventor records compiled from patent documents can be linked to individual-level census data. In our application, this linkage allows us to distinguish actual Black inventors from white inventors with Black-sounding surnames. In addition, linking patent data to the census provides detailed demographic characteristics of inventors. These variables enable a richer analysis of the stylized fact we document. For instance, Fryer and Levitt (2004) argues that, once one controls for proxies for income, occupation, and geographic location, the negative correlation between Black-sounding names and later-life outcomes among Black individuals disappears. The linked inventor sample allows us to address this concern in our setting.

To construct the linked inventor sample, Bazzi et al. (2022) proceed as follows in broad terms. Consider a patent  $p$ , filed in year  $y_p$ , with inventors  $i_j^p \in I^p = \{i_1^p, \dots, i_N^p\}$ . In our setting, most patents have a single inventor, so for simplicity, consider a single-authored patent with inventor  $i^p$ . Fix a radius of  $\tau$  kilometers around the geocoded coordinates of  $i^p$  and consider the census year closest to  $y_p$ . Let  $N_i^\tau$  denote the set of census individuals living within a circle of radius  $\tau$  centered at those coordinates. For each  $n_i^\tau \in N_i^\tau$ , compute a similarity score  $\sigma_{n_i^\tau, i^p}$  between the first and last names of  $n_i^\tau$  and those of  $i^p$ . Then fix a threshold  $\alpha$ : if  $\sigma_{n_i^\tau, i^p} > \alpha$ , retain the match; otherwise, discard it. If the set of matches satisfying  $\sigma_{n_i^\tau, i^p} > \alpha$  is non-empty, retain that set of matches. Otherwise, increase  $\tau$  and repeat the procedure.

This procedure yields, for each inventor, a set of possible census matches. In our application, we weight each inventor by the inverse of the number of matches to which the inventor is linked to avoid multiple counting. We refer interested readers to Berkes (2018) for a more detailed discussion of the dataset and to Bazzi et al. (2022) for details on the linking algorithm, which we only sketch here, as well as related descriptive statistics. The algorithm covers all patents granted between 1895 and 1935 and links them to one census between 1900 and 1930.

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<sup>1</sup>We do not describe the census microdata here because (i) many researchers are likely to be familiar with the IPUMS project, and (ii) extensive documentation is readily available on the IPUMS website.

Even though this exercise yields uniquely informative data on linked inventors, the resulting sample may be subject to several potential concerns. Successful inventors, for instance, may be less likely to be linked if they are more geographically mobile. Moreover, Black inventors, who on average may have less distinctive first and last names, may be linked to a larger set of potential matches. More generally, there are plausible sources of non-classical measurement error in the probability of being linked. In the paper, we therefore use the linked dataset only to corroborate and enrich the baseline evidence, which we document using data constructed directly from patent records.

#### *A.I.2 Data on Lynching Episodes*

We assemble data on lynching episodes from Hines and Steelwater (2006) and Seguin and Rigby (2019). Both datasets rely primarily on historical newspapers. Among other variables, they report the county in which the episode occurred, the day, month, and year of the episode, the race of the victim, the alleged crime committed by the victim, and the victim’s first and last name. We augment these data by geocoding each lynching episode to the centroid of the county in which it occurred, which allows us to compute approximate geodesic distances between inventors and lynching episodes.

Cook (2012) discusses the quality and coverage of available lynching data. In particular, Hines and Steelwater (2006) focuses on lynching episodes that occurred in Southern states, whereas Seguin and Rigby (2019) covers the rest of the country, where a larger share of lynching episodes involved non-Black victims.

We use a large language model (`gpt-4`) to classify the recorded criminal allegations into broad categories. The prompt is as follows: “You are a research assistant. Within this list of categories (Arson and Property Damage, Aid Criminals, Insult and Offensive Behavior, Miscellaneous Non-Violent Offenses, Murder or Attempted Murder, Non-Sexual Assault, Political Activities, Sexual Assault or Violence, Theft and Robbery), return the name of the category that is closest to the following entry: `< Recorded crime >`.” The categories themselves are generated by the same large language model using the following prompt: “You are a research assistant. You do not make inference. This is a list of entries. Provide a list of at most 10 elements to categorize them. `< List of all recorded crimes >`.”

#### *A.I.3 Consistent County Borders*

All data refer to historical county borders. By the 1890s, most county geographies had stabilized. However, some Western counties still underwent major boundary changes. This is unlikely to drive our results, since the vast majority of inventors resided in the Midwest and Eastern states. Nonetheless, to ensure consistency, we harmonize the geography to 1900 county borders following the simple method proposed by Eckert, Gvirtz, Liang and Peters (2020).

#### *A.I.4 Construction of the Sample*

The core of the analysis relies on a balanced surname-level dataset with information on the number and importance of patents granted to inventors with each surname, the Black Surname Index, and a set of variables constructed from the census (e.g., the share of Black individuals with a given surname). To construct this dataset, we exclude very rare surnames that appear fewer than 100 times in the 1900

census, following Fouka (2020). We apply this sample restriction because these surnames would be associated with very few patents, and their BSI would be less informative about their racial content because it would be constructed from such a small population.<sup>2</sup>

To conduct the difference-in-differences analysis, we winsorize the top 1% of the distribution of the outcome variables.<sup>3</sup> This procedure ensures that particularly common surnames do not drive our results. Moreover, we impose that each surname is observed over a five-year window around the match-specific treatment date. For example, consider surnames (Smith) and (Brown) in the same match. Surname (Smith) is treated in year  $\tau^*$  and (Brown) is not. In the estimation sample, we observe both surnames over the period  $[\tau^* - 5, \tau^* + 5]$ . The sample is therefore a balanced panel in which, irrespective of the specific treatment date  $\tau^*$ , each unit is observed over eleven consecutive periods.<sup>4</sup> We drop surnames without at least five pre- and post-treatment observations.

The second part of the analysis is conducted using an individual-level dataset. For the descriptive analysis, the sample contains both inventors and the rest of the population. This dataset pools individual-level census data from 1900, 1910, and 1920 to construct a cross-section with information on the number of patents granted to each individual, if any, and on demographic characteristics from the census. We need to pool different censuses because inventors are linked to the census year closest to each patent.<sup>5</sup> For computational reasons, we perform the descriptive analysis at the individual level using a 20% random sample of the resulting cross-section.

To conduct the individual-level difference-in-differences analysis, we assemble a balanced individual-level panel in which each inventor is followed over a ten-year window around the census year to which he is matched. Since we do not observe when inventors become active or when they retire, we restrict the sample to those who are granted at least one patent both before and after treatment. To construct the matched control group, we select inventors who are not treated and who are similarly granted at least one patent both before and after each treatment date.

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<sup>2</sup>In the Census surname variable, we retain only the first word. This avoids mistakenly selecting both the last and first name in the rare cases where both are erroneously recorded in the surname field. In such cases, the variable reports the last name followed by the first name.

<sup>3</sup>Winsorizing the bottom 1% of the data yields no changes as more than 1% of the observations are zeros.

<sup>4</sup>We choose this window length because (i) a longer window would exclude more surnames from the balanced sample, (ii) the longer the window, the more likely it is that a surname may be treated more than once, and (iii) it allows us to estimate flexible specifications over a symmetric five-year window around the treatment date.

<sup>5</sup>For example, inventors listed on a patent issued in 1903 are linked to the 1900 census, whereas those listed on a patent issued in 1907 are linked to the 1910 census.

## A.II Tables

**Table A.1.** Sample Black Surname Index

| Entire Population                     |                            |                  | Excluding Immigrants |                            |                  |
|---------------------------------------|----------------------------|------------------|----------------------|----------------------------|------------------|
| Surname<br>(1)                        | Black Surname Index<br>(2) | Frequency<br>(3) | Surname<br>(4)       | Black Surname Index<br>(5) | Frequency<br>(6) |
| Panel A. Unambiguously Black-Sounding |                            |                  |                      |                            |                  |
| Straughter                            | 1.000                      | 205              | Straughter           | 1.000                      | 205              |
| Ginyard                               | 1.000                      | 108              | President            | 1.000                      | 122              |
| Wineglass                             | 1.000                      | 103              | Armwood              | 1.000                      | 118              |
| Fantroy                               | 1.000                      | 106              | Lemelle              | 1.000                      | 111              |
| Brimage                               | 1.000                      | 129              | Ginyard              | 1.000                      | 108              |
| Panel B. Suggestively Black-Sounding  |                            |                  |                      |                            |                  |
| Rials                                 | 0.800                      | 356              | Mcferson             | 0.800                      | 245              |
| Grissett                              | 0.800                      | 286              | Midelton             | 0.800                      | 178              |
| Stacker                               | 0.800                      | 394              | Puryear              | 0.800                      | 1033             |
| Streams                               | 0.800                      | 108              | Alex                 | 0.800                      | 432              |
| Brunch                                | 0.800                      | 108              | Banknight            | 0.800                      | 108              |
| Panel C. Uninformative                |                            |                  |                      |                            |                  |
| Hutchens                              | 0.500                      | 2567             | Ginn                 | 0.500                      | 2176             |
| Cure                                  | 0.500                      | 477              | Packer               | 0.500                      | 3356             |
| Chain                                 | 0.500                      | 425              | Meares               | 0.500                      | 347              |
| Elliston                              | 0.500                      | 425              | Stan                 | 0.500                      | 321              |
| Fitsgerald                            | 0.500                      | 399              | Sorter               | 0.500                      | 269              |
| Panel D. Suggestively White-Sounding  |                            |                  |                      |                            |                  |
| Dawes                                 | 0.200                      | 1457             | Mullet               | 0.200                      | 412              |
| Gorrell                               | 0.200                      | 982              | Booton               | 0.200                      | 222              |
| Lanagan                               | 0.200                      | 317              | Gramling             | 0.200                      | 444              |
| Lavine                                | 0.200                      | 1585             | Pilant               | 0.200                      | 222              |
| Parmley                               | 0.200                      | 571              | Abels                | 0.200                      | 444              |
| Panel E. Unambiguously White-Sounding |                            |                  |                      |                            |                  |
| Groark                                | 0.000                      | 101              | Fadeley              | 0.000                      | 101              |
| Perkey                                | 0.000                      | 240              | Preis                | 0.000                      | 193              |
| Glatt                                 | 0.000                      | 240              | Vanetten             | 0.000                      | 296              |
| Weiskopf                              | 0.000                      | 240              | Collopy              | 0.000                      | 246              |
| Wingrove                              | 0.000                      | 240              | Rubinstein           | 0.000                      | 116              |

*Notes.* This table reports a set of sample surnames together with their estimated Black Surname Index (BSI). Columns (1)–(3) report the BSI estimated using the full population, whereas columns (4)–(6) restrict the sample to non-migrants. Panel A reports a set of unambiguously Black-sounding surnames, all with a BSI above 0.99. Panel B reports surnames with a BSI around 0.8, which implies that they are approximately four times more common among Black individuals than among white individuals. Panel C reports surnames for which the BSI is uninformative because they are equally common among Black and white individuals. Panel D presents the opposite case, showing surnames that are approximately four times more common among white individuals. Finally, Panel E reports surnames that are unambiguously white, in the sense that no Black individual in the full population has them. Referenced on page: 8.

**Table A.2.** Descriptive Statistics on Lynching Episodes

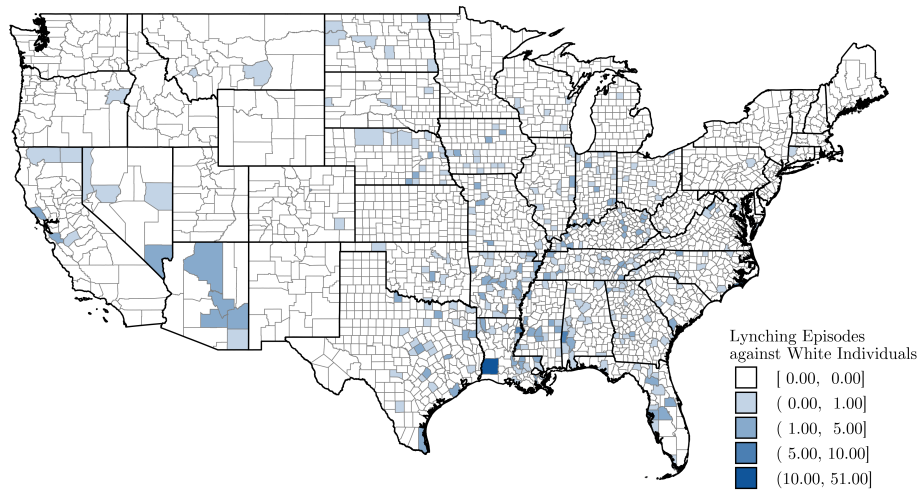
|                                     | Mean<br>(1) | Standard Deviation<br>(2) | Median<br>(3) | Min<br>(4) | Max<br>(5) | N. Observations<br>(6) |
|-------------------------------------|-------------|---------------------------|---------------|------------|------------|------------------------|
| Panel A. Number of Episodes         |             |                           |               |            |            |                        |
| Number of Episodes                  | 1.749       | 2.712                     | 1             | 1          | 39         | 2,043                  |
| Number of Black victims             | 1.793       | 4.649                     | 1             | 0          | 95         | 2,043                  |
| Number of White victims             | 0.442       | 0.915                     | 0             | 0          | 17         | 2,043                  |
| Distance from D.C.                  | 1290.681    | 595.965                   | 1,230         | 0          | 3,968      | 2,043                  |
| Panel B. Newspaper Coverage         |             |                           |               |            |            |                        |
| Episodes Covered in Newspapers      | 1.008       | 2.787                     | 1             | 0          | 52         | 2,043                  |
| Episodes Covered in D.C. Newspapers | 0.573       | 2.280                     | 0             | 0          | 41         | 2,043                  |
| Panel C. Region                     |             |                           |               |            |            |                        |
| Midwest                             | 0.153       | 0.415                     | 0             | 0          | 4          | 2,043                  |
| Northeast                           | 0.008       | 0.091                     | 0             | 0          | 1          | 2,043                  |
| South                               | 1.480       | 2.428                     | 1             | 0          | 34         | 2,043                  |
| West                                | 0.081       | 0.289                     | 0             | 0          | 3          | 2,043                  |
| Panel D. Alleged Offense            |             |                           |               |            |            |                        |
| Aid Criminals                       | 0.065       | 0.272                     | 0             | 0          | 4          | 2,043                  |
| Arson and Property Damage           | 0.063       | 0.286                     | 0             | 0          | 4          | 2,043                  |
| Non-Sexual Assault                  | 0.059       | 0.250                     | 0             | 0          | 3          | 2,043                  |
| Insult and Offensive Behavior       | 0.062       | 0.272                     | 0             | 0          | 4          | 2,043                  |
| Murder or Attempted Murder          | 0.711       | 1.483                     | 0             | 0          | 22         | 2,043                  |
| Miscellaneous Non-Violent Offenses  | 0.025       | 0.161                     | 0             | 0          | 2          | 2,043                  |
| Political Activities                | 0.039       | 0.199                     | 0             | 0          | 2          | 2,043                  |
| Sexual Assault or Violence          | 0.468       | 1.196                     | 0             | 0          | 20         | 2,043                  |
| Theft and Robbery                   | 0.110       | 0.373                     | 0             | 0          | 6          | 2,043                  |
| Unknown Allegation                  | 0.324       | 0.845                     | 0             | 0          | 14         | 2,043                  |

*Notes.* This table reports summary statistics for the sample of lynching episodes. The underlying data come from Hines and Steelwater (2006) and Seguin and Rigby (2019). We classify the alleged offense into ten mutually exclusive categories. To compute the statistics, we collapse the data to the surname level. Panel A reports the number of episodes, the number of victims by race, and the distance between each episode and Washington, D.C., where the patent office was located. Panel B reports summary statistics on newspaper coverage of the same lynching episodes. Panel C breaks down episodes by the Census Bureau region in which they occurred. Panel D reports the distribution of episodes across offense categories. The columns report the mean, standard deviation, median, minimum, maximum, and number of observations for each variable. Referenced on page: 9.

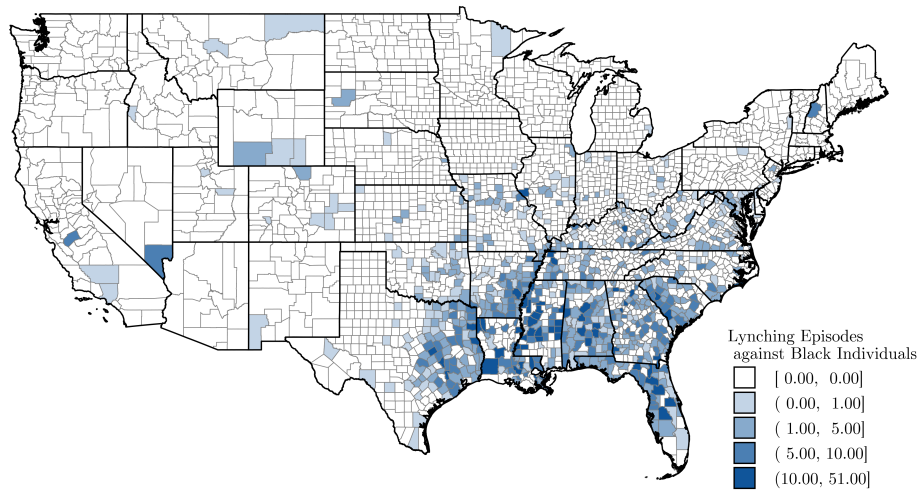
### A.III Figures

**Figure A.1.** Spatial Distribution of Lynching Episodes Involving Black and White Victims

(a) Distribution of Lynching Episodes Involving White Victims



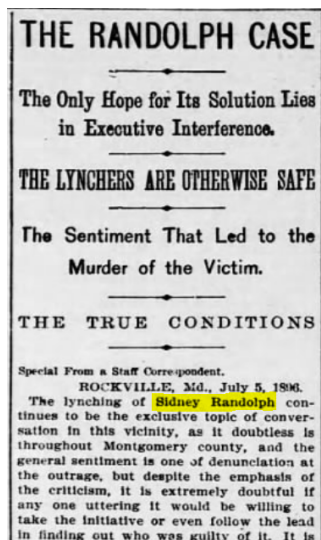
(b) Distribution of Lynching Episodes Involving Black Victims



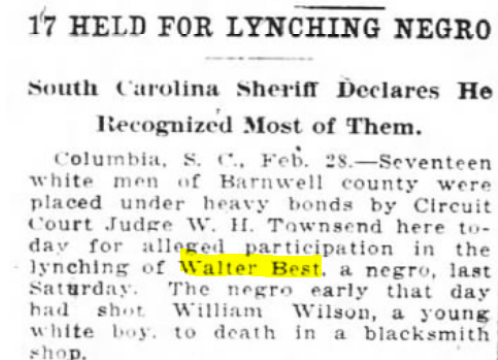
*Notes.* Panel A.1a displays the county-level spatial distribution of lynching episodes involving reportedly white victims in the United States over the period 1882–1935. Panel A.1b displays the county-level spatial distribution of lynching episodes involving reportedly Black victims in the United States over the period 1882–1935. Source: Authors' calculations based on data from Hines and Steelwater (2006) and Seguin and Rigby (2019). Referenced on page: 6.

Figure A.2. Sample Newspaper Articles

(a) Evening Star,  
July 1896



(b) Washington Post,  
March 1918



*Notes.* Examples of Washington, D.C., newspapers reporting on lynching episodes. Both newspapers were based in Washington, D.C., at the time of the reporting. Source: Authors' search of the [newspapers.com](http://newspapers.com) archive. Referenced on page: 6.

**Figure A.3.** Example of Coverage of the Same Episode Across States

(a) The Chicago Daily Tribune,  
Chicago, IL, September 1935

**DEATH SUSPECT  
IS LYNCHED AS  
JURY DEBATES**

Colored Defendant, 28,  
Dragged from Cell.

Oxford, Miss., Sept. 18.—[Special].—Judge Taylor McElroy today called before him and dismissed the jury that had been deliberating the case of Ellwood Higginbotham, 28, colored, who was charged with slaying a white planter. The services of the jury were no longer needed, for the defendant was lynched last night by a mob.

Higginbotham had been on trial for killing Glen Roberts on May 21. Witnesses said the planter was killed in "cold blood" when he went to the

(b) The New York Times, New  
York City, NY, September 1935

**MOB LYNCHES NEGRO.**  
Mississippi Prisoner Seized as Jury  
Debates Murder Case.

OXFORD, Miss., Sept. 18 (AP).—A mob lynched Ellwood Higginbotham, a 28-year-old Negro, near here last night while a jury was deliberating charges that he killed Glen Roberts, a white planter.

The screaming, terrified Negro was seized in the county jail by a crowd of between 100 and 150 persons and hanged to a tree two and a half miles from town.

The jury, which had been deliberating a verdict since late yesterday, was dismissed by Judge Taylor McElroy. It was not learned how the verdict stood, but the belief was expressed that the failure of the jury to return a quick verdict had prompted the lynching.

Members of the mob stormed the jail, overpowered Sheriff S. T. Lyles and three deputies and started smashing doors while the officers tried vainly to prevent seizure of the prisoner.

The body was left swinging from the tree. It was cut down later by the Sheriff and his aides.

Officers announced an investigation, but said none of the mob had been identified.

(c) The Atlanta Daily World,  
Atlanta, GA, September 1935

**MOB HANGS NEGRO  
WHILE JURY DE-  
LIBERATES FATE**

Deputies Overpowered. Witness  
Tells Court of Slaying of Farmer  
And Citizens Take Law Into Own  
Hands—Near University.

Oxford, Miss., Sept. 18.—While a Lafayette County jury was deliberating his fate last night, Ellwood Higginbotham, 28, a negro alleged slayer of a white man, was taken from his county jail cell and lynched.

Today, Judge Taylor McElroy, presiding over the negro's trial, called the jury into the court room and dismissed the panel.

The lynching was blamed by Sheriff S. T. Lyles, of Lafayette County, on a mob of between 100 and 150 citizens who stormed the jail and overpowered the jailer and

(d) The North Mississippi Herald,  
Water Valley, MS, September 1935

**Mob Lynches  
Man At End  
Of Trial**

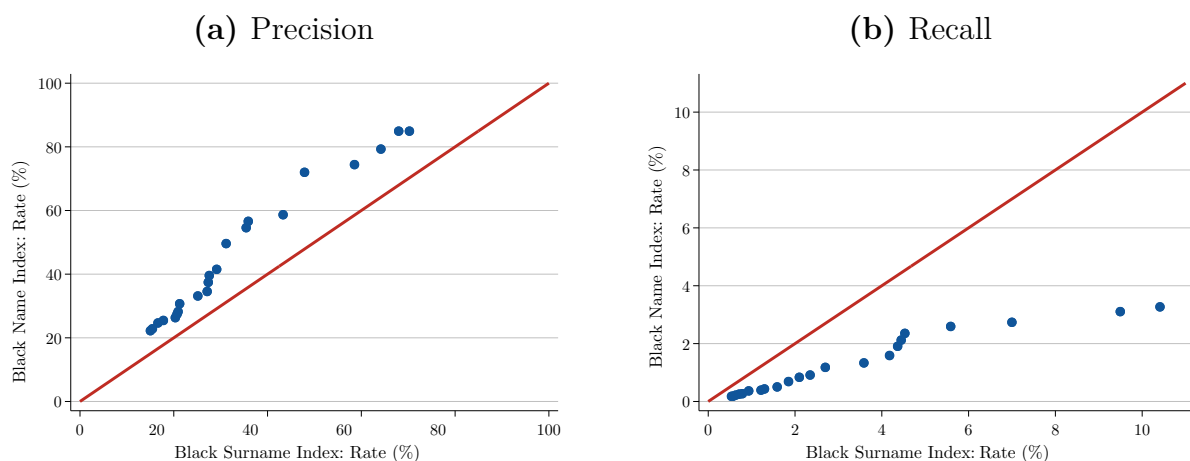
Certain Sentence Is  
No Deterrent For  
Unmasked Gang  
**JURY IS OUT**

OXFORD, Miss.—(ANP)—Another Negro has been lynched in Mississippi and again whatever investigation is made will mean naught.

Elwood Higginbotham was the victim and the lynching occurred Tuesday night as the jury deliberated his fate on a charge of murdering a white man. The procedure was as usual: a group of forty or fifty unmasked white men "overpowered" the sheriff and his THREE deputies removed the prisoner from the jail during the trial.

*Notes.* Examples of newspaper coverage of the same lynching episode across states. All articles report on the same episode, which resulted in the death of Elwood Higginbotham in September 1935. The episode took place in Oxford, Mississippi. Source: The New York Times, April 2018. Referenced on page: 6.

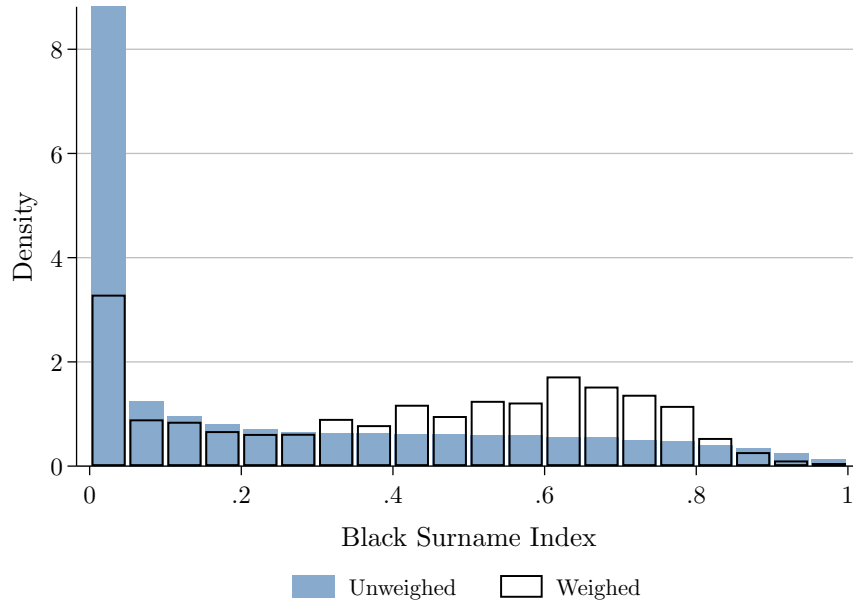
**Figure A.4.** Comparison of the Black Name and Surname Index Classifiers



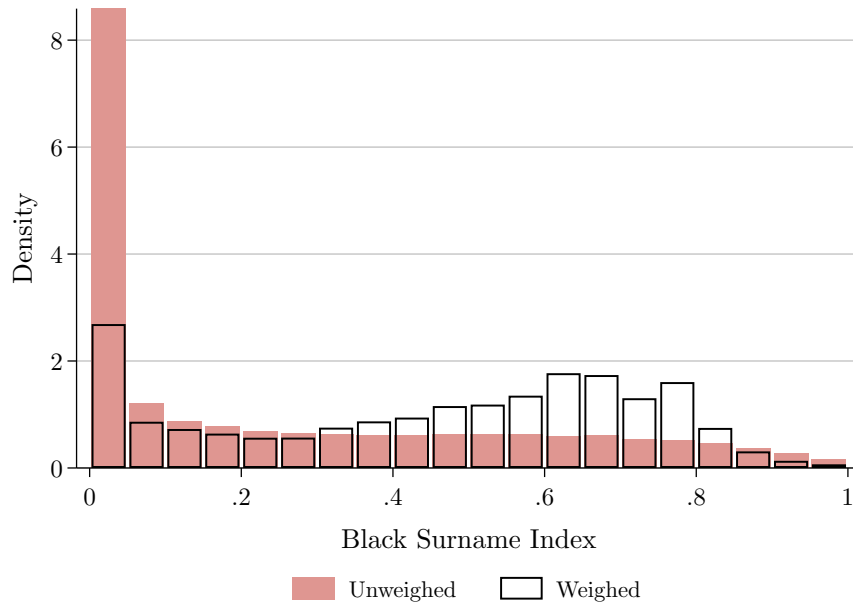
*Notes.* This figure compares classifiers based on the Black Name Index and the Black Surname Index. For threshold values between 0.4 and 1, we compute the number of individuals in the 1900–1930 censuses whom each classifier identifies as Black because their BNI or BSI is at least as large as the threshold. In Panel A.4a, we report precision, defined as the share of individuals correctly identified as Black relative to the number of individuals classified as Black by each classifier. In Panel A.4b, we report recall, defined as the share of individuals correctly identified as Black relative to the total Black population. In each panel, the  $x$ -axis reports the value for the BSI-based classifier, while the  $y$ -axis reports the value for the BNI-based classifier. The red line denotes the 45-degree line. If the two classifiers had the same precision in Panel A.4a and the same recall in Panel A.4b, all points would lie on the 45-degree line. Referenced on page: 8.

**Figure A.5.** Distribution of the Black Surname Index

**(a)** Distribution Including Immigrants

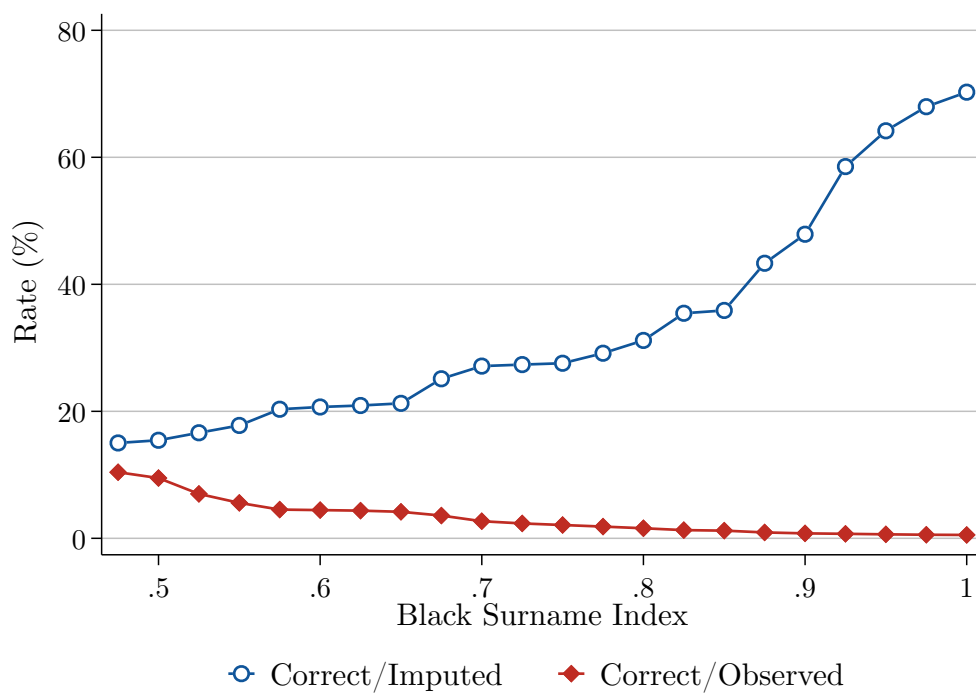


**(b)** Distribution Excluding First-Generation Immigrants



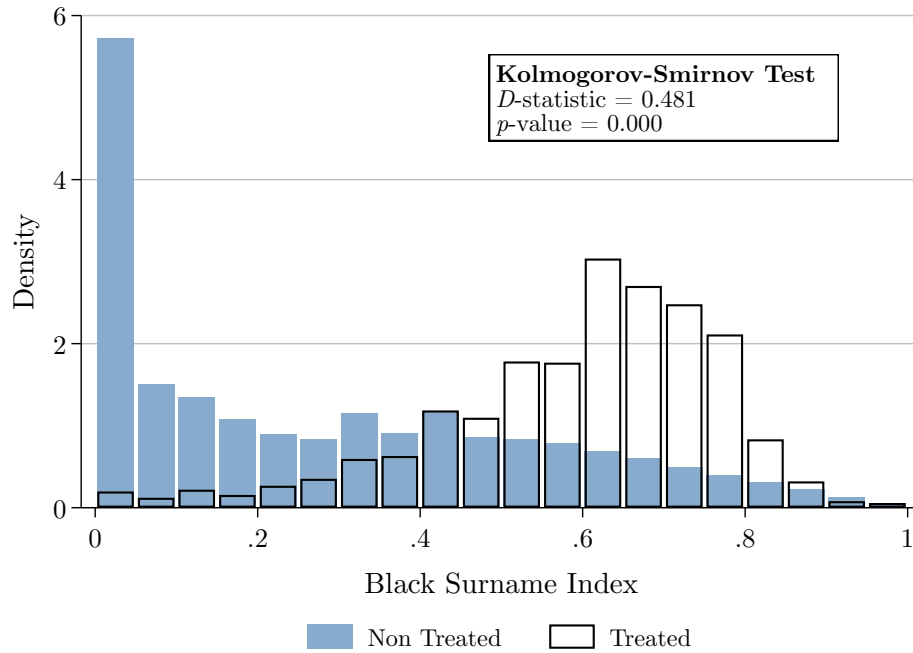
*Notes.* This figure reports the distribution of the Black Surname Index. In Panel A, the index is calculated using the full U.S. population in 1900; in Panel B, first-generation immigrants are excluded. The colored bars report the raw distribution of the two indices. The black-outlined bars report the distribution obtained when weighting each surname by the number of individuals carrying it in the 1900 full-count census. Thus, the colored bars display the distribution of estimated BSI scores across surnames, whereas the outlined bars show the distribution of those scores across individuals in the population. Referenced on page: 8.

**Figure A.6.** Precision and Recall of the Black Surname Index



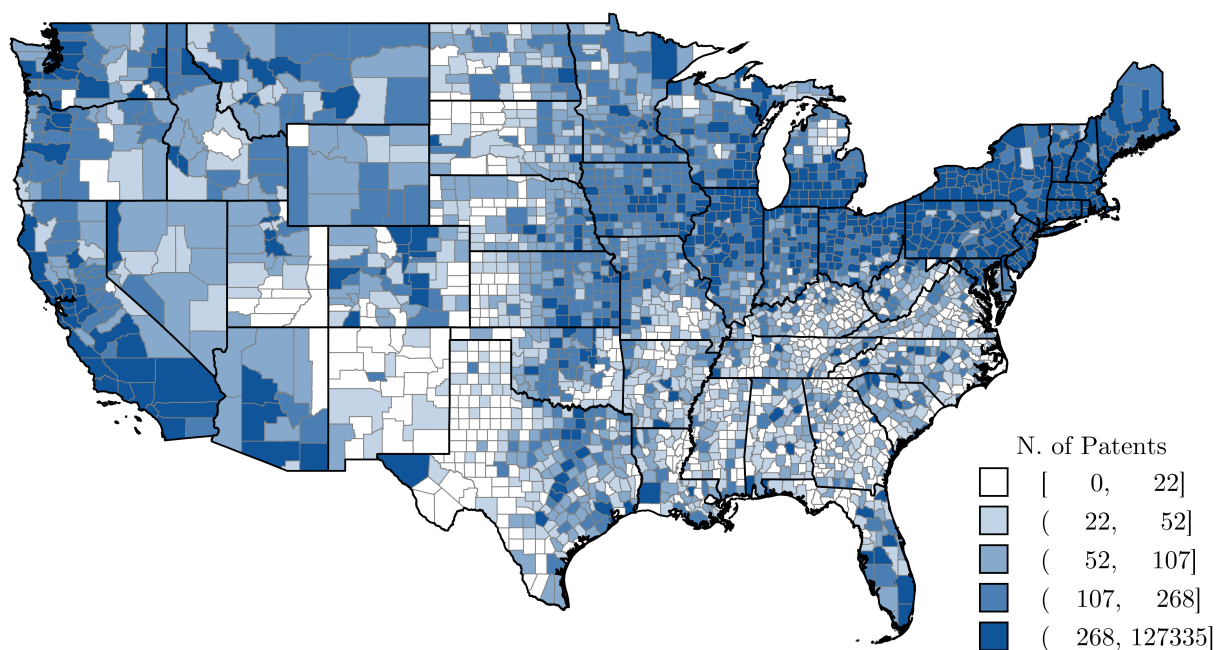
*Notes.* This figure reports two classification metrics for the Black Surname Index at different threshold values. The blue series reports precision, measured as the share of Black individuals among those with a BSI above a given threshold. For example, approximately 40% of individuals with a BSI above 0.8 are Black. The red series reports recall, measured as the share of Black individuals with a BSI above a given threshold relative to the total Black population. Referenced on page: 8.

**Figure A.7.** Distribution of the Black Surname Index of Victims of Lynching Episodes



*Notes.* This figure compares the distribution of the Black Surname Index for victims of lynching episodes with that of non-treated observations. The blue bars report the distribution for non-treated observations, while the black-outlined bars report the distribution for victims of lynching episodes. The figure also reports the  $D$ -statistic and associated  $p$ -value from a Kolmogorov–Smirnov test of equality of the two distributions. A low  $p$ -value indicates that we reject the null hypothesis that the two distributions are the same. Referenced on page: 8.

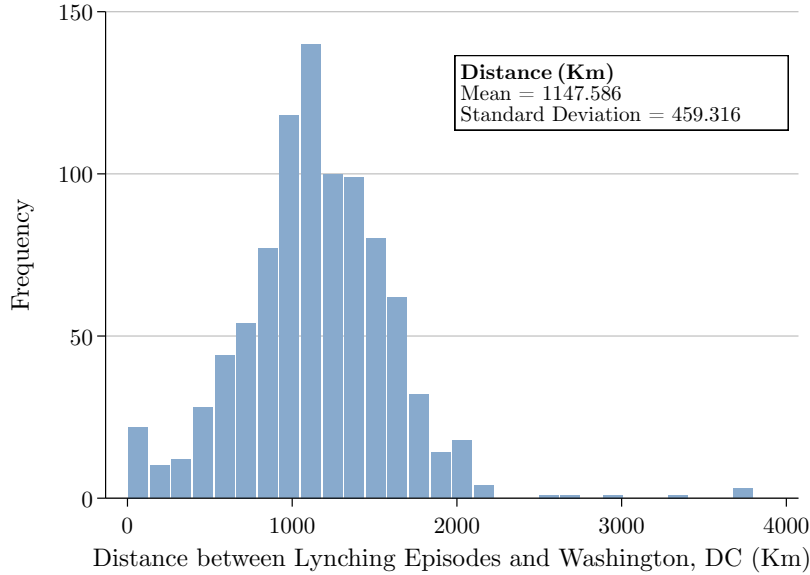
**Figure A.8.** Spatial Distribution of Granted Patents, 1882–1936



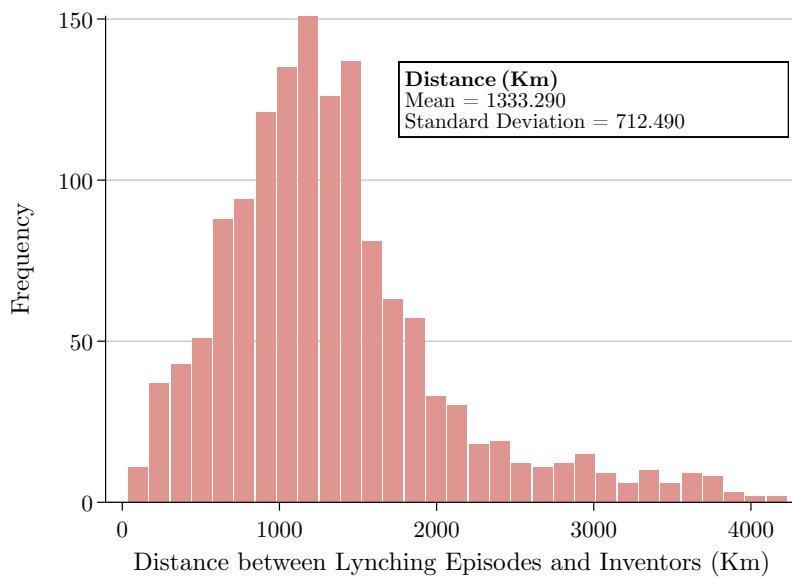
*Notes.* This figure reports the number of patents issued between 1882 and 1936, assigned to counties based on inventors' residential addresses. Patent records report each inventor's address of residence. To construct the underlying data, we assign inventors' address coordinates to counties using the 1910 GIS county shapefile. The map reports quintiles of patenting activity, with darker shades of blue indicating higher levels of patenting. Referenced on page: 9.

**Figure A.9.** Distance to Lynching Episodes

(a) Distance Between Lynching Episodes and the U.S. Patent Office in Washington, D.C.



(b) Distance Between Lynching Episodes and Inventors' Residential Address



*Notes.* Panel A reports the distribution of the distance between lynching episodes and Washington, D.C., where the patent office was located. Panel B reports the distribution of the distance between inventors and the nearest lynching episode. Distances are measured in kilometers. To construct the figure, we first geocode each lynching episode. Patent records report the latitude and longitude of each inventor's residential address. Panel A therefore reports the geodesic distance between Washington, D.C., and each lynching episode. In Panel B, we identify the nearest lynching episode for each inventor listed in the patent records and average these distances at the surname level at which the regressions are run. Thus, the frequency on the y-axis refers to surnames. We also report the mean and standard deviation of each distribution. For reference, the width of the United States is approximately 4,000 km. Referenced on page: 24.

## B Robustness of Stylized Facts

### B.I Tables

**Table B.1.** Correlation between Black Name Index and Patenting Activity

| Dependent Variable:                 | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  |
|-------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                     | Patents per Capita   |                      |                      |                      |                      |                      |                      |                      |
| Black Name Index                    | -2.116***<br>(0.463) | -2.116***<br>(0.463) | -1.468***<br>(0.341) | -1.144***<br>(0.304) |                      |                      |                      |                      |
| Black Name Index (excl. immigrants) |                      |                      |                      |                      | -2.277***<br>(0.493) | -2.277***<br>(0.494) | -1.617***<br>(0.352) | -1.257***<br>(0.313) |
| Year FE                             | No                   | Yes                  | Yes                  | Yes                  | No                   | Yes                  | Yes                  | Yes                  |
| State FE                            | No                   | No                   | Yes                  | –                    | No                   | No                   | Yes                  | –                    |
| County FE                           | No                   | No                   | No                   | Yes                  | No                   | No                   | No                   | Yes                  |
| Number of Names                     | 2,493                | 2,493                | 2,493                | 2,493                | 2,493                | 2,493                | 2,493                | 2,493                |
| Observations                        | 11,883,008           | 11,883,008           | 11,883,008           | 11,883,008           | 11,883,008           | 11,883,008           | 11,883,008           | 11,883,008           |
| R <sup>2</sup>                      | 0.002                | 0.008                | 0.042                | 0.102                | 0.002                | 0.008                | 0.042                | 0.102                |
| Dependent Variable (Mean)           | 0.000                | 0.000                | 0.000                | 0.000                | 0.000                | 0.000                | 0.000                | 0.000                |

*Notes.* This table displays the correlation between patents per capita and the Black Name Index. The unit of observation is a name-by-county pair observed annually between 1882 and 1936. The dependent variable is the number of patents obtained by inventors with a given name, divided by the frequency of that name in the 1900 census. Each regression is estimated using Poisson quasi-likelihood. Columns (2) and (6) include year fixed effects; columns (3) and (7) include state-by-year fixed effects; columns (4) and (8) include county-by-year fixed effects. In columns (1)–(4), the independent variable is the baseline Black Name Index; in columns (5)–(8), the Black Name Index is constructed excluding first-generation immigrants. Standard errors are clustered by name and reported in parentheses. Referenced on page: 8.

\*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

**Table B.2.** Correlation between the Black Surname Index and Observable Characteristics

|                                                                                  | Surname Level      |                       | Individual Level   |                       |
|----------------------------------------------------------------------------------|--------------------|-----------------------|--------------------|-----------------------|
|                                                                                  | Coefficient<br>(1) | Standard Error<br>(2) | Coefficient<br>(3) | Standard Error<br>(4) |
| Panel A. Correlation between Black Surname Index and Demographic Characteristics |                    |                       |                    |                       |
| Number of Patents                                                                | 0.199***           | (0.011)               | -0.009**           | (0.004)               |
| Surname Frequency                                                                | 0.339***           | (0.011)               | 0.356***           | (0.068)               |
| Share of Blacks                                                                  | 0.939***           | (0.007)               | 0.141***           | (0.006)               |
| Share of Immigrants                                                              | -0.378***          | (0.009)               | -0.168***          | (0.006)               |
| Share of Males                                                                   | 0.027***           | (0.009)               | -0.023***          | (0.004)               |
| Share of Married                                                                 | -0.292***          | (0.009)               | -0.008**           | (0.004)               |
| Share of Literate                                                                | -0.684***          | (0.008)               | -0.040***          | (0.004)               |
| Panel B. Correlation between Black Surname Index and Area of Residence           |                    |                       |                    |                       |
| Midwest                                                                          | -0.508***          | (0.009)               | -0.061***          | (0.005)               |
| Northeast                                                                        | -0.497***          | (0.008)               | -0.055***          | (0.005)               |
| South                                                                            | 0.799***           | (0.006)               | 0.151***           | (0.005)               |
| West                                                                             | -0.137***          | (0.009)               | 0.027***           | (0.005)               |
| Panel C. Correlation between Black Surname Index and Occupational Structure      |                    |                       |                    |                       |
| Clerical                                                                         | -0.309***          | (0.009)               | 0.010***           | (0.004)               |
| Skilled Blue Collar                                                              | -0.565***          | (0.008)               | -0.042***          | (0.004)               |
| Farmer                                                                           | 0.410***           | (0.009)               | 0.041***           | (0.004)               |
| Laborer                                                                          | 0.663***           | (0.008)               | 0.012***           | (0.004)               |
| Manager                                                                          | -0.379***          | (0.009)               | -0.016***          | (0.004)               |
| Unskilled Blue Collar                                                            | -0.358***          | (0.009)               | -0.023***          | (0.004)               |
| Professional                                                                     | -0.127***          | (0.010)               | 0.008*             | (0.004)               |
| Sales                                                                            | -0.339***          | (0.009)               | 0.006              | (0.004)               |
| Services                                                                         | 0.184***           | (0.011)               | 0.015***           | (0.004)               |

*Notes.* This table displays the correlation between the Black Surname Index and a set of demographic characteristics. In columns (1)–(2), the unit of observation is a surname. The sample is a cross-section of surnames, and the variables are constructed from the censuses between 1900 and 1930. Each variable except the number of patents and surname frequency is expressed as a share of surname frequency; all variables are expressed as zero-inflated log counts. In columns (3)–(4), the unit of observation is an individual. The sample is a cross-section of inventors linked to full-count census records between 1900 and 1920. Each row reports the standardized beta coefficient from a regression of the outcome listed in that row on the Black Surname Index. Standard errors are reported in parentheses and clustered at the surname level. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on pages: 12, 14.

**Table B.3.** Correlation between Black Surname Index and Patenting: Individual-Level

| Dependent Variable:                    | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  |
|----------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Pr(Granted at Least One Patent)        |                      |                      |                      |                      |                      |                      |                      |                      |
| Panel A. Entire Population             |                      |                      |                      |                      |                      |                      |                      |                      |
| Black Surname Index                    | -0.662***<br>(0.033) | -0.228***<br>(0.036) | -0.503***<br>(0.031) | -0.397***<br>(0.031) |                      |                      |                      |                      |
| Black Surname Index (excl. immigrants) |                      |                      |                      |                      | -0.659***<br>(0.032) | -0.244***<br>(0.035) | -0.511***<br>(0.030) | -0.399***<br>(0.030) |
| Number of Individuals                  | 24,710,401           | 24,593,771           | 13,803,690           | 24,710,401           | 24,710,274           | 24,593,644           | 13,803,587           | 24,710,274           |
| R <sup>2</sup>                         | 0.002                | 0.013                | 0.038                | 0.015                | 0.002                | 0.013                | 0.038                | 0.015                |
| Panel B. White Sub-Sample              |                      |                      |                      |                      |                      |                      |                      |                      |
| Black Surname Index                    | -0.390***<br>(0.031) | -0.133***<br>(0.035) | -0.300***<br>(0.030) | -0.388***<br>(0.031) |                      |                      |                      |                      |
| Black Surname Index (excl. immigrants) |                      |                      |                      |                      | -0.395***<br>(0.030) | -0.152***<br>(0.034) | -0.315***<br>(0.029) | -0.391***<br>(0.030) |
| Number of Individuals                  | 22,224,322           | 22,140,224           | 12,038,989           | 22,224,322           | 22,224,195           | 22,140,097           | 12,038,886           | 22,224,195           |
| R <sup>2</sup>                         | 0.001                | 0.008                | 0.032                | 0.009                | 0.001                | 0.008                | 0.032                | 0.009                |
| Panel C. Black Sub-Sample              |                      |                      |                      |                      |                      |                      |                      |                      |
| Black Surname Index                    | -1.218***<br>(0.160) | -0.924***<br>(0.163) | -1.209***<br>(0.160) | -1.166***<br>(0.160) |                      |                      |                      |                      |
| Black Surname Index (excl. immigrants) |                      |                      |                      |                      | -1.251***<br>(0.164) | -0.944***<br>(0.167) | -1.247***<br>(0.164) | -1.196***<br>(0.163) |
| Number of Individuals                  | 2,486,079            | 2,453,547            | 1,764,701            | 2,486,079            | 2,486,079            | 2,453,547            | 1,764,701            | 2,486,079            |
| R <sup>2</sup>                         | 0.003                | 0.031                | 0.016                | 0.011                | 0.003                | 0.031                | 0.016                | 0.011                |
| Region FE                              | No                   | Yes                  | No                   | No                   | No                   | Yes                  | No                   | No                   |
| Occupation FE                          | No                   | No                   | Yes                  | No                   | No                   | No                   | Yes                  | No                   |
| Demographics FE                        | No                   | No                   | No                   | Yes                  | No                   | No                   | No                   | Yes                  |

*Notes.* This table displays the correlation between inventor status and the Black Surname Index. The unit of observation is an individual. In Panels A, B, and C, the sample consists of the full stacked 1900, 1910, and 1920 census population, the sub-sample of white individuals, and the sub-sample of Black individuals, respectively. The dependent variable is Pr(Granted at Least One Patent). Inventors are weighted by the inverse of the number of matches, as detailed in the text. Each regression is estimated using a Poisson model. Columns (2) and (6) include region fixed effects; columns (3) and (7) include occupation fixed effects; columns (4) and (8) include demographic controls: marital status, race—where non-redundant—, and literacy. In columns (1)–(4), the independent variable is the baseline Black Surname Index; in columns (5)–(8), the Black Surname Index is constructed excluding first-generation immigrants. Standard errors are clustered by surname and reported in parentheses. Referenced on pages: 14, 15.

\*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

**Table B.4.** Correlation between Black Surname Index and Patenting Activity

| Dependent Variable:                    | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  |
|----------------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                        | Patents per Capita   |                      |                      |                      |                      |                      |                      |                      |
| Black Surname Index                    | -4.514***<br>(0.250) | -4.514***<br>(0.250) | -4.175***<br>(0.211) | -3.402***<br>(0.138) |                      |                      |                      |                      |
| Black Surname Index (excl. immigrants) |                      |                      |                      |                      | -4.468***<br>(0.240) | -4.468***<br>(0.240) | -4.117***<br>(0.202) | -3.358***<br>(0.131) |
| Year FE                                | No                   | Yes                  | Yes                  | Yes                  | No                   | Yes                  | Yes                  | Yes                  |
| State FE                               | No                   | No                   | Yes                  | –                    | No                   | No                   | Yes                  | –                    |
| County FE                              | No                   | No                   | No                   | Yes                  | No                   | No                   | No                   | Yes                  |
| Number of Surnames                     | 93,276               | 93,276               | 93,276               | 93,276               | 90,197               | 90,197               | 90,197               | 90,197               |
| Observations                           | 33,824,320           | 33,824,320           | 33,824,320           | 33,824,320           | 33,569,920           | 33,569,920           | 33,569,920           | 33,569,920           |
| R <sup>2</sup>                         | 0.046                | 0.054                | 0.065                | 0.090                | 0.045                | 0.052                | 0.063                | 0.087                |
| Dependent Variable (Mean)              | 0.001                | 0.001                | 0.001                | 0.001                | 0.000                | 0.000                | 0.000                | 0.000                |

*Notes.* This table displays the correlation between patents per capita and the Black Surname Index. The unit of observation is a surname-by-county pair observed annually between 1882 and 1936. The dependent variable is the number of patents obtained by inventors with a given surname, divided by the frequency of that surname in the 1900 census. Each regression is estimated using Poisson quasi-likelihood. Columns (2) and (6) include year fixed effects; columns (3) and (7) include state-by-year fixed effects; columns (4) and (8) include county-by-year fixed effects. In columns (1)–(4), the independent variable is the baseline Black Surname Index; in columns (5)–(8), the Black Surname Index is constructed excluding first-generation immigrants. Standard errors are clustered by surname and reported in parentheses. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 14.

**Table B.5.** Correlation between Black Surname Index and High-Importance Patents

|                                        | (1)                                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 |
|----------------------------------------|-------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Panel A.                               |                                     |                     |                     |                     |                     |                     |                     |                     |
| Dependent Variable:                    | Share of Top 20% Importance Patents |                     |                     |                     |                     |                     |                     |                     |
| Black Surname Index                    | 0.461***<br>(0.076)                 | 0.461***<br>(0.076) | 0.921***<br>(0.123) | 1.644***<br>(0.206) |                     |                     |                     |                     |
| Black Surname Index (excl. immigrants) |                                     |                     |                     |                     | 0.449***<br>(0.077) | 0.449***<br>(0.077) | 0.932***<br>(0.124) | 1.679***<br>(0.216) |
| Observations                           | 33,824,320                          | 33,824,320          | 33,824,320          | 33,687,104          | 33,569,920          | 33,569,920          | 33,220,499          | 23,702,642          |
| R <sup>2</sup>                         | 0.001                               | 0.015               | 0.057               | 0.130               | 0.001               | 0.014               | 0.068               | 0.171               |
| Panel B.                               |                                     |                     |                     |                     |                     |                     |                     |                     |
| Dependent Variable:                    | Share of Top 10% Importance Patents |                     |                     |                     |                     |                     |                     |                     |
| Black Surname Index                    | 0.428***<br>(0.064)                 | 0.428***<br>(0.064) | 0.930***<br>(0.113) | 1.673***<br>(0.197) |                     |                     |                     |                     |
| Black Surname Index (excl. immigrants) |                                     |                     |                     |                     | 0.410***<br>(0.063) | 0.410***<br>(0.063) | 0.938***<br>(0.113) | 1.702***<br>(0.205) |
| Observations                           | 33,824,320                          | 33,824,320          | 33,824,320          | 33,233,536          | 33,569,920          | 33,569,920          | 32,824,152          | 20,071,720          |
| R <sup>2</sup>                         | 0.000                               | 0.022               | 0.068               | 0.144               | 0.000               | 0.022               | 0.083               | 0.172               |
| Panel C.                               |                                     |                     |                     |                     |                     |                     |                     |                     |
| Dependent Variable:                    | Share of Top 5% Importance Patents  |                     |                     |                     |                     |                     |                     |                     |
| Black Surname Index                    | 0.373***<br>(0.070)                 | 0.373***<br>(0.070) | 0.904***<br>(0.104) | 1.632***<br>(0.174) |                     |                     |                     |                     |
| Black Surname Index (excl. immigrants) |                                     |                     |                     |                     | 0.351***<br>(0.070) | 0.351***<br>(0.070) | 0.910***<br>(0.101) | 1.657***<br>(0.180) |
| Observations                           | 33,824,320                          | 33,824,320          | 33,824,320          | 32,158,656          | 33,569,920          | 33,569,920          | 31,579,242          | 16,286,044          |
| R <sup>2</sup>                         | 0.000                               | 0.034               | 0.082               | 0.154               | 0.000               | 0.034               | 0.099               | 0.174               |
| Year FE                                | No                                  | Yes                 | Yes                 | Yes                 | No                  | Yes                 | Yes                 | Yes                 |
| State FE                               | No                                  | No                  | Yes                 | –                   | No                  | No                  | Yes                 | –                   |
| County FE                              | No                                  | No                  | No                  | Yes                 | No                  | No                  | No                  | Yes                 |

*Notes.* This table displays the correlation between the share of highly important patents and the Black Surname Index. The unit of observation is a surname-by-county pair observed annually between 1882 and 1936. In Panels A, B, and C, the dependent variable is the number of patents in the top 20%, 10%, and 5% of the patent importance distribution, respectively, expressed as a share of the total number of patents. Zero-patent cells are coded as zeros. Each regression is estimated using Poisson quasi-likelihood. Columns (2) and (6) include year fixed effects; columns (3) and (7) include state-by-year fixed effects; columns (4) and (8) include county-by-year fixed effects. In columns (1)–(4), the independent variable is the baseline Black Surname Index; in columns (5)–(8), the Black Surname Index is constructed excluding first-generation immigrants. Standard errors are clustered by surname and reported in parentheses. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 14.

**Table B.6.** Correlation between Black Surname Index and Patent Importance: Alternative Measures

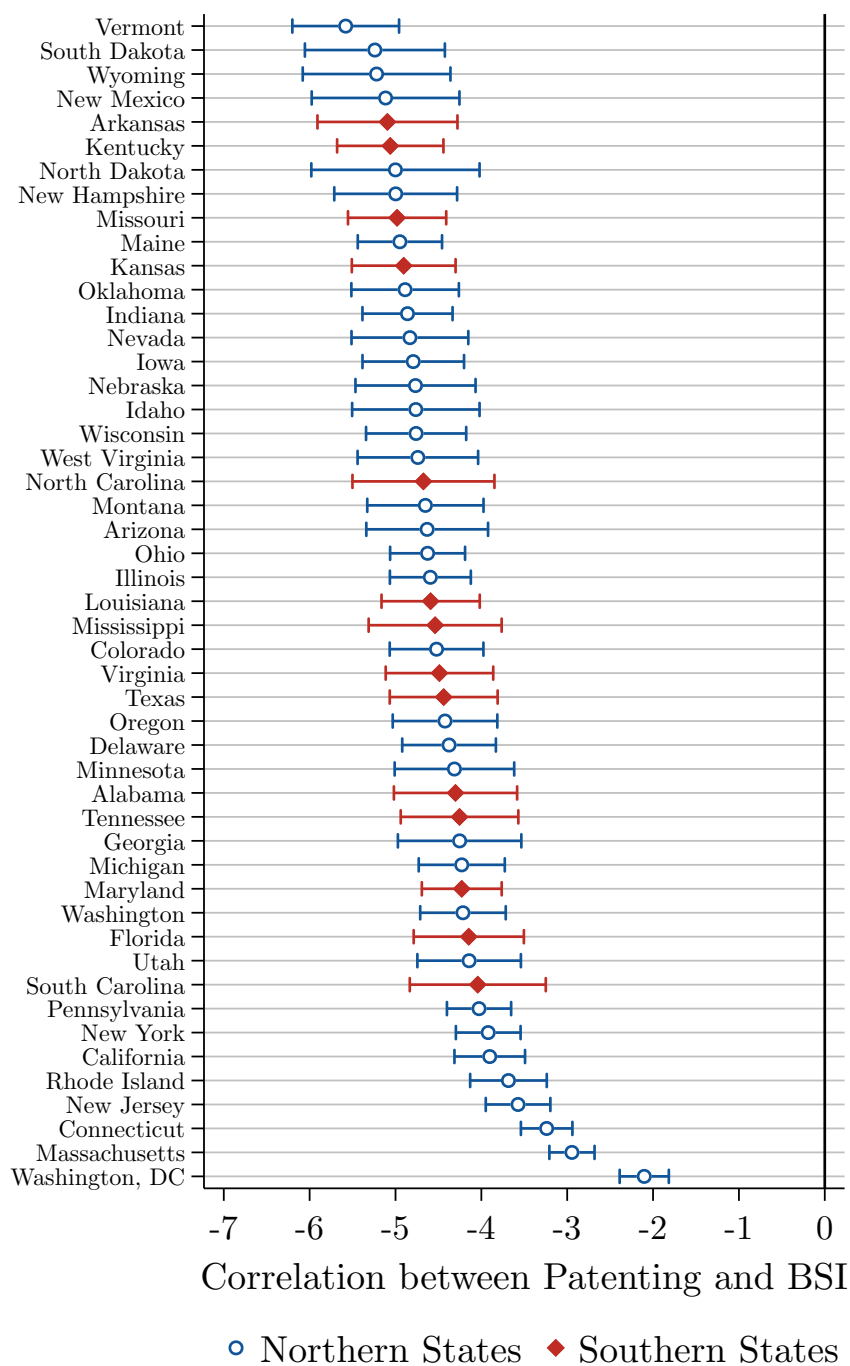
|                           | Year Fixed Effect Correction |                     |                     | No Year Fixed Effect Correction |                     |                     |
|---------------------------|------------------------------|---------------------|---------------------|---------------------------------|---------------------|---------------------|
|                           | (1)                          | (2)                 | (3)                 | (4)                             | (5)                 | (6)                 |
|                           | 1 Year                       | 5 Years             | 10 Years            | 1 Year                          | 5 Years             | 10 Years            |
| Black Surname Index       | 1.638***<br>(0.198)          | 1.673***<br>(0.197) | 1.670***<br>(0.192) | 1.655***<br>(0.224)             | 1.669***<br>(0.220) | 1.688***<br>(0.218) |
| Year FE                   | Yes                          | Yes                 | Yes                 | Yes                             | Yes                 | Yes                 |
| County FE                 | Yes                          | Yes                 | Yes                 | Yes                             | Yes                 | Yes                 |
| Number of Surnames        | 92,865                       | 92,697              | 92,615              | 92,922                          | 92,913              | 92,878              |
| Observations              | 33,377,600                   | 33,233,536          | 33,157,056          | 33,463,296                      | 33,436,736          | 33,399,936          |
| R <sup>2</sup>            | 0.129                        | 0.144               | 0.150               | 0.182                           | 0.191               | 0.180               |
| Dependent Variable (Mean) | 0.002                        | 0.002               | 0.002               | 0.002                           | 0.002               | 0.002               |

*Notes.* This table displays the correlation between the share of highly important patents and the Black Surname Index. The unit of observation is a surname-by-county pair observed annually between 1882 and 1936. In columns (1)–(3), we residualize the patent measure with respect to year fixed effects before aggregating across patents; in columns (4)–(6), we use the raw indicator. In columns (1) and (4), importance is constructed by comparing patents with those issued in the previous year; in columns (2) and (5), the comparison is with patents issued within the previous five years; in columns (3) and (6), it is with patents issued within the previous ten years. The dependent variable is the number of patents in the top 10% of each patent importance distribution, normalized by the annual number of patents. Zero-patent cells are coded as zeros. The independent variable is the baseline Black Surname Index. Each regression is estimated using Poisson quasi-likelihood. Standard errors are clustered by surname and reported in parentheses. Referenced on page: 14.

\*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$

## B.II Figures

**Figure B.1.** Correlation between Patenting Activity and Black Surname Index, by State



*Notes.* This figure reports the correlation between patenting activity and the Black Surname Index, estimated according to equation (2). The unit of observation is a surname-year cell observed between 1882 and 1936. The dependent variable is the number of patents obtained by inventors with that surname, normalized by surname frequency. Each row reports the estimated correlation using the subset of patents granted in a given state. Red diamonds remark the coefficients when the excluded state is in the South. The horizontal bars report 95% confidence intervals, with standard errors clustered by surname. Referenced on page: 14.

## C Robustness of Difference-in-Differences Results

### C.I Tables

**Table C.1.** Comparison Between Treated and Untreated Surnames Conditional on Surnames Included in the Estimation Sample

|                                             | Treated<br>Surnames | All Non-Treated<br>Surnames |            |         | Matched Non-Treated<br>Surnames |            |         |
|---------------------------------------------|---------------------|-----------------------------|------------|---------|---------------------------------|------------|---------|
|                                             | Mean                | Mean                        | Difference |         | Mean                            | Difference |         |
|                                             | (1)                 | (2)                         | (3)        | (4)     | (5)                             | (6)        | (7)     |
| <b>Panel A. Demographic Characteristics</b> |                     |                             |            |         |                                 |            |         |
| Black Surname Index                         | 0.546               | 0.235                       | 0.311***   | (0.009) | 0.501                           | 0.045***   | (0.014) |
| Share of Blacks Individuals                 | 0.154               | 0.056                       | 0.098***   | (0.004) | 0.135                           | 0.019***   | (0.007) |
| Number of Individuals                       | 9.178               | 7.169                       | 2.009***   | (0.047) | 9.104                           | 0.074      | (0.129) |
| Total Number of Patents                     | 2.470               | 1.310                       | 1.160***   | (0.047) | 2.500                           | -0.030     | (0.135) |
| Share of Immigrants                         | 0.065               | 0.112                       | -0.047***  | (0.002) | 0.073                           | -0.008**   | (0.004) |
| Share of Males                              | 0.409               | 0.410                       | -0.001**   | (0.000) | 0.408                           | 0.001**    | (0.001) |
| Share of Young                              | 0.307               | 0.294                       | 0.014***   | (0.001) | 0.302                           | 0.005***   | (0.002) |
| Share of Married                            | 0.313               | 0.322                       | -0.008***  | (0.001) | 0.315                           | -0.002     | (0.001) |
| Share of Literate                           | 0.507               | 0.540                       | -0.033***  | (0.002) | 0.514                           | -0.007***  | (0.003) |
| <b>Panel B. Area of Residence</b>           |                     |                             |            |         |                                 |            |         |
| South                                       | 0.349               | 0.197                       | 0.153***   | (0.006) | 0.320                           | 0.029***   | (0.009) |
| West                                        | 0.046               | 0.053                       | -0.007***  | (0.001) | 0.049                           | -0.003*    | (0.002) |
| Midwest                                     | 0.251               | 0.317                       | -0.066***  | (0.003) | 0.262                           | -0.011**   | (0.005) |
| North-East                                  | 0.200               | 0.273                       | -0.073***  | (0.004) | 0.217                           | -0.017***  | (0.007) |
| <b>Panel C. Occupational Structure</b>      |                     |                             |            |         |                                 |            |         |
| Professionals                               | 0.016               | 0.018                       | -0.002***  | (0.000) | 0.017                           | -0.001*    | (0.000) |
| Farmers                                     | 0.083               | 0.072                       | 0.011***   | (0.001) | 0.081                           | 0.002*     | (0.001) |
| Managers                                    | 0.017               | 0.021                       | -0.004***  | (0.000) | 0.018                           | -0.001***  | (0.000) |
| Clerical                                    | 0.011               | 0.013                       | -0.002***  | (0.000) | 0.012                           | -0.001***  | (0.000) |
| Sales                                       | 0.014               | 0.017                       | -0.003***  | (0.000) | 0.015                           | -0.001**   | (0.000) |
| Skilled Blue Collar                         | 0.034               | 0.044                       | -0.010***  | (0.000) | 0.036                           | -0.002***  | (0.001) |
| Unskilled Blue Collar                       | 0.040               | 0.048                       | -0.009***  | (0.001) | 0.041                           | -0.001     | (0.001) |
| Services                                    | 0.032               | 0.028                       | 0.004***   | (0.000) | 0.031                           | 0.001      | (0.001) |
| Laborers                                    | 0.120               | 0.102                       | 0.018***   | (0.001) | 0.116                           | 0.004***   | (0.001) |
| Number of Observations                      | 652                 | 7,870                       | 8,522      |         | 652                             | 1,304      |         |

*Notes.* This table reports sample means for surname-level characteristics. The sample of surnames includes only surnames included in the estimation sample of column (1) of Table III. One observation is one surname that is held by at least 100 individuals in the 1900 U.S. Census. Column (1) reports statistics for the surnames of victims of lynching episodes. Columns (2) and (3–4) report, respectively, statistics for the universe of untreated surnames and the differences between treated and untreated surnames. Columns (5) and (6–7) restrict the sample to untreated surnames matched to treated surnames. Standard errors are reported in parentheses and clustered by surname. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 16.

**Table C.2.** Comparison Between Treated and Untreated Inventors

|                                      | Treated Inventors | All Non-Treated Inventors |            |         | Matched Non-Treated Inventors |            |         |
|--------------------------------------|-------------------|---------------------------|------------|---------|-------------------------------|------------|---------|
|                                      | Mean              | Mean                      | Difference |         | Mean                          | Difference |         |
|                                      | (1)               | (2)                       | (3)        | (4)     | (5)                           | (6)        | (7)     |
| Panel A. Demographic Characteristics |                   |                           |            |         |                               |            |         |
| Black Surname Index                  | 0.614             | 0.286                     | 0.329***   | (0.011) | 0.438                         | 0.176***   | (0.014) |
| Is Black                             | 0.059             | 0.024                     | 0.035***   | (0.004) | 0.055                         | 0.004      | (0.008) |
| Number of Individuals                | 11.377            | 7.805                     | 3.573***   | (0.124) | 10.714                        | 0.663***   | (0.210) |
| Total Number of Patents              | 3.735             | 1.882                     | 1.853***   | (0.110) | 4.116                         | -0.381*    | (0.217) |
| Is Immigrant                         | 0.161             | 0.187                     | -0.026***  | (0.010) | 0.156                         | 0.005      | (0.016) |
| Is Male                              | 0.989             | 0.970                     | 0.019***   | (0.002) | 0.934                         | 0.054***   | (0.010) |
| Is Young                             | 0.195             | 0.298                     | -0.103***  | (0.016) | 0.186                         | 0.009      | (0.021) |
| Is Married                           | 0.727             | 0.724                     | 0.003      | (0.007) | 0.649                         | 0.078***   | (0.012) |
| Is Literate                          | 0.979             | 0.983                     | -0.004**   | (0.002) | 0.915                         | 0.064***   | (0.010) |
| Panel B. Area of Residence           |                   |                           |            |         |                               |            |         |
| South                                | 0.087             | 0.111                     | -0.024***  | (0.005) | 0.097                         | -0.010     | (0.009) |
| West                                 | 0.064             | 0.113                     | -0.049***  | (0.005) | 0.070                         | -0.006     | (0.008) |
| Midwest                              | 0.337             | 0.394                     | -0.056***  | (0.011) | 0.339                         | -0.002     | (0.017) |
| North-East                           | 0.504             | 0.375                     | 0.128***   | (0.013) | 0.462                         | 0.041**    | (0.019) |
| Panel C. Occupation                  |                   |                           |            |         |                               |            |         |
| Professional                         | 0.104             | 0.108                     | -0.005     | (0.004) | 0.088                         | 0.015**    | (0.007) |
| Farmer                               | 0.057             | 0.072                     | -0.016***  | (0.004) | 0.056                         | 0.001      | (0.006) |
| Manager                              | 0.170             | 0.162                     | 0.008      | (0.005) | 0.152                         | 0.017*     | (0.009) |
| Clerical                             | 0.048             | 0.040                     | 0.009***   | (0.003) | 0.047                         | 0.001      | (0.005) |
| Sales                                | 0.052             | 0.058                     | -0.006**   | (0.003) | 0.052                         | -0.000     | (0.005) |
| Skilled Blue Collar                  | 0.202             | 0.212                     | -0.010**   | (0.005) | 0.196                         | 0.006      | (0.010) |
| Unskilled Blue Collar                | 0.100             | 0.085                     | 0.015***   | (0.005) | 0.090                         | 0.010      | (0.007) |
| Services                             | 0.035             | 0.026                     | 0.009***   | (0.003) | 0.035                         | -0.000     | (0.005) |
| Laborer                              | 0.164             | 0.146                     | 0.017***   | (0.006) | 0.150                         | 0.014      | (0.010) |
| Number of Inventors                  | 5497              | 374469                    | 379966     |         | 5497                          | 10994      |         |

*Notes.* This table reports the sample means of a set of inventor-level characteristics. Column (1) reports statistics for treated inventors, defined as inventors whose surname is shared by at least one victim of a lynching episode. Columns (2)–(4) report, respectively, the mean among all non-treated inventors and the difference between treated and non-treated inventors. Columns (5)–(7) report, respectively, the mean among matched non-treated inventors and the difference between treated inventors and their matched non-treated counterparts. Standard errors are reported in parentheses and clustered by surname. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 18.

**Table C.3.** Surname-Level Effect of Lynching Episodes on Innovation: Alternative Standard Error Estimators

|                                | Number of Patents    |                      |                      |                      |                     |                     |                     |                    |
|--------------------------------|----------------------|----------------------|----------------------|----------------------|---------------------|---------------------|---------------------|--------------------|
|                                | No Adjust.           | White                | Clustered            |                      |                     |                     | Bootstrap           |                    |
|                                | (1)                  | (2)                  | (3)                  | (4)                  | (5)                 | (6)                 | (7)                 | (8)                |
|                                |                      |                      | Surname              | Surname-Year         | Surname, BSI        | Surname, Frequency  | No Adjust.          | Surname            |
| Lynching Episode $\times$ Post | -0.066***<br>(0.015) | -0.066***<br>(0.015) | -0.066***<br>(0.020) | -0.066***<br>(0.025) | -0.066**<br>(0.027) | -0.066**<br>(0.032) | -0.066**<br>(0.031) | -0.066*<br>(0.035) |
| Surname FE                     | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                |
| Match-Year FE                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                 | Yes                 | Yes                |
| Number of Surnames             | 1,304                | 1,304                | 1,304                | 1,304                | 1,304               | 1,304               | 1,304               | 1,304              |
| Observations                   | 13,262               | 13,262               | 13,262               | 13,262               | 13,262              | 13,262              | 13,262              | 13,262             |
| R <sup>2</sup>                 | 0.701                | 0.701                | 0.701                | 0.701                | 0.701               | 0.701               | 0.701               | 0.701              |

*Notes.* This table displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable is the number of patents. Each regression is estimated using Poisson quasi-likelihood, as in (3). All regressions include surname and match-by-year fixed effects. Each column reports results using a different standard error estimator: in column (1), we report unadjusted standard errors; in column (2), heteroskedasticity-robust standard errors; in columns (3)–(6), standard errors clustered at alternative levels: surname, two-way surname and year, and two-way surname and quantile of BSI; and in columns (7)–(8), bootstrapped standard errors without adjustment and clustered by surname, respectively. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 23.

**Table C.4.** Surname-Level Effect of Lynching on Innovation: Patents Filed Before Lynching Episode

|                                 | Number of Patents  |                    |                     |                     | Novelty           |
|---------------------------------|--------------------|--------------------|---------------------|---------------------|-------------------|
|                                 | (1)                | (2)                | (3)                 | (4)                 | (5)               |
| Lynching Episode $\times$ Post  | -0.056*<br>(0.029) | -0.050*<br>(0.028) | -0.067**<br>(0.031) | -0.063**<br>(0.031) | -0.143<br>(0.160) |
| Surname FE                      | Yes                | Yes                | Yes                 | Yes                 | Yes               |
| Match-Year FE                   | Yes                | Yes                | Yes                 | Yes                 | Yes               |
| BSI $\times$ Post               | No                 | No                 | Yes                 | No                  | No                |
| Surname Frequency $\times$ Post | No                 | Yes                | No                  | No                  | No                |
| Controls $\times$ Post          | No                 | No                 | No                  | Yes                 | No                |
| Number of Surnames              | 1,214              | 1,214              | 1,214               | 1,214               | 280               |
| Observations                    | 9,282              | 9,282              | 9,282               | 9,264               | 1,124             |
| R <sup>2</sup>                  | 0.663              | 0.664              | 0.663               | 0.667               | 0.229             |

*Notes.* This table displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable in columns (1)–(4) is the number of patents; in column (5), it is the number of patents in the top 5% of the patent importance distribution. To construct the dependent variables, we restrict the sample to patents filed before the lynching episode. The treatment variable is an interaction between a surname-specific lynching indicator and a match-specific post-treatment indicator. Each regression is estimated using Poisson quasi-likelihood, as in (3). All regressions include surname and match-by-year fixed effects. In columns (2), (3), and (4), we additionally include, respectively, quantiles of surname frequency, quantiles of the Black Surname Index, and a set of surname-level controls measured in 1900, each interacted with a post-episode indicator. Standard errors are reported in parentheses and clustered by surname. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 23.

**Table C.5.** Surname-Level Effect of Lynching Episodes on Innovation: Episodes with a White Victim

|                                              | Number of Patents |                   |                   |                   | Breakthrough      |
|----------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                              | (1)               | (2)               | (3)               | (4)               | (5)               |
| Lynching Episode against White $\times$ Post | -0.018<br>(0.033) | -0.027<br>(0.033) | -0.010<br>(0.037) | -0.009<br>(0.036) | -0.195<br>(0.119) |
| Surname FE                                   | Yes               | Yes               | Yes               | Yes               | Yes               |
| Match-Year FE                                | Yes               | Yes               | Yes               | Yes               | Yes               |
| BSI $\times$ Post                            | No                | No                | Yes               | No                | No                |
| Surname Frequency $\times$ Post              | No                | Yes               | No                | No                | No                |
| Controls $\times$ Post                       | No                | No                | No                | Yes               | No                |
| Number of Surnames                           | 380               | 380               | 380               | 378               | 166               |
| Observations                                 | 3,878             | 3,878             | 3,878             | 3,832             | 1,066             |
| R <sup>2</sup>                               | 0.736             | 0.737             | 0.736             | 0.741             | 0.284             |

*Notes.* This table displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable in columns (1)–(4) is the number of patents; in column (5), it is the number of patents in the top 5% of the patent importance distribution. The treatment variable is an interaction between a surname-specific indicator for lynching episodes involving white victims and a match-specific post-treatment indicator. Each regression is estimated using Poisson quasi-likelihood, as in (3). All regressions include surname and match-by-year fixed effects. In columns (2), (3), and (4), we additionally control for, respectively, quintiles of the Black Surname Index, quintiles of surname frequency, and a set of surname-level controls measured in 1900, each interacted with a post-lynching indicator. Standard errors are reported in parentheses and clustered by surname. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 23.

**Table C.6.** Surname-Level Effect of Lynching on Innovation: Alternative Importance Measures

|                                    | Year Fixed Effect Correction                                       |                      |                      | No Year Fixed Effect Correction |                      |                   |
|------------------------------------|--------------------------------------------------------------------|----------------------|----------------------|---------------------------------|----------------------|-------------------|
|                                    | (1)                                                                | (2)                  | (3)                  | (4)                             | (5)                  | (6)               |
| Time Window of Importance Measure: | 1 Yr                                                               | 5 Yrs                | 10 Yrs               | 1 Yr                            | 5 Yrs                | 10 Yrs            |
| Panel A.                           |                                                                    |                      |                      |                                 |                      |                   |
| Dependent Variable:                | Number of Patents in the Bottom 10% of the Importance Distribution |                      |                      |                                 |                      |                   |
| Lynching Episode $\times$ Post     | -0.252***<br>(0.060)                                               | -0.245***<br>(0.067) | -0.274***<br>(0.074) | -0.476***<br>(0.069)            | -0.433***<br>(0.134) | 0.022<br>(0.198)  |
| Number of Surnames                 | 764                                                                | 768                  | 694                  | 656                             | 446                  | 220               |
| Observations                       | 4,816                                                              | 4,514                | 4,148                | 3,212                           | 1,896                | 884               |
| R <sup>2</sup>                     | 0.292                                                              | 0.262                | 0.275                | 0.388                           | 0.315                | 0.357             |
| Panel B.                           |                                                                    |                      |                      |                                 |                      |                   |
| Dependent Variable:                | Number of Patents in the Top 10% of the Importance Distribution    |                      |                      |                                 |                      |                   |
| Lynching Episode $\times$ Post     | 0.077<br>(0.054)                                                   | -0.102*<br>(0.059)   | -0.069<br>(0.064)    | 0.009<br>(0.054)                | -0.155***<br>(0.058) | -0.065<br>(0.055) |
| Number of Surnames                 | 800                                                                | 744                  | 716                  | 852                             | 806                  | 830               |
| Observations                       | 5,238                                                              | 4,826                | 4,646                | 5,084                           | 5,030                | 5,246             |
| R <sup>2</sup>                     | 0.313                                                              | 0.320                | 0.330                | 0.423                           | 0.443                | 0.428             |
| Surname FE                         | Yes                                                                | Yes                  | Yes                  | Yes                             | Yes                  | Yes               |
| Match-Year FE                      | Yes                                                                | Yes                  | Yes                  | Yes                             | Yes                  | Yes               |

*Notes.* This table displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variables are various measures of patent importance. In columns (1)–(3), we residualize the measure with respect to year fixed effects before aggregating across patents; in columns (4)–(6), we use the raw measure. In columns (1) and (4), importance is constructed by comparing patents with those issued in the previous year; in columns (2) and (5), the comparison is with patents issued within the previous five years; in columns (3) and (6), it is with patents issued within the previous ten years. In Panels A and B, the dependent variables are the numbers of patents in the bottom and top 10% of each importance distribution, respectively. Each regression is estimated using Poisson quasi-likelihood, as in (3). All regressions include surname and match-by-year fixed effects. Standard errors are clustered by surname and reported in parentheses. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 24.

**Table C.7.** Effect of Lynching Episodes on Innovation: Individual-Level Evidence, Robustness Analysis

| Dependent Variable:            | (1)                  | (2)                 | (3)                  | (4)                      | (5)                 | (6)                  | (7)                      | (8)                | (9)               |
|--------------------------------|----------------------|---------------------|----------------------|--------------------------|---------------------|----------------------|--------------------------|--------------------|-------------------|
|                                | Number of Patents    |                     |                      | I(Number of Patents > 0) |                     |                      | Number of Top 5% Patents |                    |                   |
| Panel A. All Inventors         |                      |                     |                      |                          |                     |                      |                          |                    |                   |
| Lynching Episode $\times$ Post | -0.079***<br>(0.026) | -0.095**<br>(0.040) | -0.055***<br>(0.021) | -0.038*<br>(0.020)       | -0.070**<br>(0.034) | -0.055***<br>(0.021) | -0.497<br>(0.303)        | 1.104**<br>(0.515) | -0.325<br>(0.308) |
| Number of Individuals          | 9,015                | 6,336               | 8,888                | 8,888                    | 6,106               | 8,888                | 507                      | 63                 | 507               |
| Observations                   | 64,038               | 40,810              | 60,108               | 60,108                   | 37,622              | 60,108               | 1,878                    | 252                | 1,878             |
| R <sup>2</sup>                 | 0.312                | 0.368               | 0.082                | 0.081                    | 0.139               | 0.082                | 0.411                    | 0.564              | 0.420             |
| Panel B. White Inventors       |                      |                     |                      |                          |                     |                      |                          |                    |                   |
| Lynching Episode $\times$ Post | -0.090***<br>(0.026) | -0.093**<br>(0.043) | -0.058***<br>(0.021) | -0.047**<br>(0.020)      | -0.073**<br>(0.035) | -0.058***<br>(0.021) | -0.505*<br>(0.271)       | 1.003**<br>(0.452) | -0.200<br>(0.283) |
| Number of Individuals          | 8,097                | 5,624               | 7,984                | 7,984                    | 5,398               | 7,984                | 451                      | 44                 | 451               |
| Observations                   | 57,236               | 35,928              | 53,782               | 53,782                   | 33,054              | 53,782               | 1,662                    | 198                | 1,662             |
| R <sup>2</sup>                 | 0.312                | 0.370               | 0.082                | 0.082                    | 0.139               | 0.082                | 0.413                    | 0.589              | 0.423             |
| Individual FE                  | Yes                  | Yes                 | Yes                  | Yes                      | Yes                 | Yes                  | Yes                      | Yes                | Yes               |
| Match-Year FE                  | Yes                  | Yes                 | Yes                  | Yes                      | Yes                 | Yes                  | Yes                      | Yes                | Yes               |
| County-Year FE                 | No                   | Yes                 | No                   | No                       | Yes                 | No                   | No                       | Yes                | No                |
| Controls $\times$ Post         | No                   | No                  | Yes                  | No                       | No                  | Yes                  | No                       | No                 | Yes               |

*Notes.* This table displays the effect of lynching episodes on patented innovation. The unit of observation is an individual inventor, observed annually over a ten-year window around the census to which he is linked. Each inventor is matched to a non-treated control inventor. The dependent variable in columns (1)–(3) is the number of patents; in columns (4)–(6), it is an indicator equal to one if the inventor produces at least one patent and zero otherwise; in columns (7)–(9), it is the number of patents in the top 5% of the patent impact distribution. The treatment variable is an interaction between a surname-specific lynching indicator and a match-specific post-treatment indicator. In Panel A, the sample consists of all inventors who produced at least one patent before and after someone with their surname was exposed to a lynching episode between 1895 and 1925; in Panel B, we restrict the sample to white inventors only. The sample of Black inventors is too small to allow the same exercise. We draw a 20% random sample of white inventors. Each regression is estimated using Poisson quasi-likelihood. All regressions include inventor and match-by-year fixed effects. Standard errors are reported in parentheses and clustered by surname. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 24.

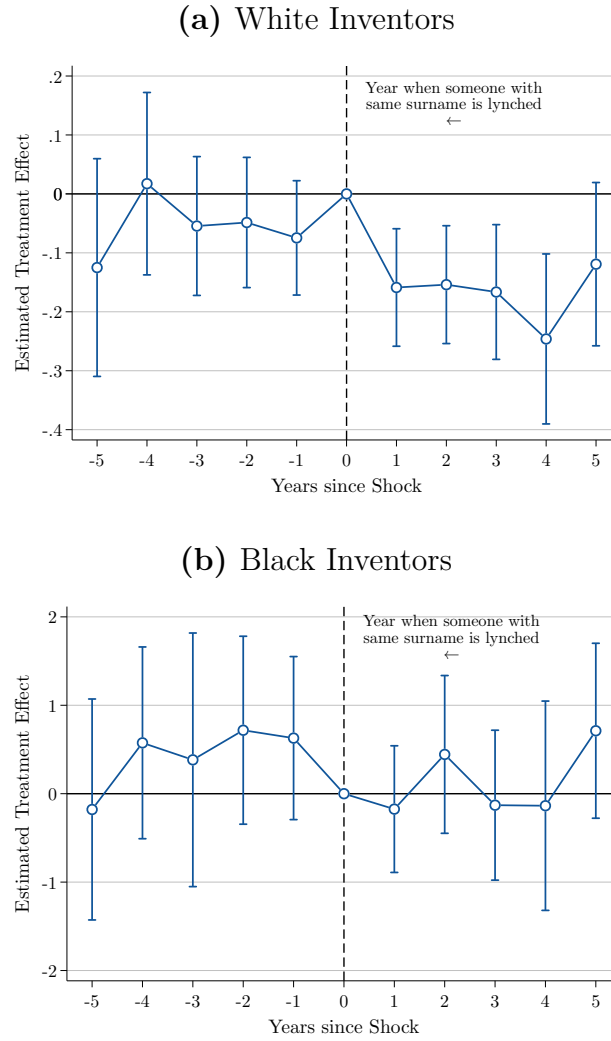
**Table C.8.** Surname-Level Effect of Lynching Episodes on Innovation: Newspaper Coverage

| Dependent Variable:                                                 | (1)               | (2)       | (3)       | (4)     | (5)     | (6)     |
|---------------------------------------------------------------------|-------------------|-----------|-----------|---------|---------|---------|
|                                                                     | Number of Patents |           |           |         |         |         |
| Episode Covered in News $\times$ Post                               | -0.030*           |           |           |         |         |         |
|                                                                     | (0.016)           |           |           |         |         |         |
| Episode Covered in News in DC, VA, or MD $\times$ Post              |                   | -0.060*** |           |         |         |         |
|                                                                     |                   | (0.021)   |           |         |         |         |
| Episode Covered in News in DC $\times$ Post                         |                   |           | -0.048*** |         |         |         |
|                                                                     |                   |           | (0.017)   |         |         |         |
| Episode Not Covered in News $\times$ Post                           |                   |           |           | -0.056  |         |         |
|                                                                     |                   |           |           | (0.048) |         |         |
| Episode Not Covered in News in neither DC, VA, nor MD $\times$ Post |                   |           |           |         | 0.006   |         |
|                                                                     |                   |           |           |         | (0.026) |         |
| Episode Not Covered in News in DC $\times$ Post                     |                   |           |           |         |         | -0.007  |
|                                                                     |                   |           |           |         |         | (0.020) |
| Surname FE                                                          | Yes               | Yes       | Yes       | Yes     | Yes     | Yes     |
| Match-Year FE                                                       | Yes               | Yes       | Yes       | Yes     | Yes     | Yes     |
| Number of Surnames                                                  | 1,226             | 946       | 814       | 100     | 522     | 724     |
| Observations                                                        | 12,798            | 9,004     | 8,674     | 1,078   | 4,948   | 7,540   |
| R <sup>2</sup>                                                      | 0.765             | 0.761     | 0.760     | 0.813   | 0.811   | 0.811   |

*Notes.* This Table displays how alternative measures of newspaper coverage of lynching episodes impact their effect on innovation. The unit of observation is a surname, observed at a yearly frequency over a ten-year window around the time when someone with the given surname is the victim of a lynching episode. Each treated surname is matched to a control non-treated last name. The dependent variable is the number of patents by surname. In columns (1), (2), and (3), the treatment variable codes an interaction between a post-lynching dummy and an indicator that returns a value of one if the lynching is covered in, respectively, any newspaper, a newspaper in DC and its neighboring states (Virginia and Maryland), or a newspaper in DC. In columns (4), (5), and (6), the treatment variable codes an interaction between a post-lynching dummy and an indicator which returns a value of one if the lynching is not covered in, respectively, any newspaper, any newspaper in DC, Virginia, or Maryland, or any newspaper in DC. Each regression is estimated as a Poisson quasi-likelihood regression, as in (3). All regressions include surname and match-by-year fixed effects. Standard errors are shown in parentheses and are clustered by surname. \*\*\*:  $p < 0.01$ , \*\*:  $p < 0.05$ , \*:  $p < 0.10$ . Referenced on page: 26.

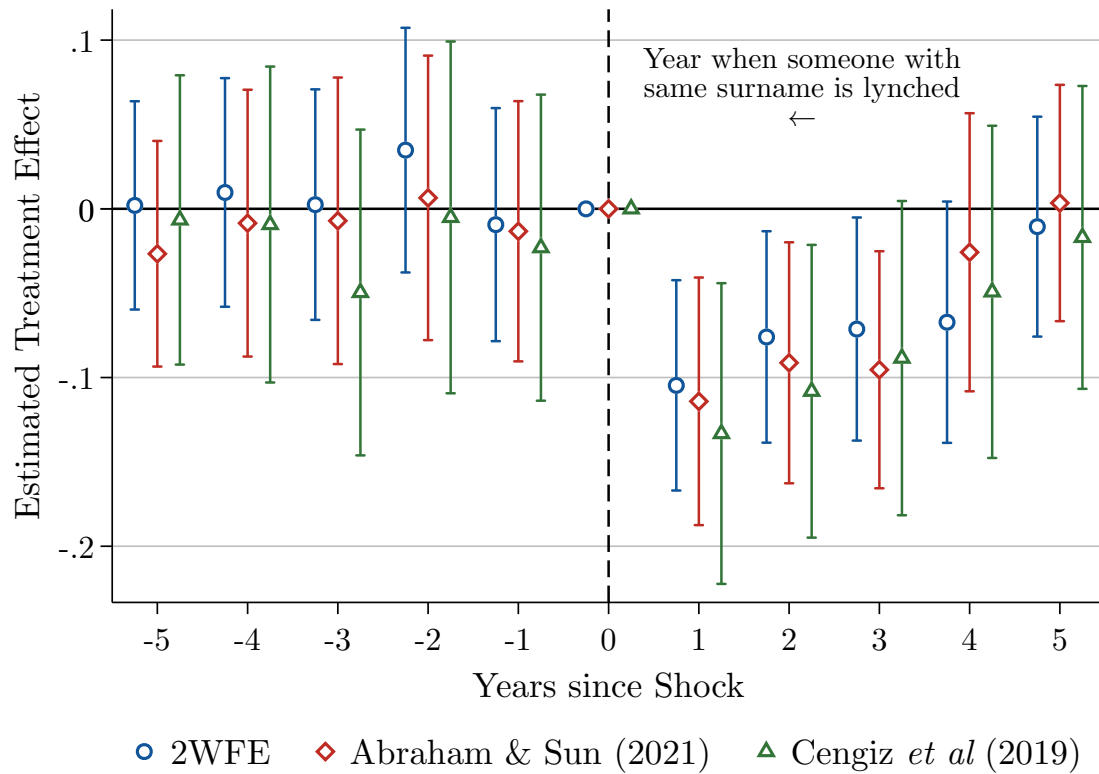
## C.II Figures

**Figure C.1.** Dynamic Individual-Level Difference-in-Differences Effect of Lynching Episodes on Innovation by Inventor Race



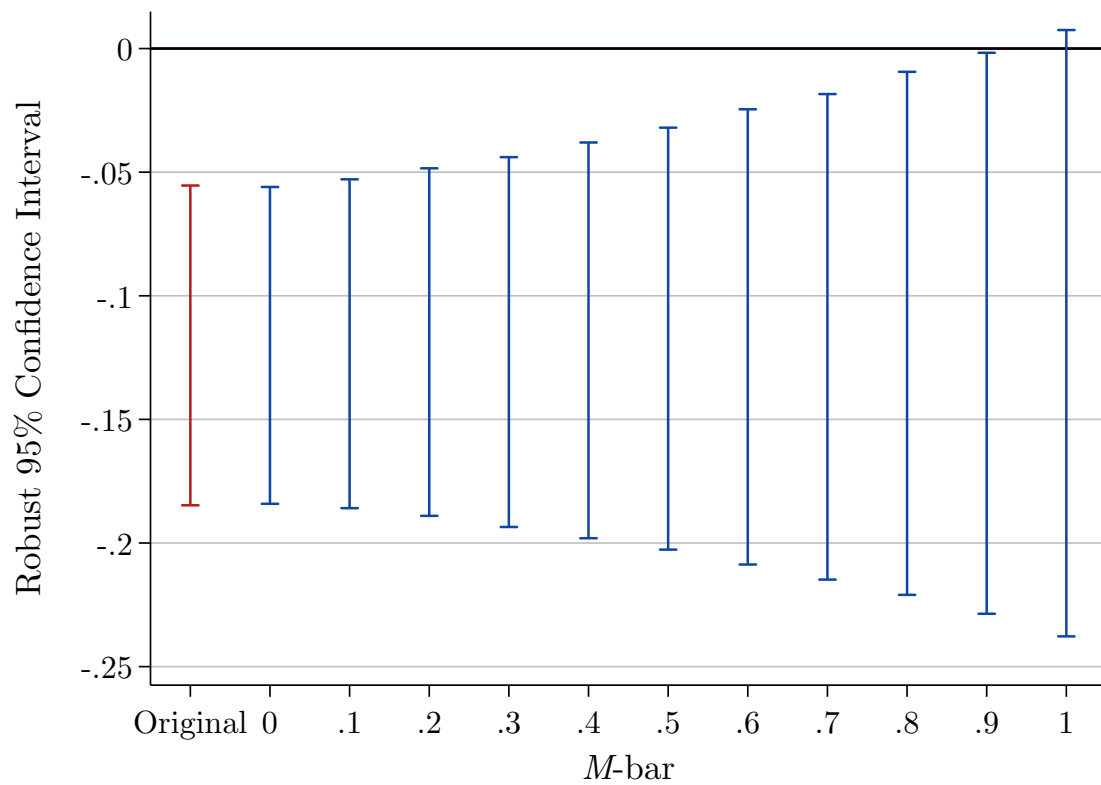
*Notes.* This figure displays the effect of lynching episodes on patented innovation. The unit of observation is an individual inventor, observed annually over a ten-year window around the census to which he is linked. Each treated inventor is matched to a non-treated control inventor. The dependent variable is the number of patents. The dots report the estimated coefficients on a set of interaction terms between a surname-specific lynching indicator and event-time dummies corresponding to the number of years since the lynching episode. Event time is shown on the x-axis. The omitted category is the year when someone with a given surname is treated. In Panel C.1a, the sample includes only white inventors; in Panel C.1b, the sample includes only Black inventors. The model is estimated using Poisson quasi-likelihood, as in (3), and includes individual and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 23.

**Figure C.2.** Dynamic Difference-in-Differences Effect of Lynching Episodes on Innovation: Alternative Estimators



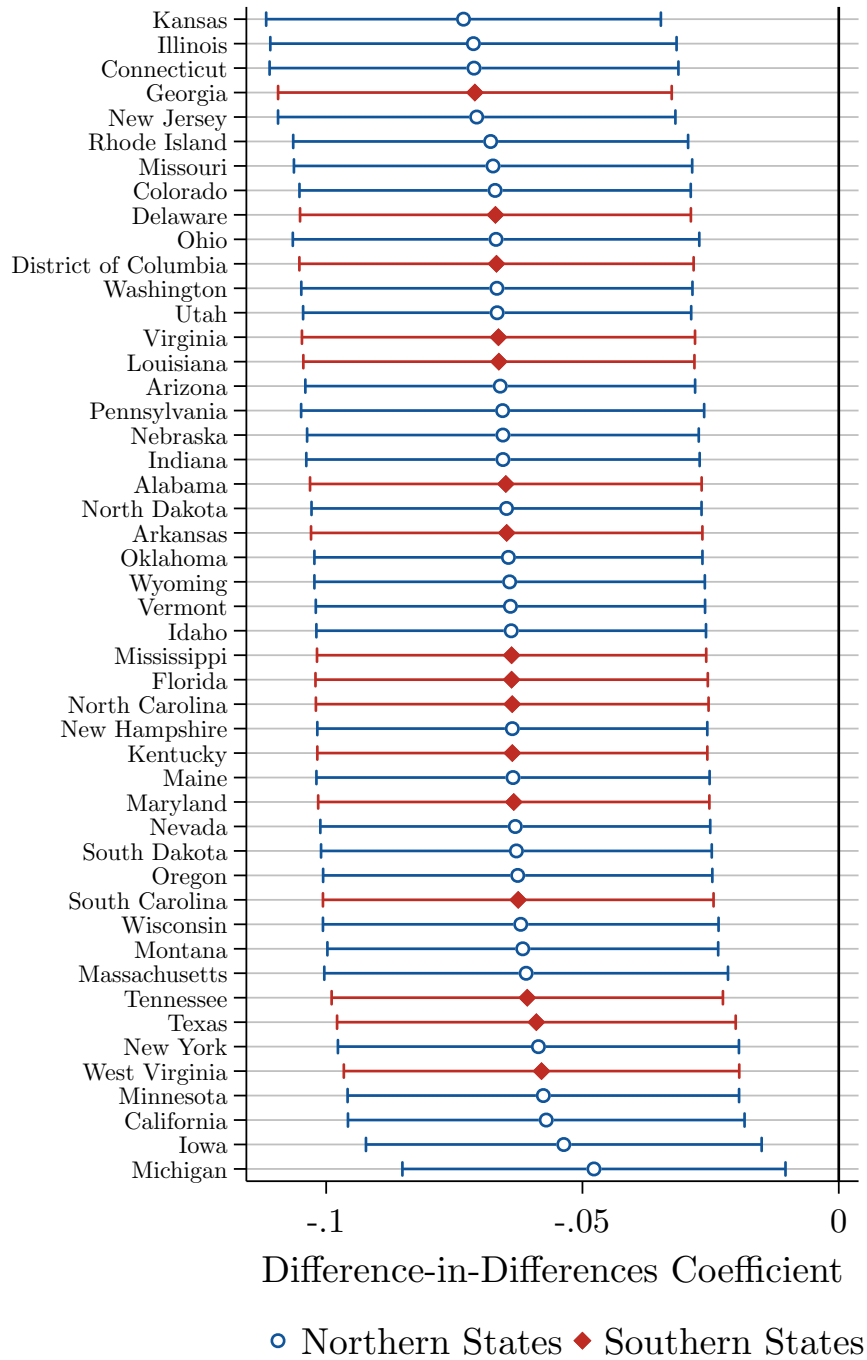
*Notes.* This figure displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable is the number of patents. The dots report the estimated coefficients on a set of interaction terms between a surname-specific lynching indicator and event-time dummies corresponding to the number of years since the lynching episode. The blue dots report the baseline two-way fixed effects estimates; the red dots report estimates using the estimator developed in Sun and Abraham (2021); the green dots report estimates using the estimator developed in Cengiz et al. (2019). Event time is shown on the x-axis. The omitted category is the year in which someone with a given surname is treated. All models are estimated using Poisson quasi-likelihood, as in (3), and include surname and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 24.

**Figure C.3.** Relative Magnitude of Parallel Trends Violation



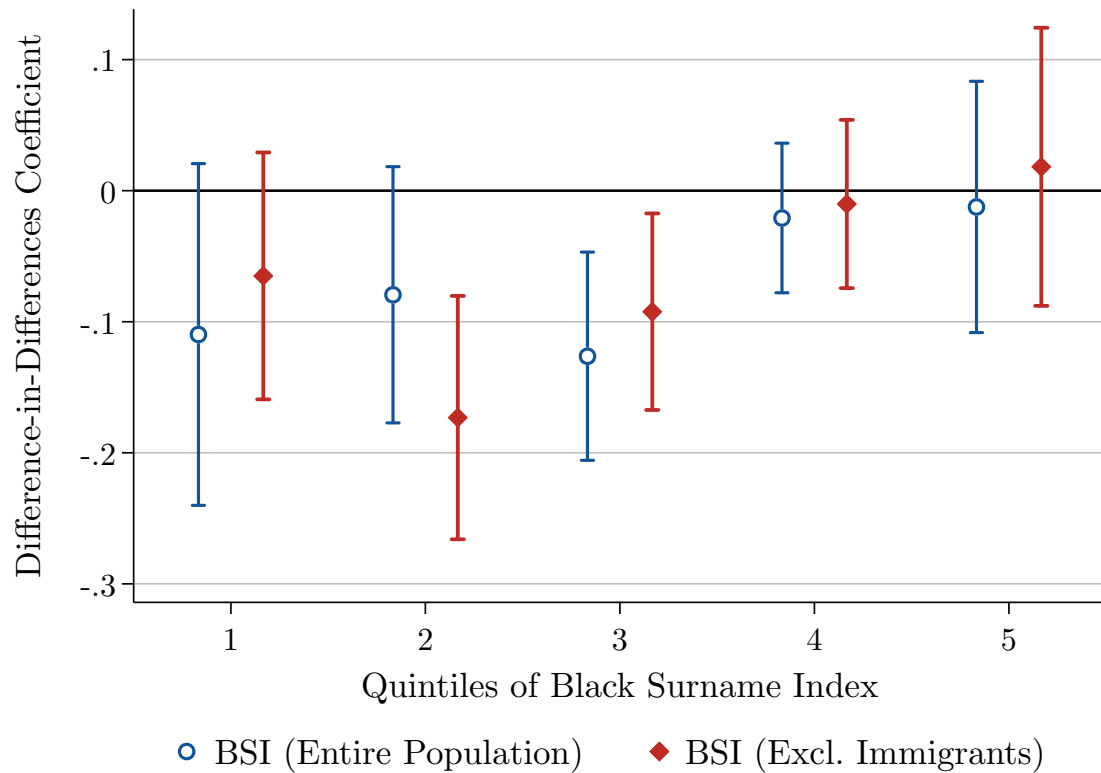
*Notes.* This figure reports alternative confidence intervals for the effect of lynching episodes on the volume of innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable is the number of patents. The model is estimated using a Poisson pseudo-likelihood specification, as in (3), and includes surname and match-by-year fixed effects. The bands report 95% robust confidence intervals constructed using the method developed by Rambachan and Roth (2023).  $\bar{M}$  bounds the magnitude of post-treatment violations of parallel trends relative to the largest pre-treatment violation; the breakdown value is the largest  $\bar{M}$  under which the result remains statistically significant. Referenced on page: 24.

**Figure C.4.** Lynching Episodes and Innovation: State-by-State Leave-out



*Notes.* This figure displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the year in which someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable is the number of patents. The treatment variable is the interaction between a surname-specific lynching indicator and a post-treatment, match-specific indicator. Each regression is estimated using a Poisson quasi-likelihood model, as in (3). Each row reports the estimated treatment effect from a separate regression excluding all inventors in a given state. Red diamonds denote the coefficients when the excluded state is in the South. All regressions include surname and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 24.

**Figure C.5.** Heterogeneous Effects of Lynching Episodes on Innovation across the BSI Distribution



*Notes.* This figure displays the effect of lynching episodes on patented innovation. The unit of observation is a surname, observed annually over a ten-year window around the time when someone with that surname is the victim of a lynching episode. Each treated surname is matched to a non-treated control surname. The dependent variable is the number of patents. The treatment variable is an interaction between a surname-specific lynching indicator, a match-specific post-treatment indicator, and a set of indicators for the quintiles of the Black Surname Index distribution. The blue dots display the coefficients using the BSI computed over the full population, whereas the red dots display the coefficients using the BSI constructed excluding first-generation immigrants. Each regression is estimated using Poisson quasi-likelihood, as in (3). All regressions include surname and match-by-year fixed effects. The bands report 95% confidence intervals. Standard errors are clustered by surname. Referenced on page: 27.

## D Newspaper Coverage of Lynching Episodes

### D.I Automated Construction of Newspaper-Coverage Measures

We measure newspaper coverage using automated searches of the `newspapers.com` archive. For each lynching episode in the analysis sample with a recorded victim name and year, an R script using `RSelenium` searches for the victim’s exact full name in the year of the episode. Our preferred search combines the name with the terms `(lynched OR lynching)` and `(negro OR colored)`. The script records the number of matching results across the full newspaper archive and after restricting the publication location to the District of Columbia. We classify an episode as covered in the relevant geographic area when the corresponding search returns at least one result.

The resulting indicators are used in the newspaper-coverage heterogeneity analyses. Because automated search results may include references to other individuals with the same name or articles that do not provide enough information to identify the event, we assess the procedure using the manual validation exercise described below.

### D.II Manual Validation of Newspaper Coverage

In this appendix, we report a simple validation exercise to assess newspaper coverage of lynching episodes and the informativeness of the automated searches. We find that the vast majority of episodes were reported in newspapers, and that articles often mentioned the victim’s name and race. Moreover, a substantial share of episodes was covered by newspapers based in Washington, D.C.

We draw a random sample of 128 lynching episodes involving Black victims recorded in Seguin and Rigby (2019) and Hines and Steelwater (2006). We sample from all states, drawing at most five episodes per state.<sup>6</sup>

**Table D.1.** Newspaper Coverage of Sampled Lynching Episodes

| Coverage criterion                                      | Episodes | Denominator | Share |
|---------------------------------------------------------|----------|-------------|-------|
| Victim’s name appears in any newspaper                  | 114      | 128         | 89.1% |
| Victim’s name appears in a Washington, D.C. newspaper   | 71       | 128         | 55.5% |
| Victim’s name appears in a Washington, D.C. newspaper   | 71       | 114         | 62.3% |
| Washington, D.C. article confirms Black lynching victim | 44       | 71          | 62.0% |

*Notes.* The table reports results from a manual validation exercise based on a random sample of 128 lynching episodes involving Black victims recorded in Seguin and Rigby (2019) and Hines and Steelwater (2006). The first row reports whether the victim’s full name appears in any newspaper archived in `newspapers.com` in the year of the lynching episode. The second and third rows report Washington, D.C. newspaper coverage using two different denominators: all sampled episodes and episodes covered anywhere in the newspaper corpus. The final row reports the share of Washington, D.C.-covered episodes for which the article itself confirms that the named person was Black and was the victim of a lynching episode.

For each episode, we first manually verify whether at least one newspaper archived in `newspapers.com` mentions the victim’s full name in the year of the lynching episode. In nearly 90% of cases (114 out

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<sup>6</sup>Existing scholarship provides a more comprehensive validation of this hypothesis (Perloff, 2000). Nonetheless, because newspaper coverage of lynching episodes is a central assumption underlying our empirical strategy, we conduct this manual check.

of 128), the victim's name is reported in a newspaper somewhere in the country that is part of their corpus. This is not surprising, since the underlying lynching data are themselves drawn from historical newspapers, but it nonetheless illustrates the broad coverage available in the digitized newspaper sample.

Second, we examine whether newspapers in Washington, D.C., where the patent office was located, also covered these lynching episodes and identified the victims by name. For 71 episodes, at least one newspaper based in Washington, D.C., mentioned the victim's name in the year of the lynching episode. This corresponds to more than half of the sample (55.5% of all 128 episodes) and 62% of the relevant sample of lynchings covered anywhere, and supports our assessment that newspapers in Washington, D.C., reported on lynching episodes from across the country.

Lastly, we assess the accuracy of this approach. To do so, we read the individual articles for each of the 71 lynching episodes covered by Washington, D.C., newspapers and verify whether at least one article indicates that a person with that name was a Black individual and was the victim of a lynching episode. This is the case for 44 episodes (61% of the 71 cases).

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